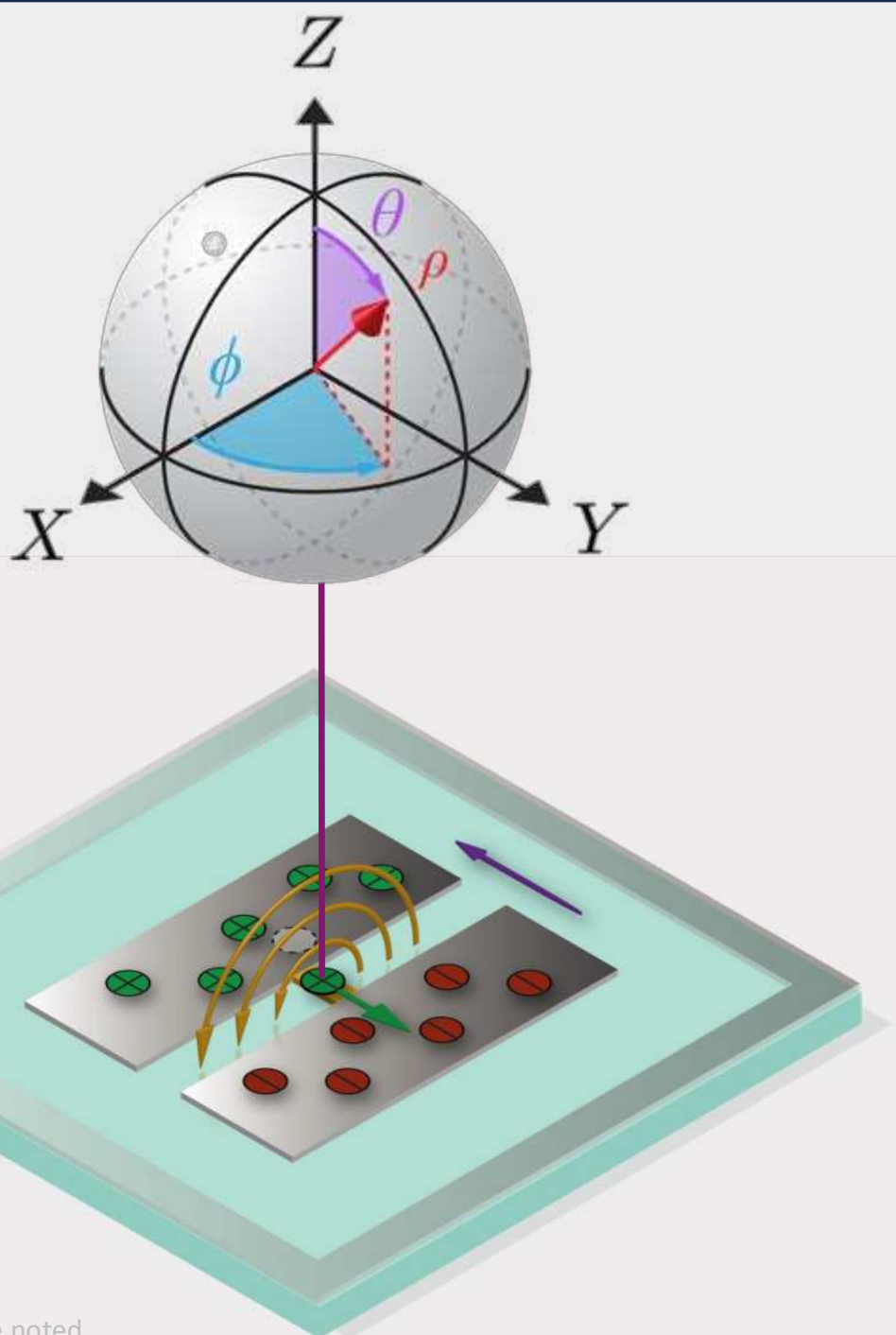




Your First Superconducting Qubit

Introduction to **Circuit** Quantum Electrodynamics (cQED)

Zlatko K. Mineev

Current Affiliation:

Google Quantum AI; Assoc. Fellow @ CIFAR

Former Affiliation: IBM Quantum; Yale University



@zlatko_mineev



zlatko-mineev.com/blog

Huge THANK YOU to the organizers and broader team!

Meet the Team

This is the organizational team responsible for making this workshop happen. The combined efforts between representatives from UCLA, USC and Google Quantum AI is responsible for the stellar lineup and broad range of topics. If you have any questions or concerns please reach out to:

quantum.ucla@gmail.com



Eli Levenson-Falk



Zlatko Minev



Mark Gyure



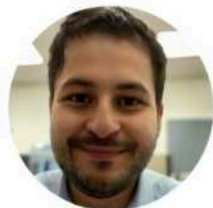
Alex Jürgens



Nicolas Dirneger



Cody Fan



Murat C. Sarihan



Samuel Oh



Drew Downing



Meet Our Speakers



Dr. Ofer Naaman
Ofer Naaman (he/him) is a research scientist with Google Quantum AI, where he leads the readout hardware team. Ofer holds a BSc degree from Tel Aviv University and a PhD in physics from UC San Diego. Prior to joining Google, Ofer was at MIT, Boulder, UC Berkeley, and Northrop Grumman, where he has worked on topics ranging from single-electron transistors, quantum computing, and parametric amplifiers, to superconducting logic and cryogenic memory.



Professor Jens Koch
Professor Jens Koch is a theoretical physicist specializing in quantum information and condensed matter physics. He is a faculty member at Northwestern University, where his research focuses on superconducting circuits, quantum electronics (QED), and the design of qubits for quantum computing. Prof. Koch is widely recognized for co-developing the transmon qubit, a more robust type of superconducting qubit that has become a foundational element in many quantum computing architectures.



Benjamin Jarvis-Frain
Ben is a quantum integrated circuit designer at Rigetti Computing where he works on modeling and design of superconducting quantum processors. Ben's work at Rigetti has focused on developing scalable simulation techniques for quantum processors in 3D integrated environments, and understanding parasitic, multi-qubit interactions that arise in these systems.



David Pahl
David is a PhD student at MIT under Prof. Will Oliver. Before coming to MIT, David studied Electrical Engineering and Information Technology at ETH Zurich, where he received a BSc in 2022 with distinction. At MIT, his research aims to develop scalable hardware architectures for fast and high-fidelity qubit readout. He is also designing quantum devices, capable of near-term quantum error correction.



Meet Our Speakers



Dr. Murat Can Sarihan
Dr. Murat Can Sarihan is a research scientist at Google Quantum AI, working on NISQ applications of superconducting quantum devices. He completed his PhD as a Fulbright scholar at University of California, Los Angeles (UCLA) with a thesis on quantum networking. He worked on all the aspects of a practical quantum network, i.e. generation, transmission, and storage of quantum information, plus efficient coding and error correction schemes for quantum key generation.



Lukas Pahl
Lukas is a graduate student researcher in the Engineering Quantum Systems group at MIT, advised by Prof. Hakan Türeci. He received his BSc in Electrical Engineering and Information Technology at ETH Zurich with distinction. At MIT, Lukas aims to explore techniques that will deliver superconducting quantum computers at scale. His research involves improving qubit gate fidelities through realistic control as well as the development and demonstration of low-overhead quantum error correction architectures.



Dung Pham
Dung (Strong) is a Ph.D. student in the Quantum Engineering group at Princeton University, advised by Prof. Hakan Türeci. He earned his B.Sc. degree in Physics and Electrical & Computer Engineering from Worcester Polytechnic Institute (WPI) at Princeton. His research focuses on developing theoretical techniques for modeling charged quantum fluids and superconducting quantum circuits, along with building computational tools for large-scale, efficient simulation of these devices.



Dr. Daniel Sank
Dr. Dan Sank is a physicist and researcher at Google Quantum AI, where he works on the development of superconducting qubits and scalable quantum computing systems. He has been a key contributor to the engineering of Google's quantum processors, including the Sycamore chip used in their demonstration of quantum supremacy. Dr. Sank focuses on improving qubit coherence, microwave control, and system integration, bridging the gap between quantum hardware and practical computation.



Meet Our Speakers



Professor Eli Levenson-Falk
Professor Eli Levenson-Falk is a physicist and researcher at Google Quantum AI, where he works on advancing the field of quantum computing through innovations in superconducting qubit technologies and quantum control. He is renowned for his pioneering experiments that captured and realized a quantum state in real time, offering profound insight into the nature of quantum state quantum physics research.



Dr. Zlatko Minev
Dr. Zlatko Minev is a physicist and researcher at Google Quantum AI, where he works on advancing the field of quantum computing through innovations in superconducting qubit technologies and quantum control. He is renowned for his pioneering experiments that captured and realized a quantum state in real time, offering profound insight into the nature of quantum state quantum physics research.



Dr. Jin-Sung Kim
An experimental quantum physicist by training, Dr. Kim is currently responsible for developing the ecosystem of the NISQ Quantum Platform. He manages technical collaborations and external partnerships, and ensures that users have everything they need to develop, provision, and manage their quantum applications. He received his PhD from Princeton University studying electron spin control in metal-oxide-semiconductor quantum dots.



Professor Hakan Türeci
Professor Hakan Türeci is a theoretical physicist at Princeton University, specializing in quantum optics, quantum information, and the physics of open quantum systems. His research explores the fundamental behavior of light-matter interactions, non-equilibrium systems, and quantum technologies such as superconducting circuits. Prof. Türeci has made significant contributions in understanding how coherence and dissipation interplay in complex quantum systems, with applications in quantum computing and photonics.



Professor Michel Devoret
Professor Michel Devoret is a renowned experimental physicist who recently joined UC Santa Barbara and Google Quantum AI, following a distinguished career at Yale University. He is a pioneer in the field of superconducting quantum circuits, best known for co-developing the transmon qubit and advancing quantum circuit architectures. His research has been instrumental in shaping the architecture of today's quantum processors, blending deep theoretical insight with experimental innovation.



Meet Our Speakers



Dr. Loren D. Aiegria
Dr. Loren D. Aiegria is a staff scientist at Lawrence Livermore National Laboratory, specializing in the materials science understanding of low-dissipation information processing technologies. As part of the Quantum Coherent Device Physics Group, he focuses on the design, microfabrication, and testing of innovative superconducting architectures to explore the fundamental limits of scalability in quantum information processors.



Sadman Ahmed Shanto
Sadman Ahmed Shanto is a Ph.D. candidate in Physics at the University of Southern California, working in the Levenson-Falk Lab on superconducting quantum hardware. His research focuses on quasiparticle dynamics in superconducting circuits, and the optimization of quantum device design workflows.



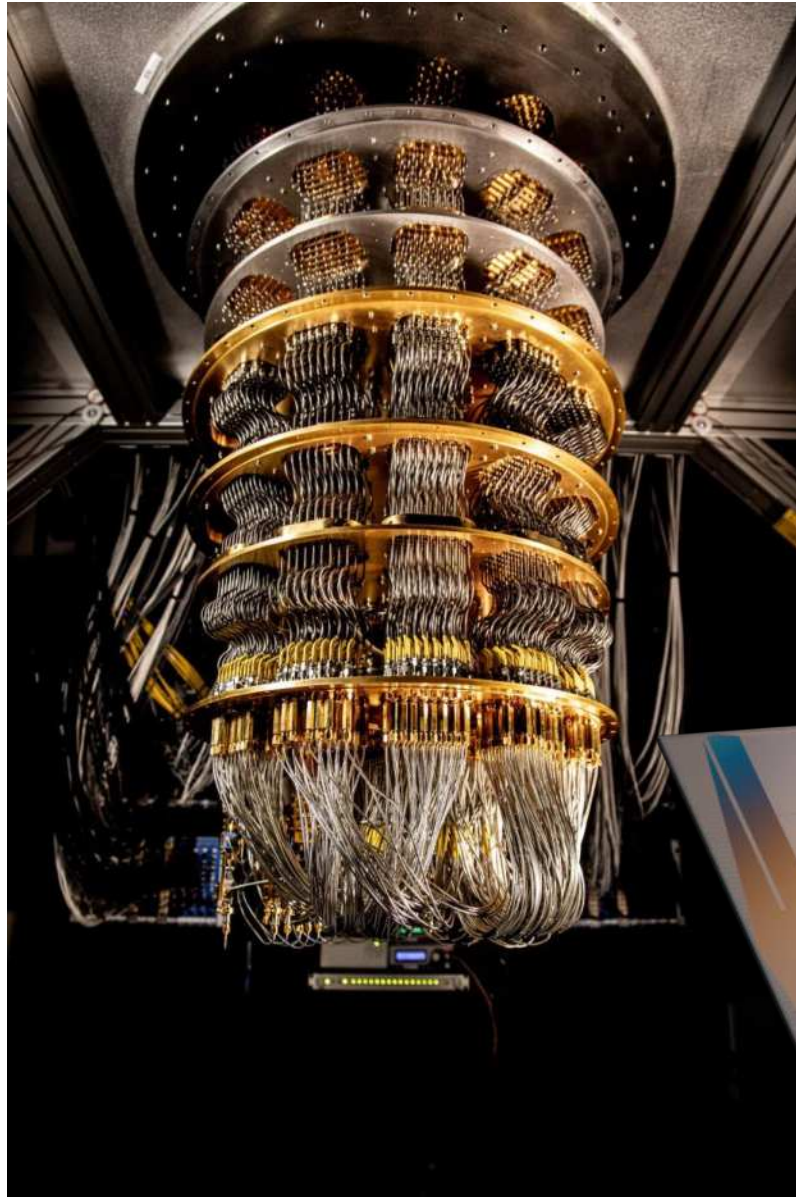
Dr. Arpit Arora
Arpit is a quantum scientist at NVIDIA, working at the interface of quantum materials, devices, and computing. His research focuses on understanding emergent phenomena in novel materials to inform the design of next-generation components for utility-scale quantum computers. Currently, he investigates photon-matter hybrid systems to advance coherent quantum technologies and enable new capabilities in quantum control and information processing.



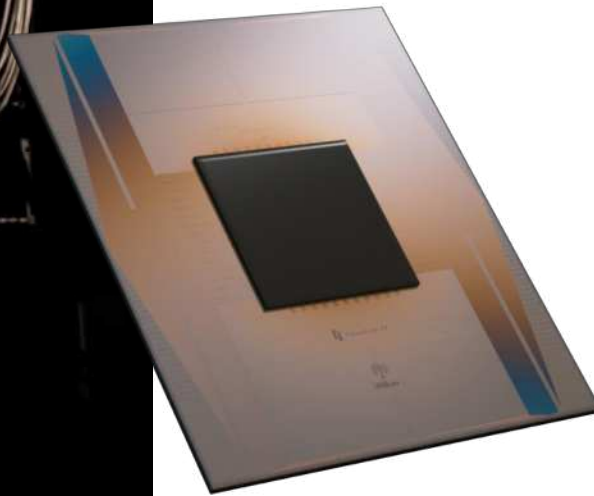
Professor Andreas Waltraff
Prof. Andreas Waltraff conducts research in quantum information processing and quantum optics. He has taught as a professor at ETH Zurich in Zurich, Switzerland since 2008. He worked as a research scientist with Robert J. Schoelkopf at Yale University from 2002 to 2005. His current work at ETH Zurich focuses on hybrid quantum systems combining superconducting electronic circuits with semiconductor quantum dots and individual Rydberg atoms as well as quantum error



Let's get to know each other 😊



1. Have you ever used a quantum device?
2. Analyzed/designed/made one?



This Lecture



Introductory and skill reaffirming

Don't need to know much going in, but we will go far

Examples: simplest, most practical examples

Step by step

Ask questions!

Lab work with Dr. Murat Sarihan,
Sadman Shanto, and more!



Advanced
material

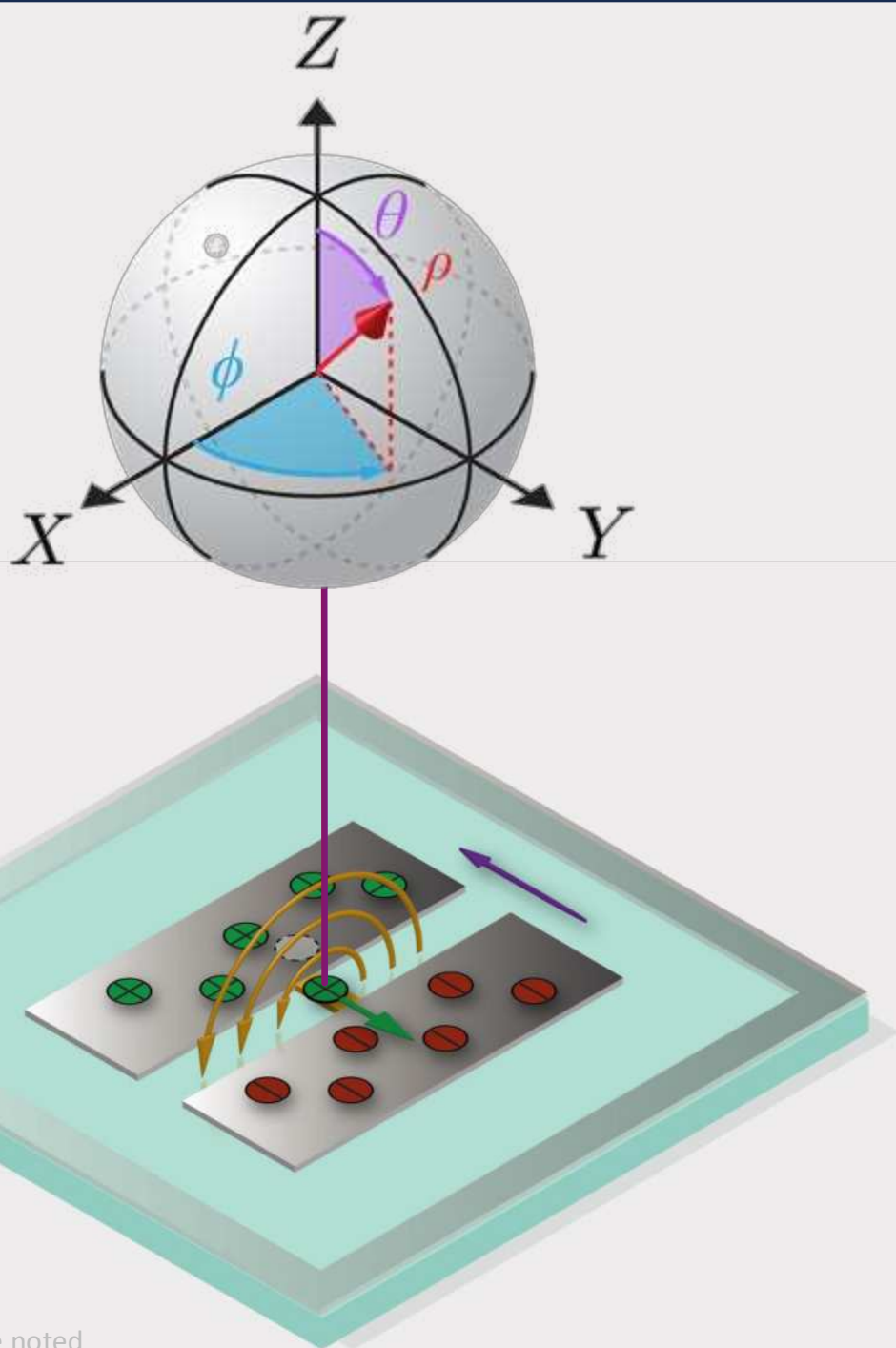


Slides!

Qubit

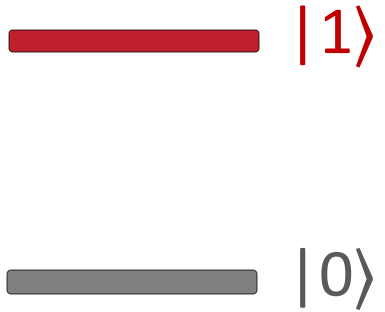
From Idea to Reality

THE BIG PICTURE
before calculations

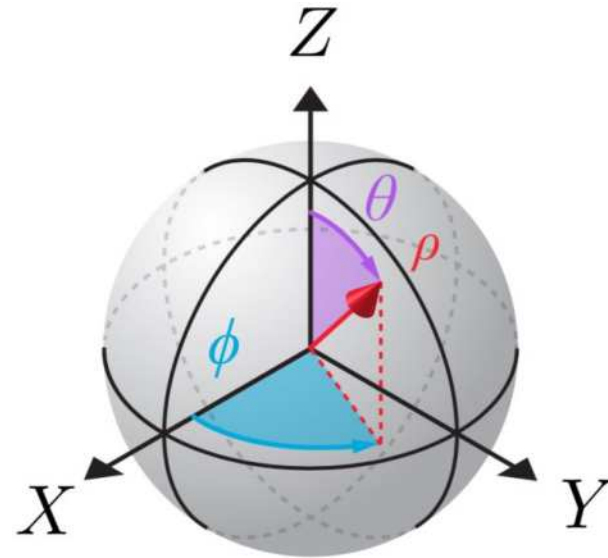


Qubit: idea

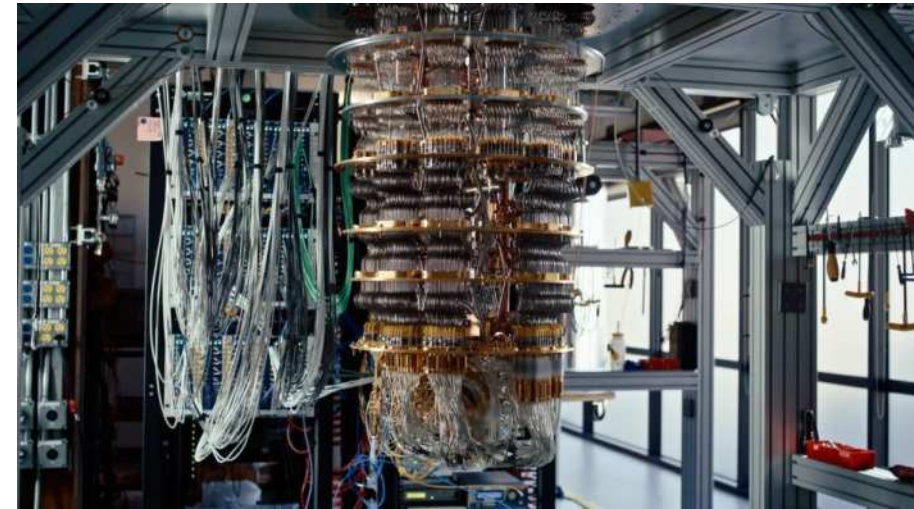
Energy levels

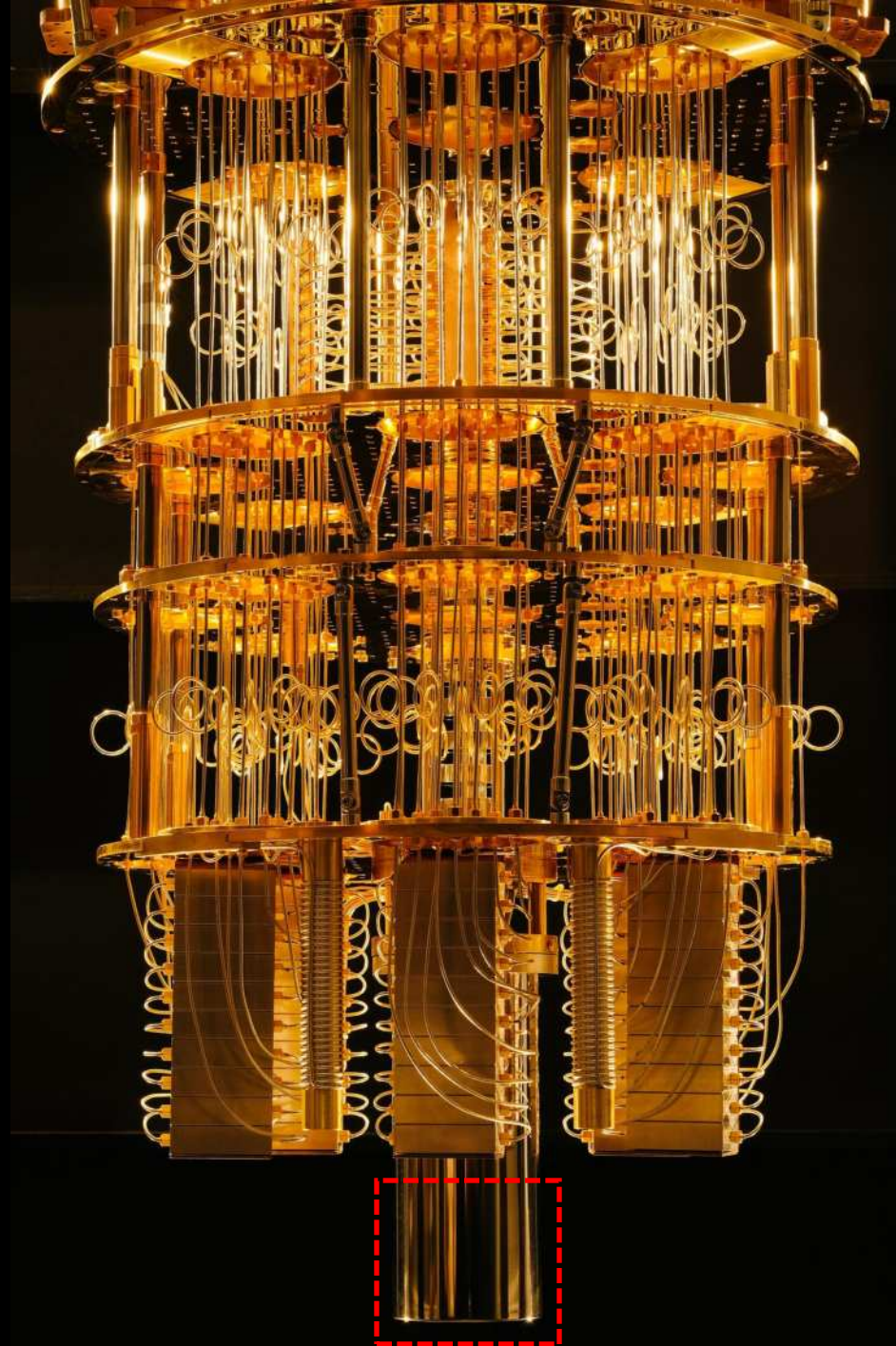


Hilbert space

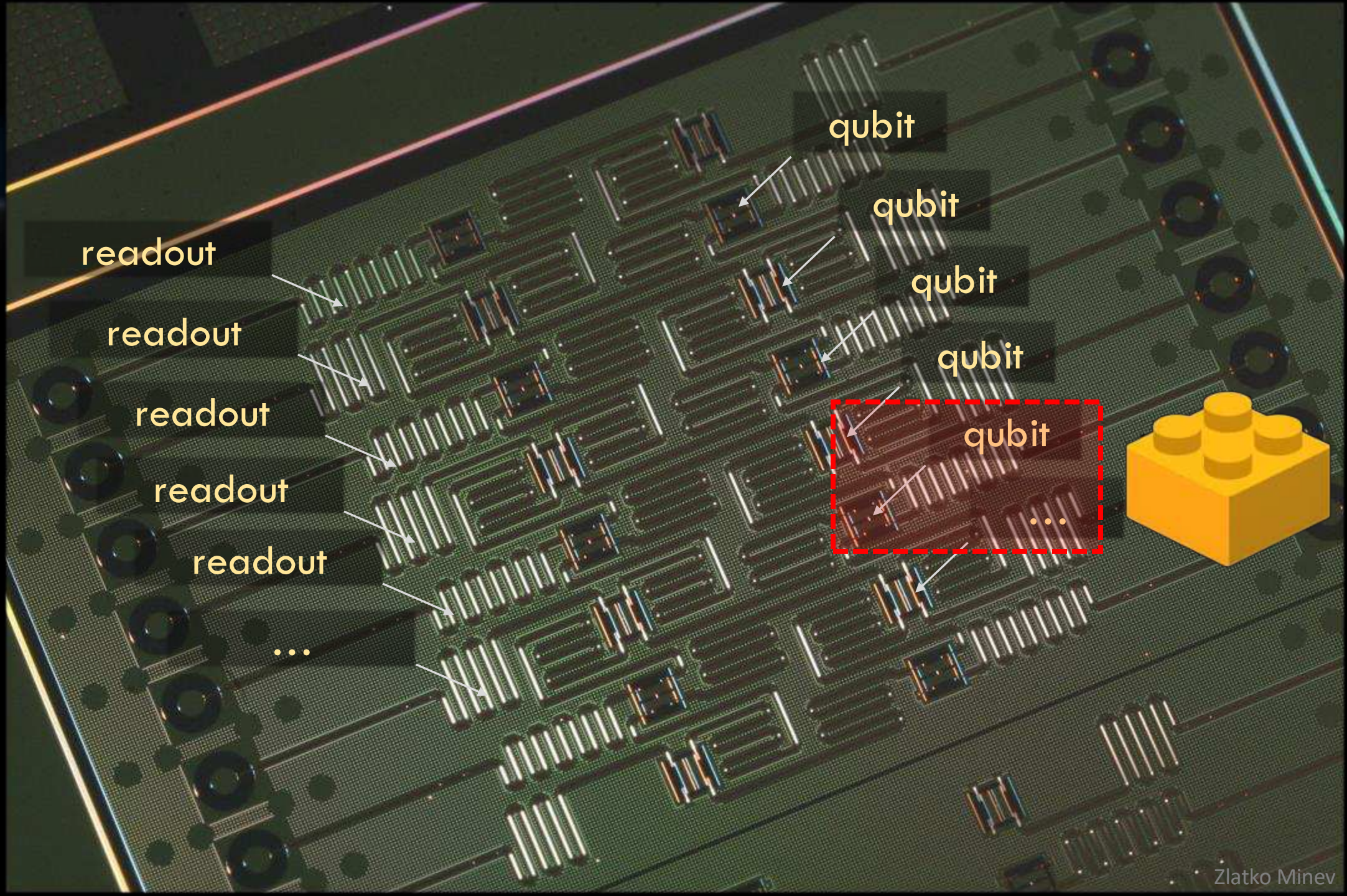
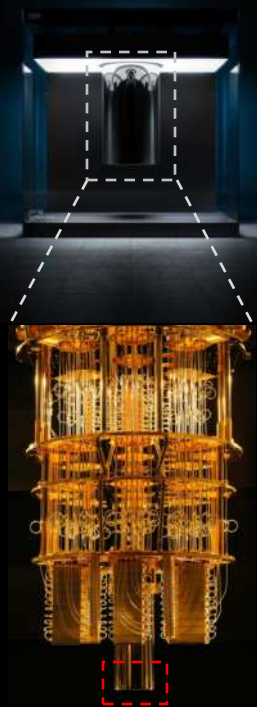


Quantum cloud?



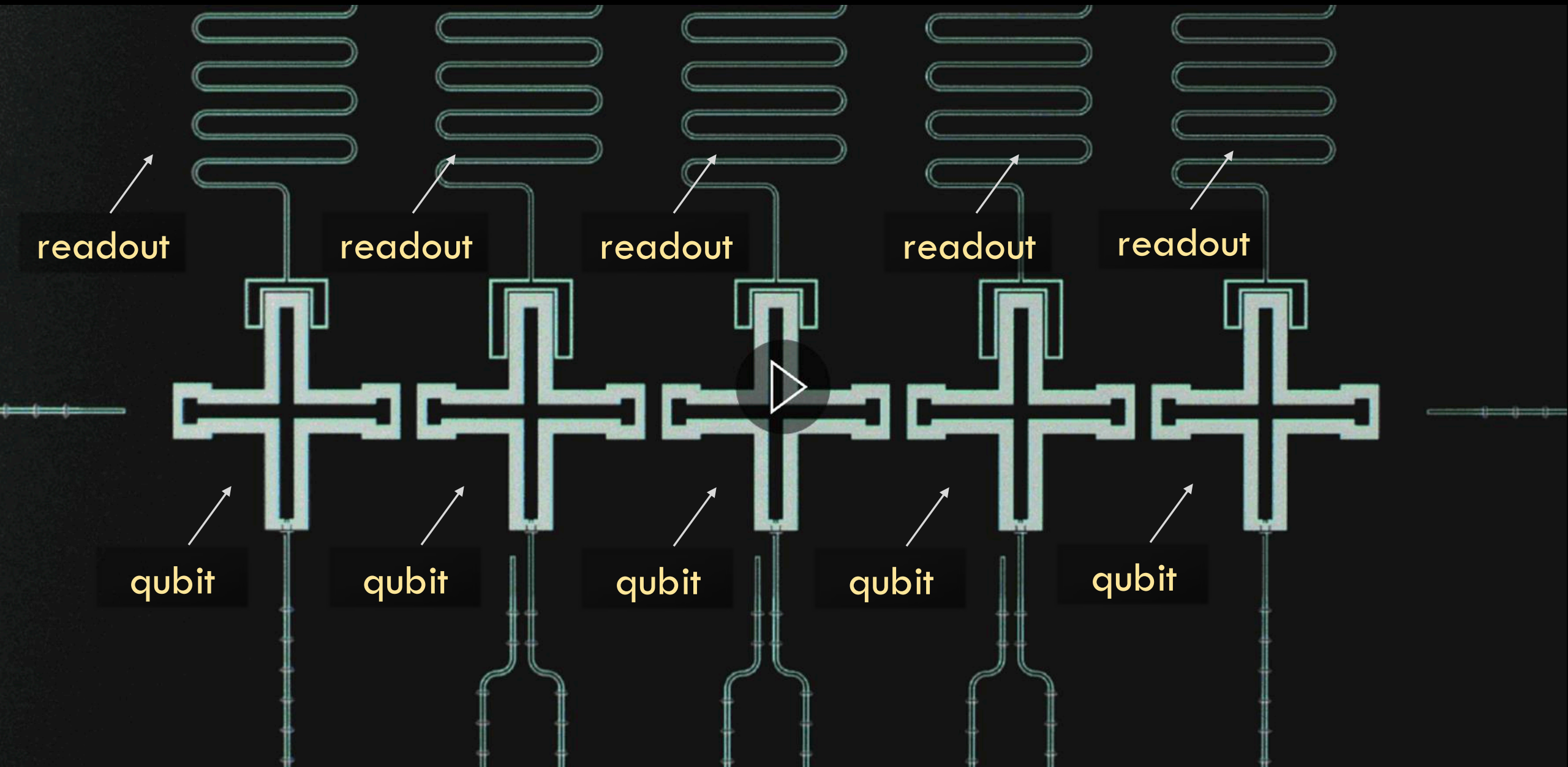


Operation at
15 mK (-273.13 °C)

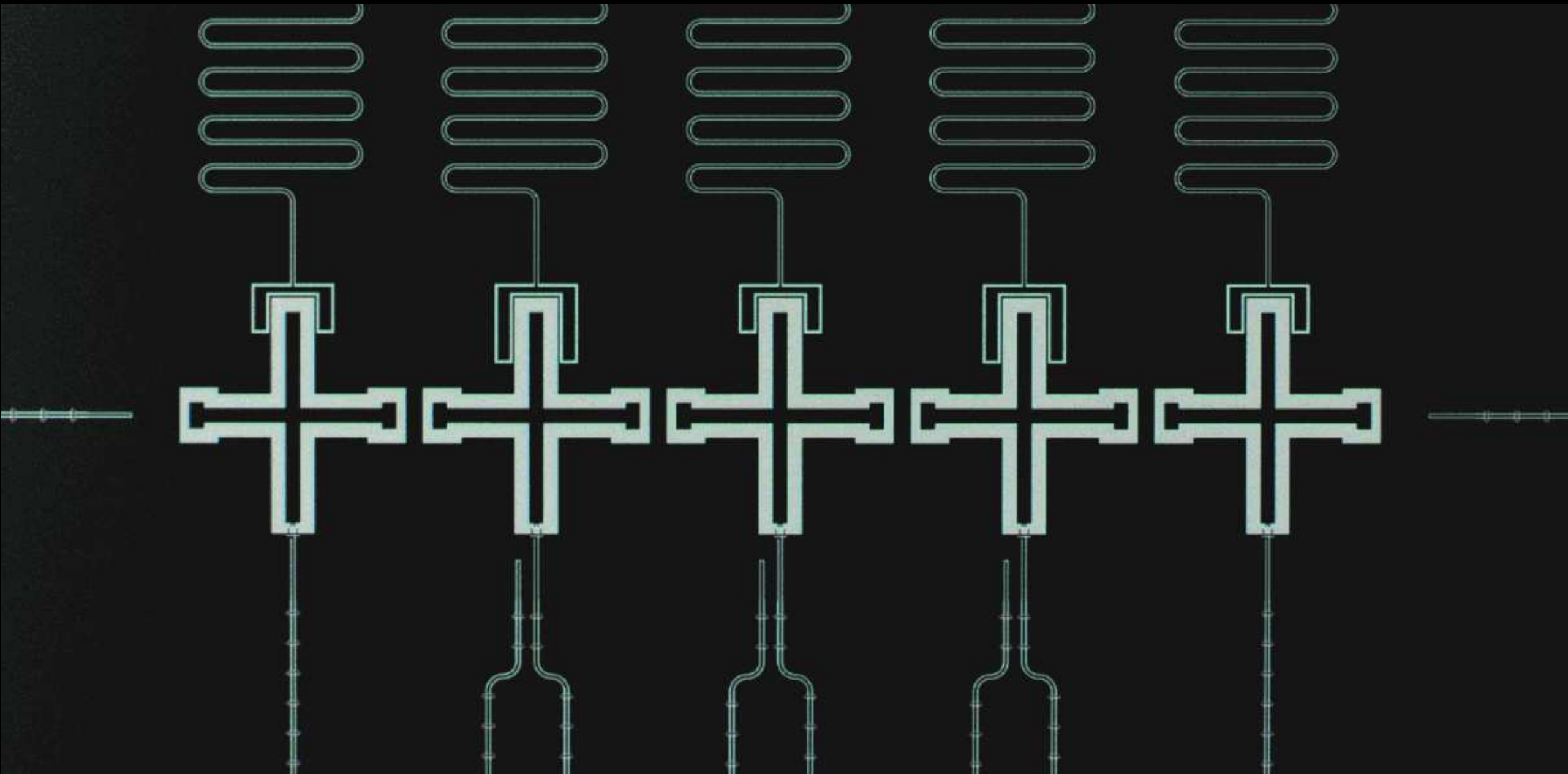


Example
IBM Architecture

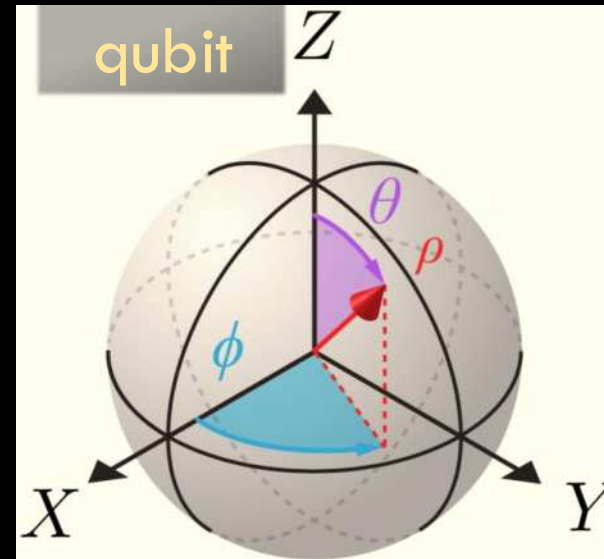
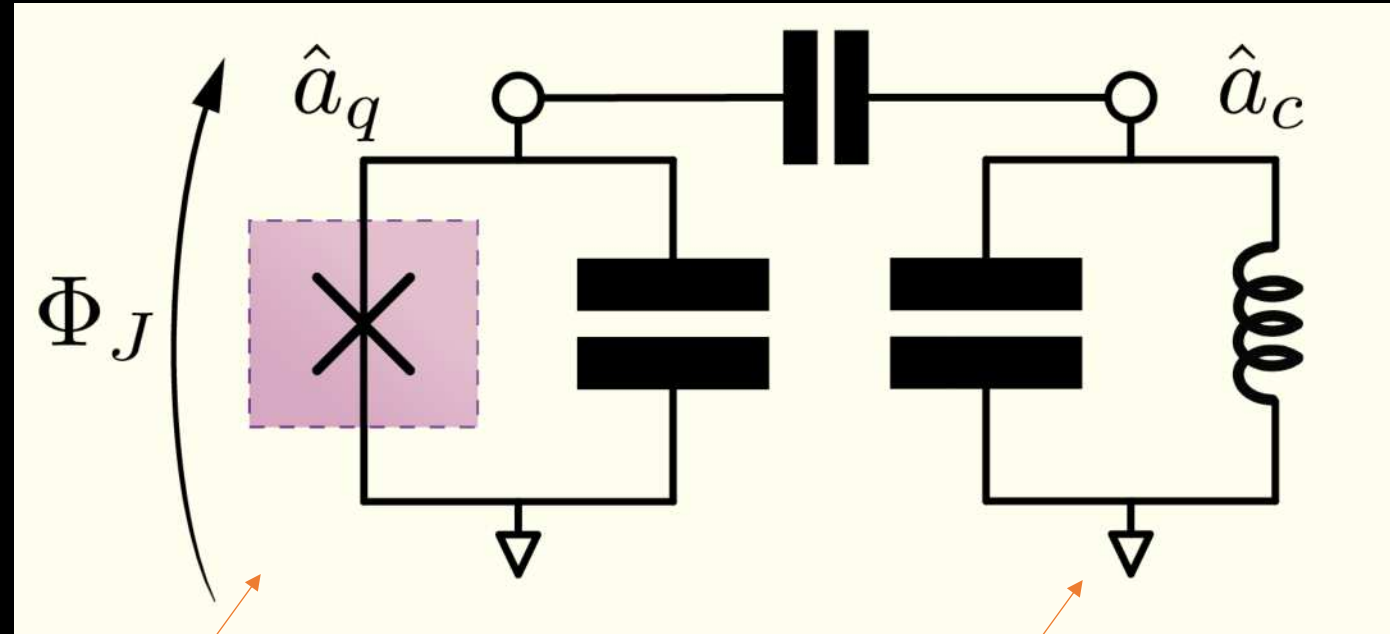
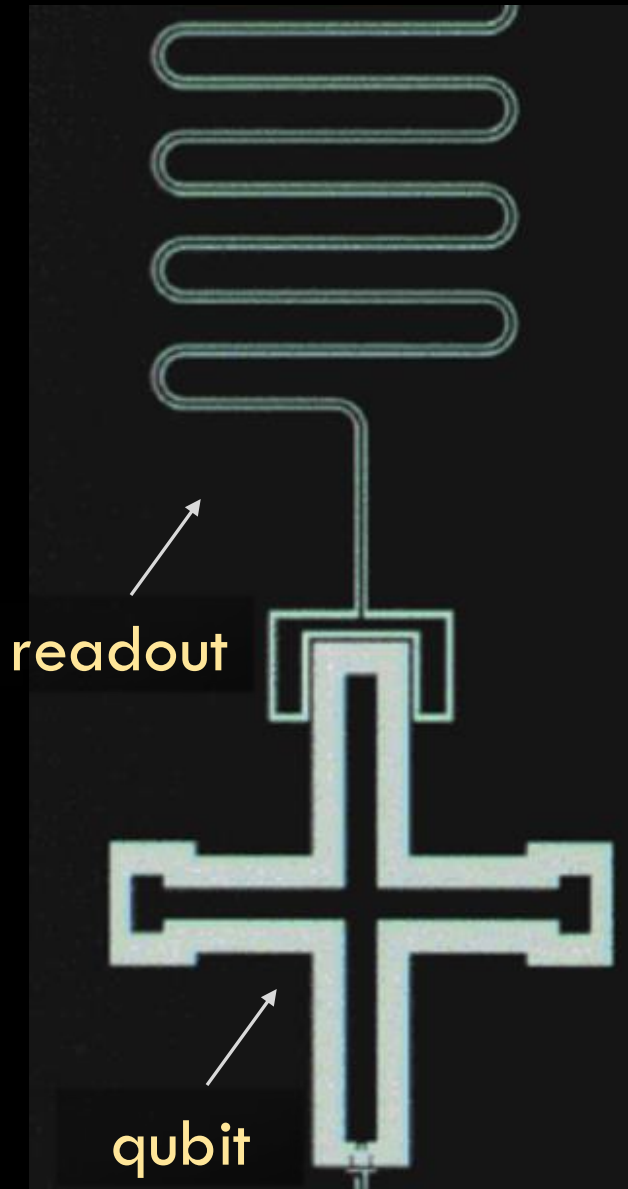
Example Google Architecture



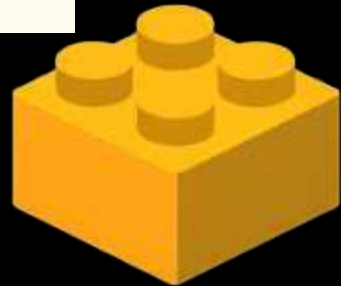
Example Google Architecture



Physical and abstract circuits and Hilbert space

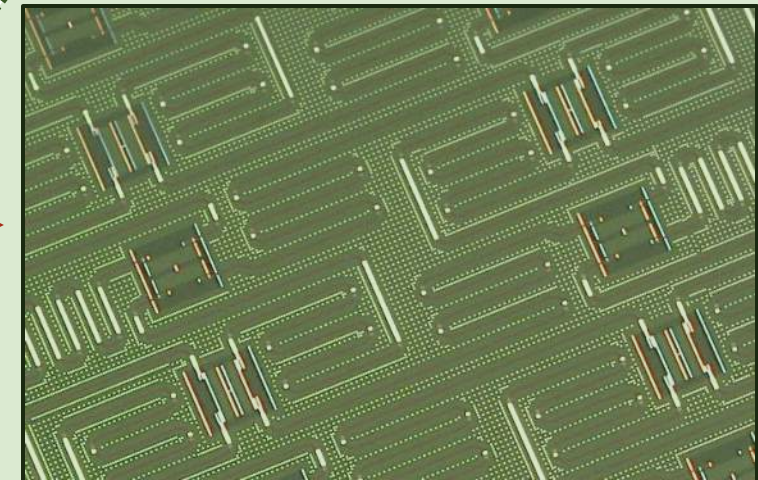
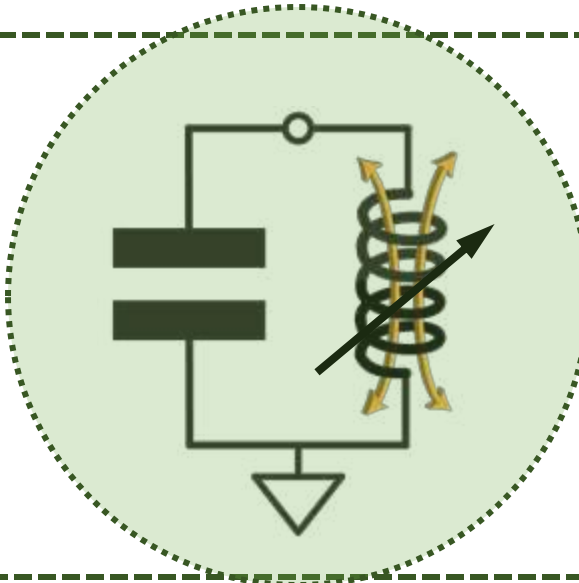
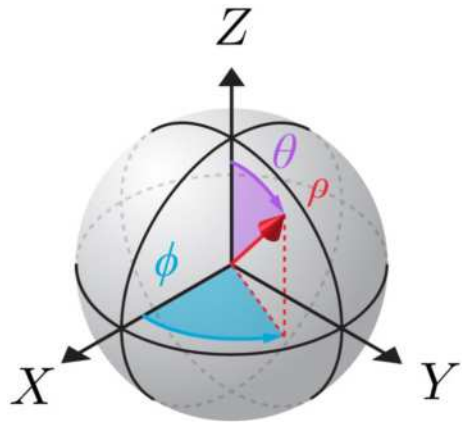
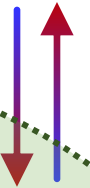
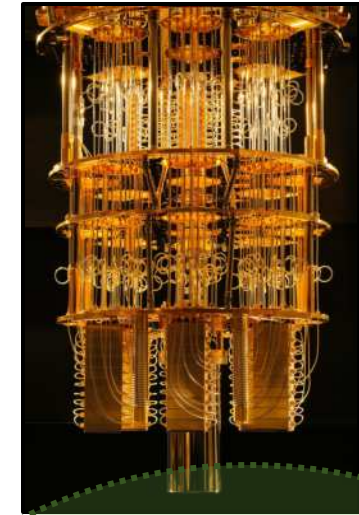


readout



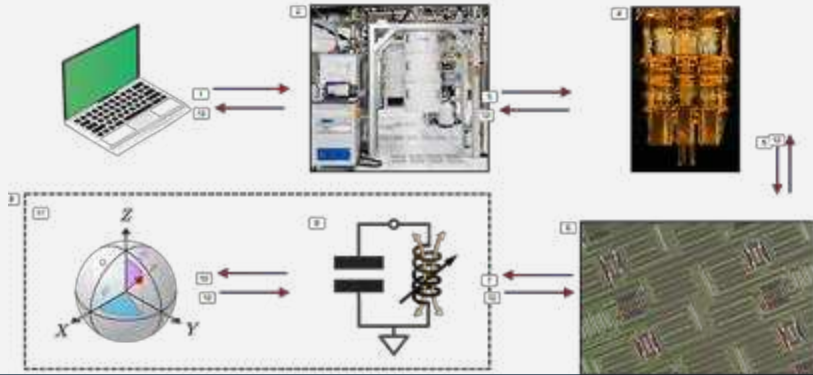
$$\hat{H} \approx \omega_0 \hat{a}^\dagger \hat{a} - \frac{\alpha}{2} \hat{a}^{\dagger 2} \hat{a}^2$$

cQED qubit in the cloud: Summary of flow

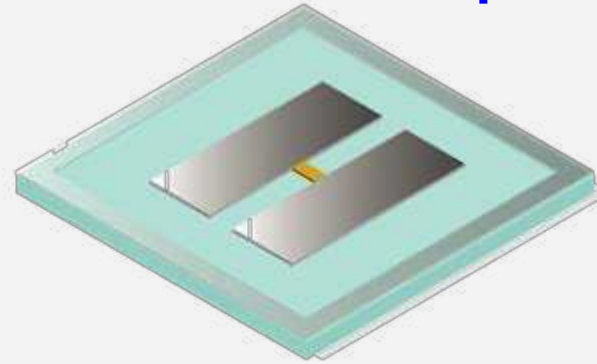


On the road ahead

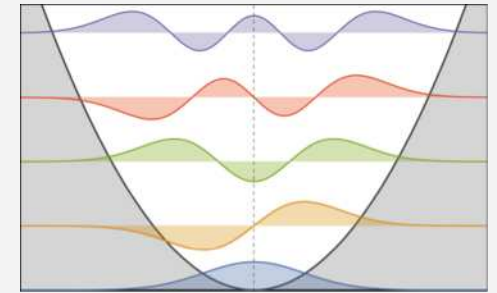
Qubit in the cloud



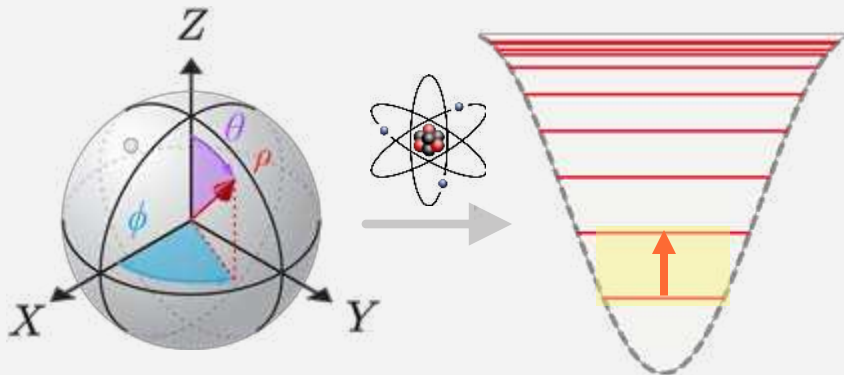
cQED: Transmon qubit



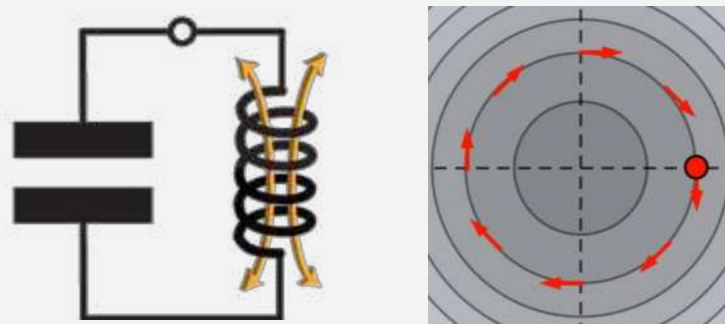
Unveiling the quantum oscillator



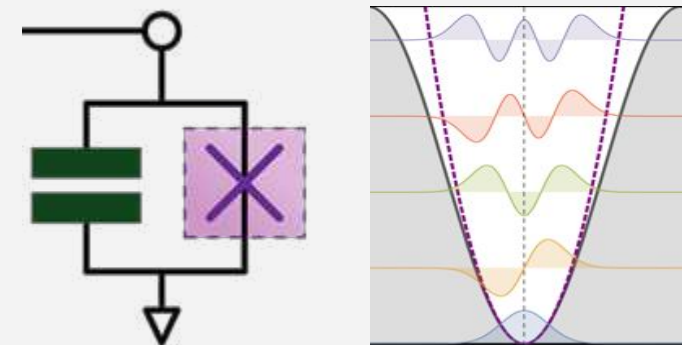
Qubit from atom / oscillator



Classical circuits & the LC



Transmon qubit

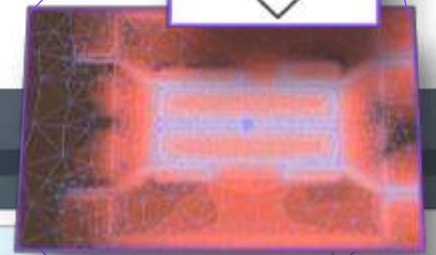
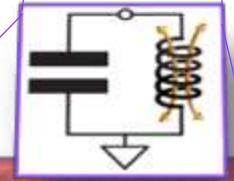
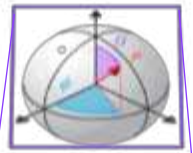
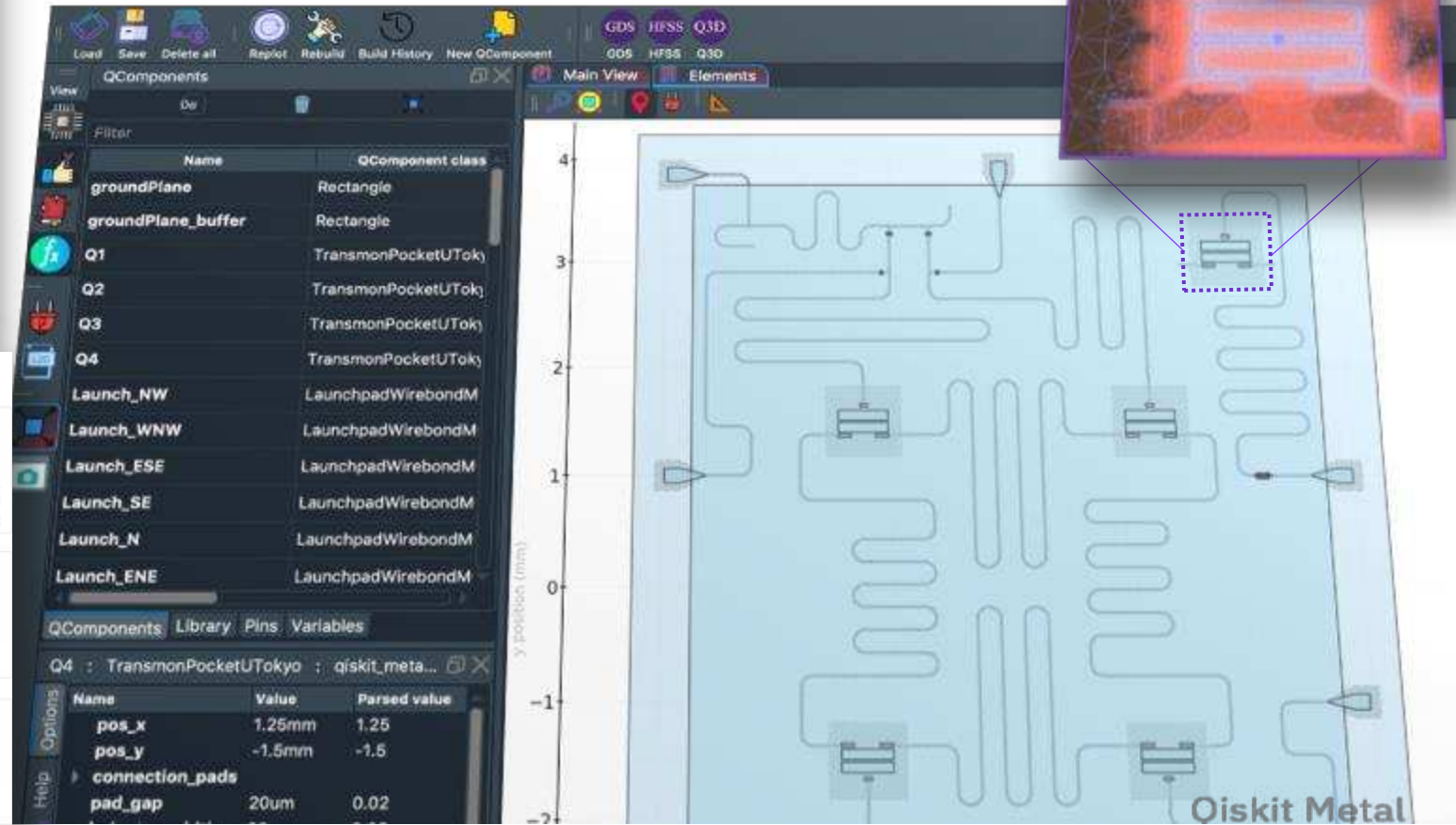
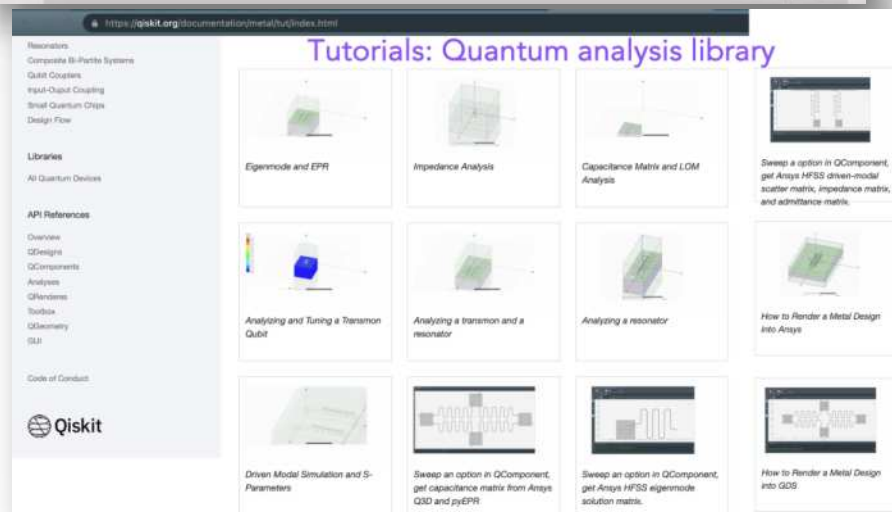
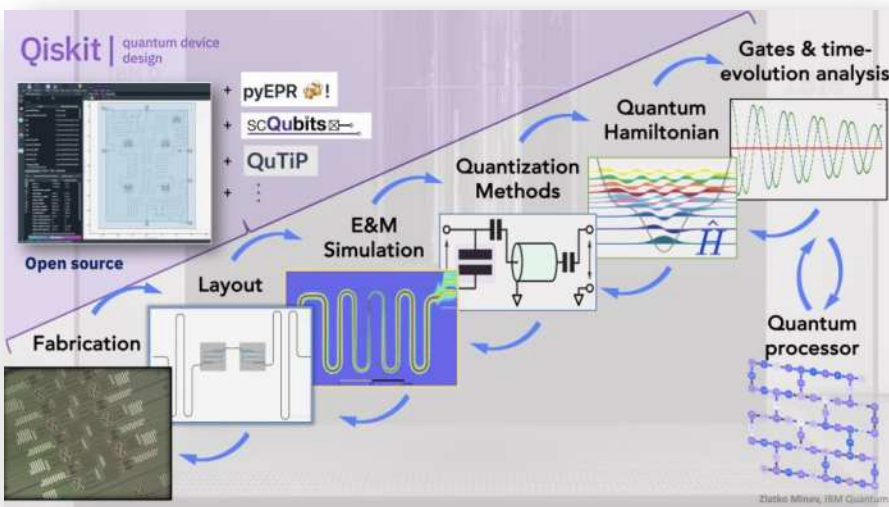


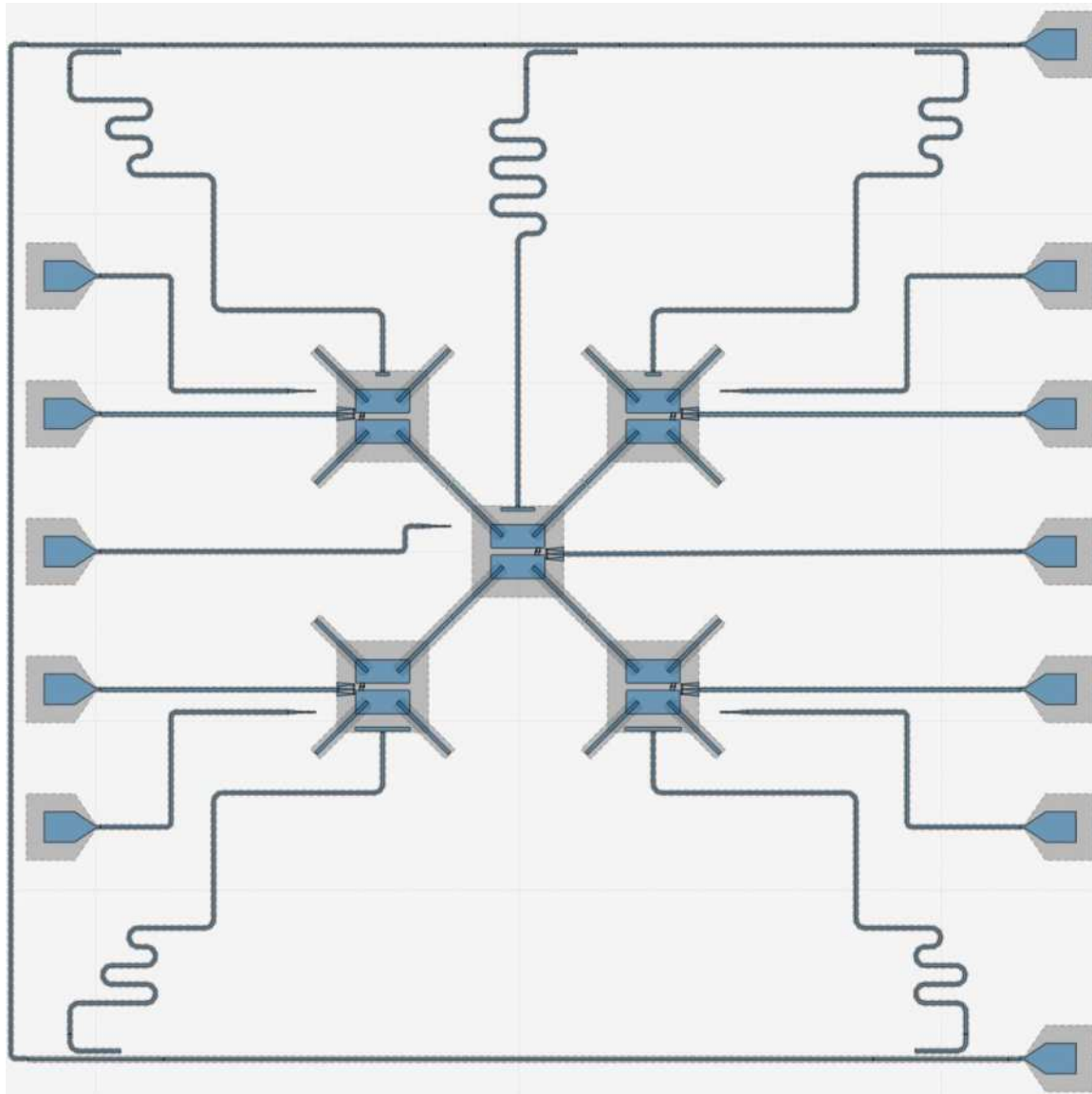
Automation in Practice: Qiskit Metal + Community



- + advanced talk (today)
- + workshops (tomorrow)
- + lunch meeting discussion (today)

Community ecosystem driven (Apache-2.0)
github.com/qiskit-community/qiskit-metal





Example designs from the community, folks like you!

NQCP
NIELS BOHR INSTITUTE



5 tunable transmon weight-4 devices
(direct coupling and Delft/ETH style)

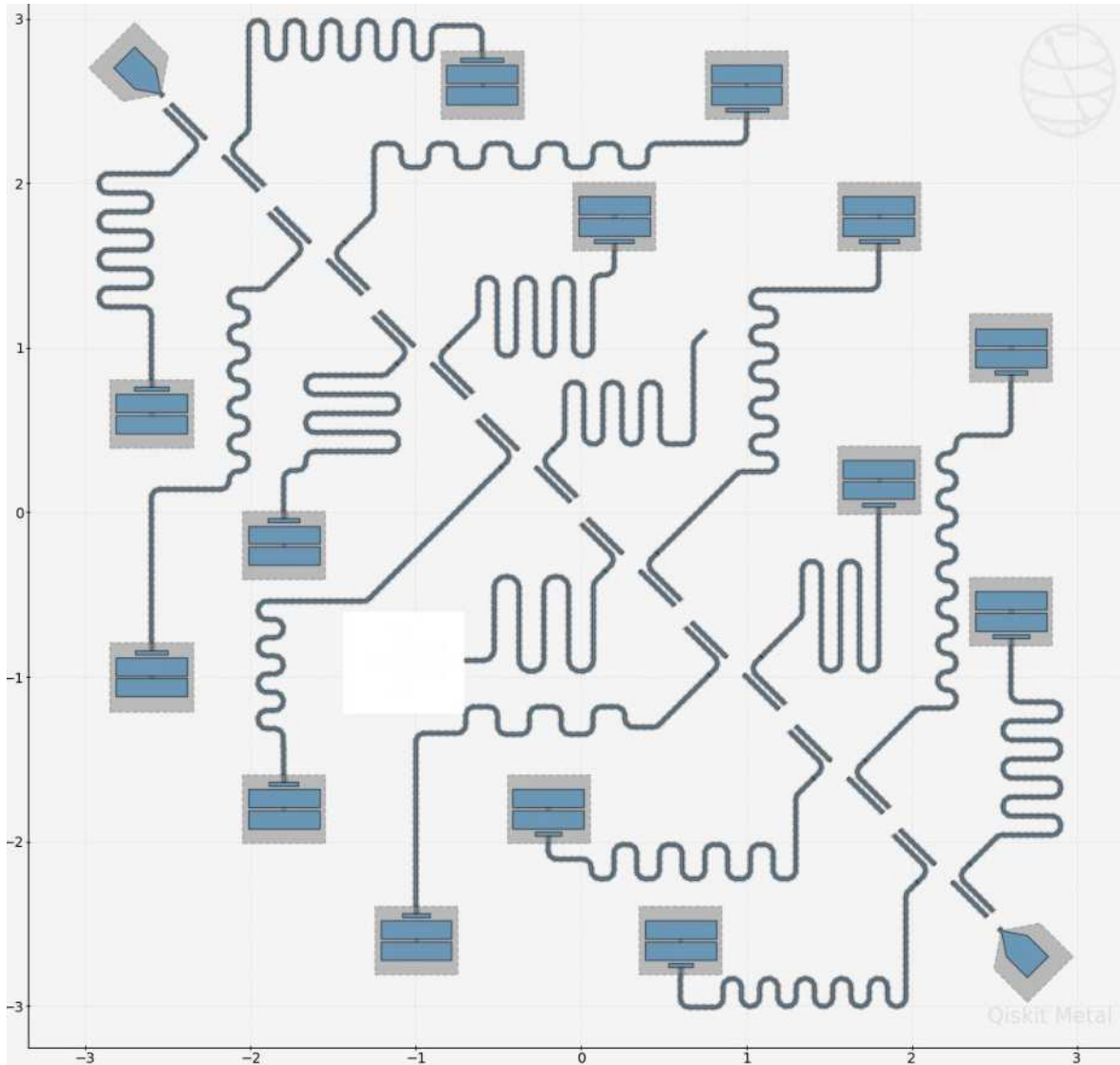
Designs:

- Benjamin Joecker, PhD
- Thue Christian Thann, PhD Candidate



Affiliations:

- Niels Bohr Institute, University of Copenhagen
- Novo Nordisk Quantum Computing Programme (NQCP)



Example designs from the community, folks like you!

NQCP
NIELS BOHR INSTITUTE



Standard candle devices for tracing reproducibility

Designs:

-Thue Christian Thann, PhD Candidate



Affiliations:

- Niels Bohr Institute, University of Copenhagen
- Novo Nordisk Quantum Computing Programme (NQCP)

Qiskit | quantum device
design
Metal

Example designs from the community, folks like you!

NQCP
NIELS BOHR INSTITUTE



Christopher. W. Warren

Niels Bohr Institute
Verified email at nbi.ku.dk

quantum computing quantum simulation
quantum information

12 Tunable qubit chip; directly coupled (dubbed “Adamite”)

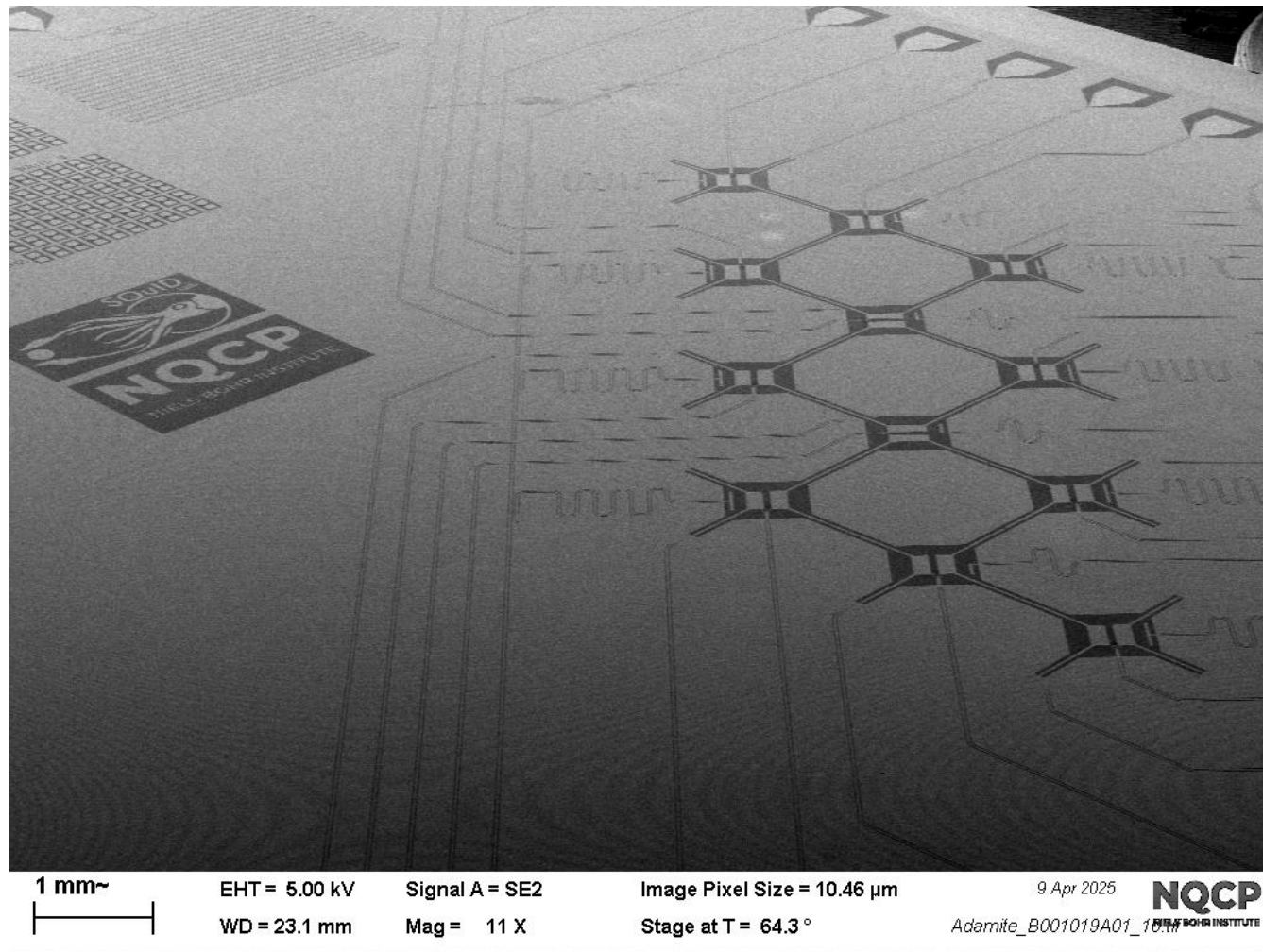
Design:

- Christopher W. Warren, PhD

Affiliations:

- SQuID Lab, Niels Bohr Institute, University of Copenhagen
- Novo Nordisk Quantum Computing Programme (NQCP)

Zlatko Minev



Example designs from the community, folks like you!

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quantum computing quantum simulation
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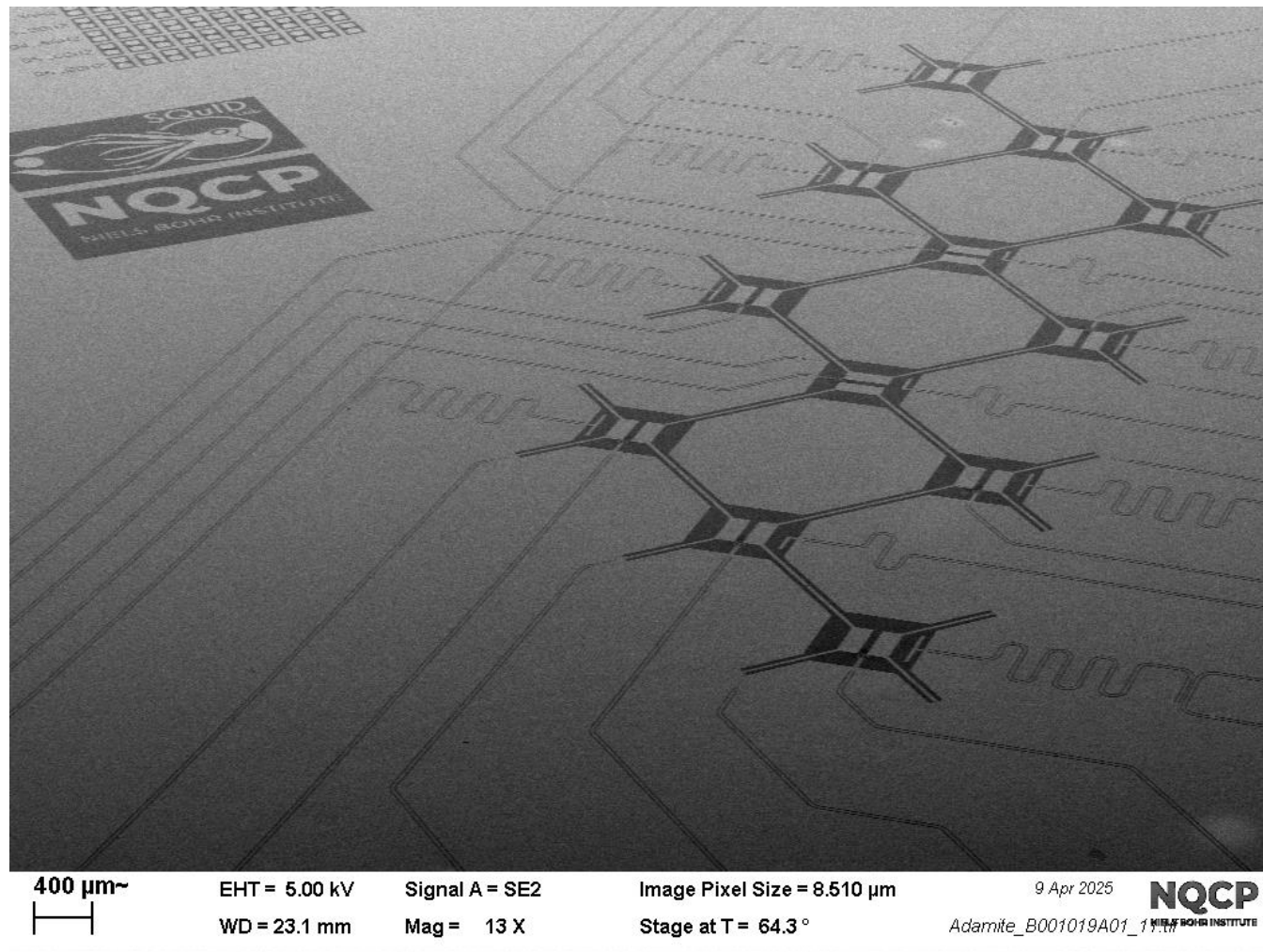
Design:

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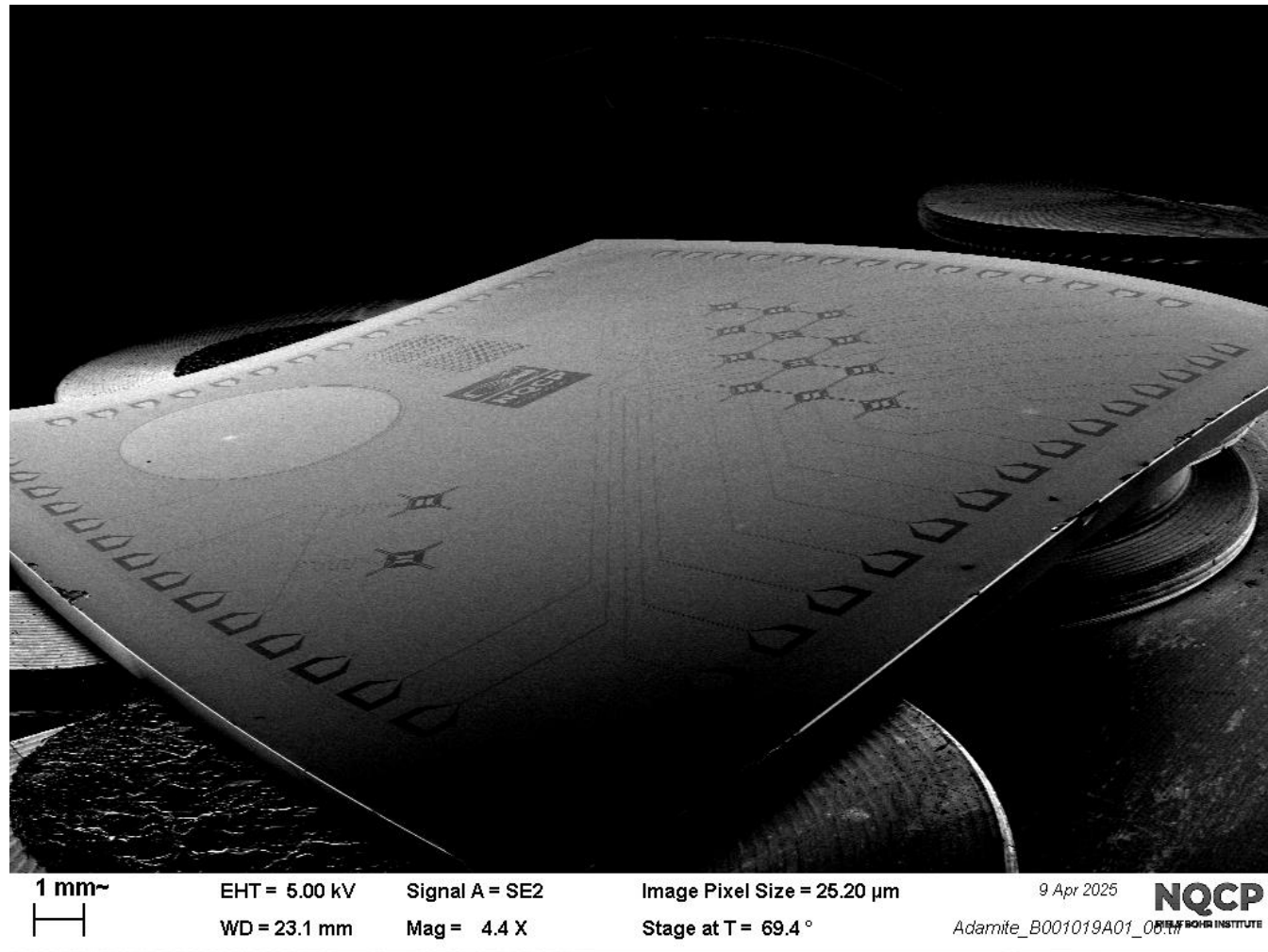
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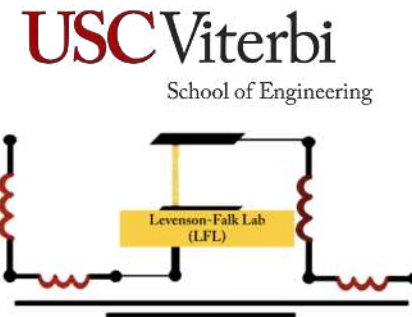
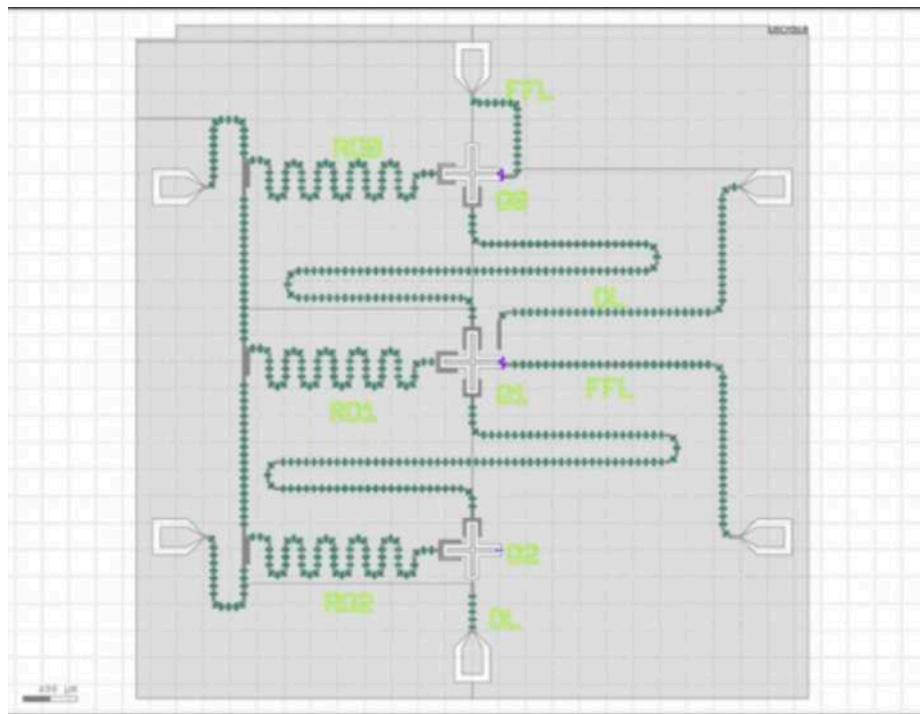
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Zlatko Minev



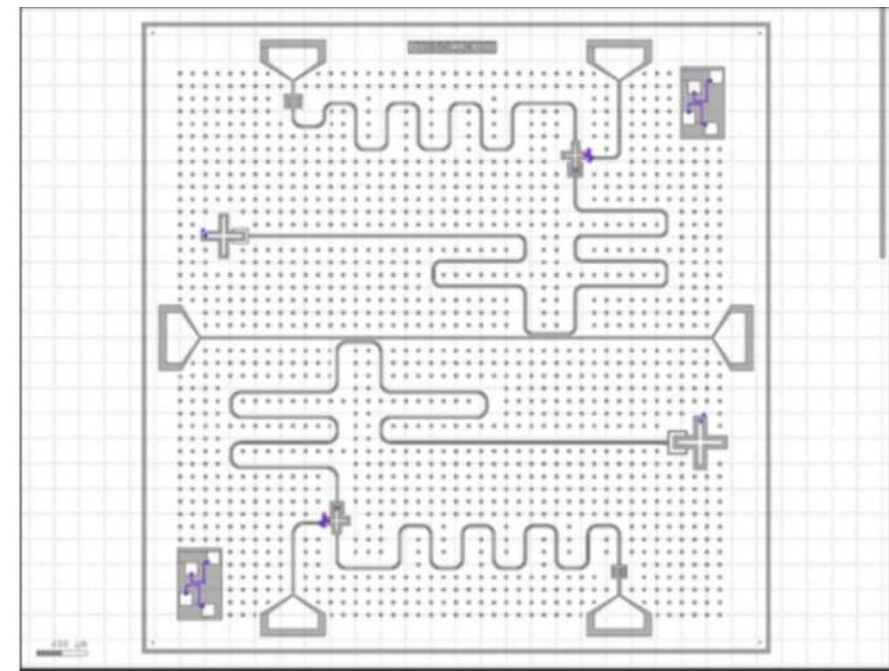
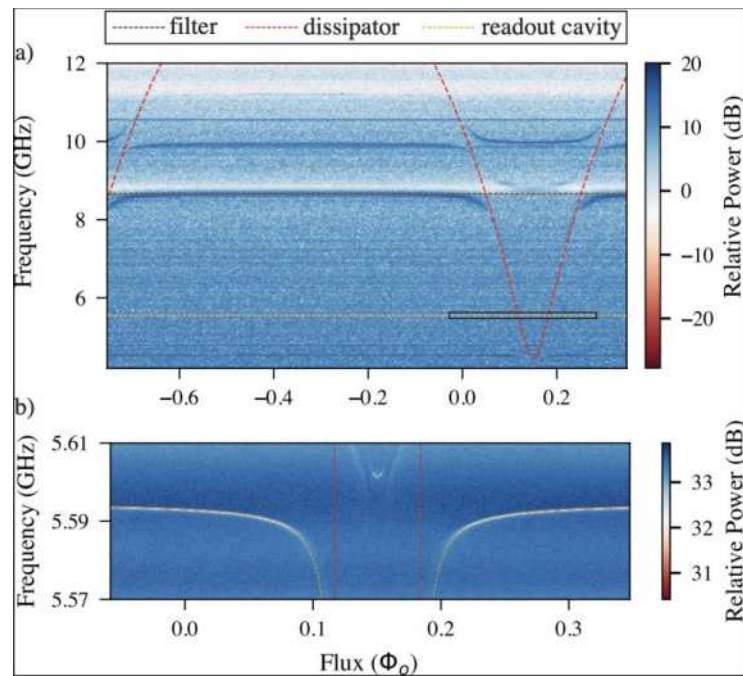
Example designs from the community, folks like you!



Sadman Ahmed Shanto



Eli Levenson-Falk



Zlatko Minev

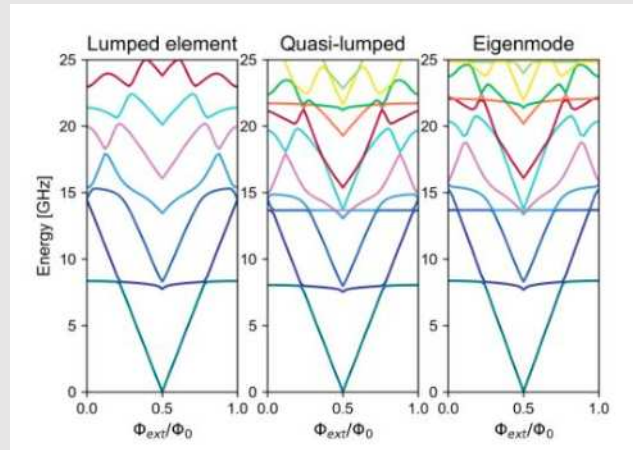
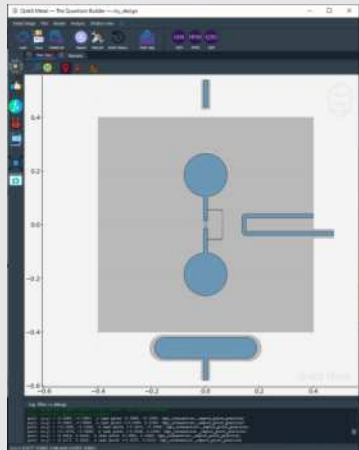
Example designs from the community, folks like you!

QuTech / Technische Universiteit Delft

Figen Yilmaz, C. Andersen Lab: Fluxonium chip design

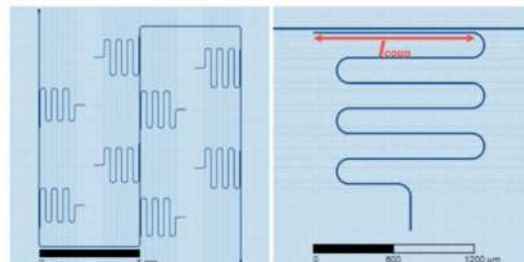
Roald van den Boogaart-Andersen Lab: Fluxonium spectrum (eigenmode = pyEPR) to show how they extended the analysis methods to be able to handle fluxonium qubits

Sara Buhktari-Andersen Lab: Fabricated resonator design being measured



3.1. Final design

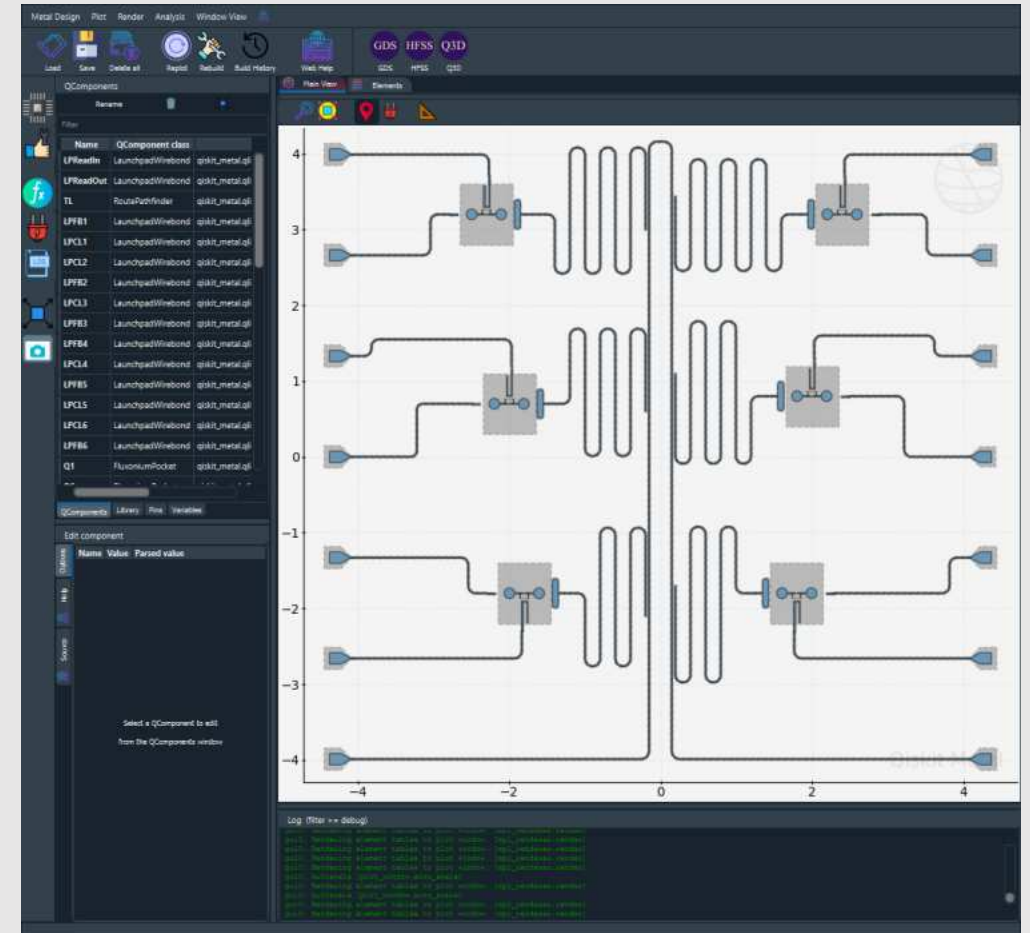
The final design consists of an S-shaped feedline routed between two launchpads. Eight resonators are capacitively coupled to this feedline. Every component on this 9 x 9 mm device is implemented by a coplanar waveguide transmission line. The schematic representations of the full device and single resonator geometry are depicted in Figure 6.



“Qiskit-Metal is the best tool if one studies superconducting qubits” -- Figen Yilmaz

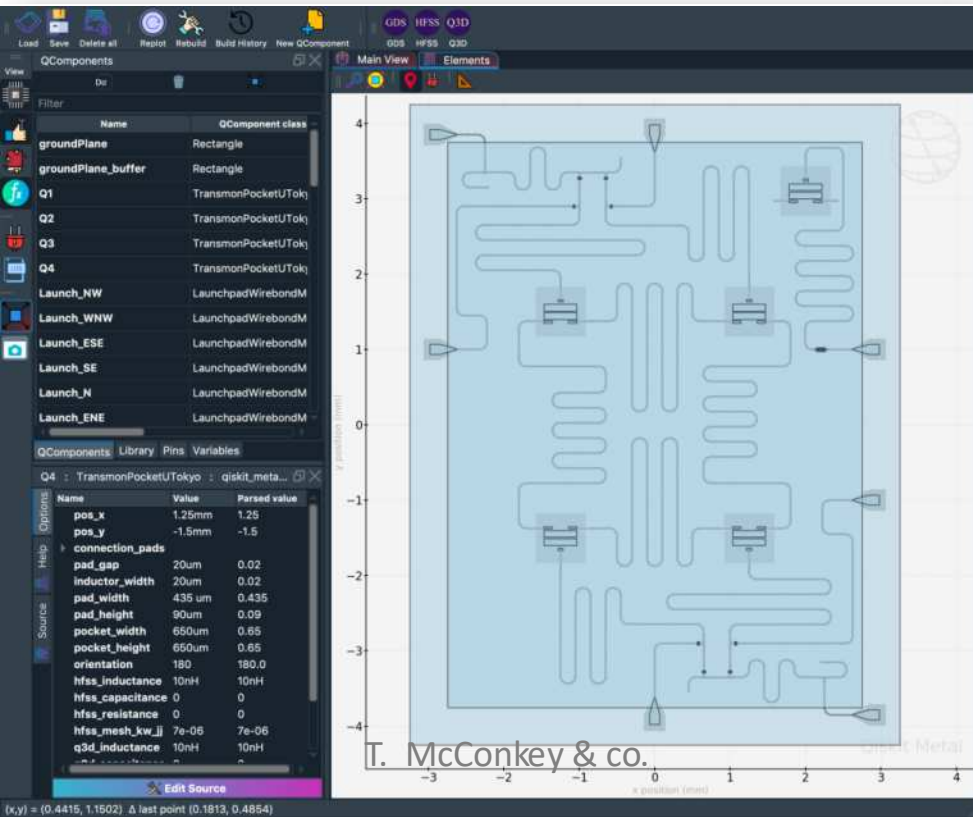
QuTech / Technische Universiteit Delft

Intermediate chip design.

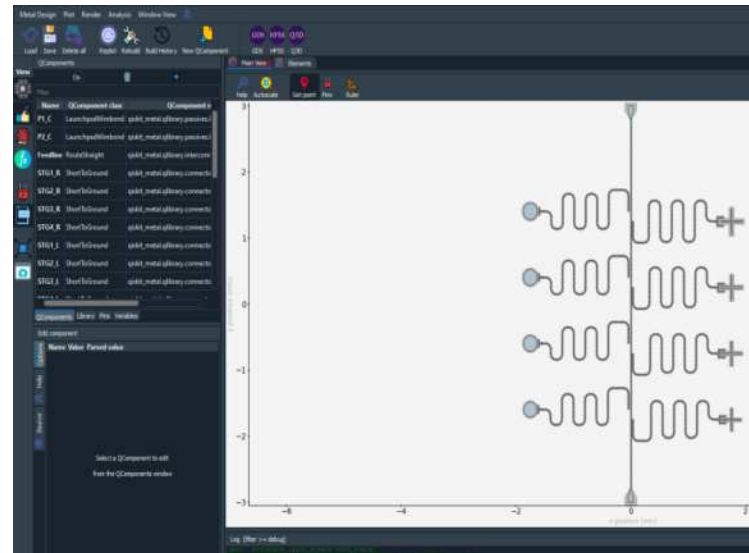


Example designs from the community, folks like you!

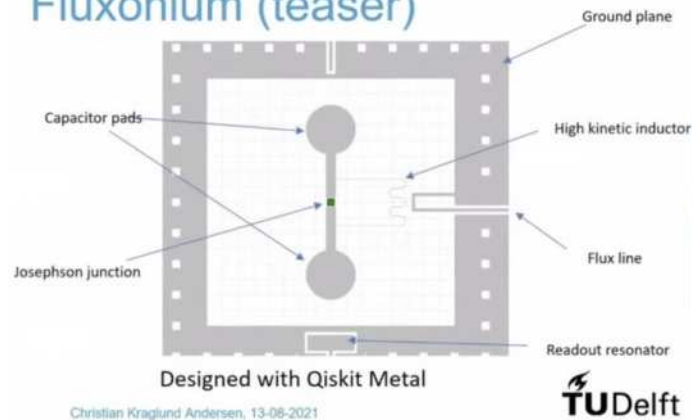
IBM 5Q Tsuru U Tokyo



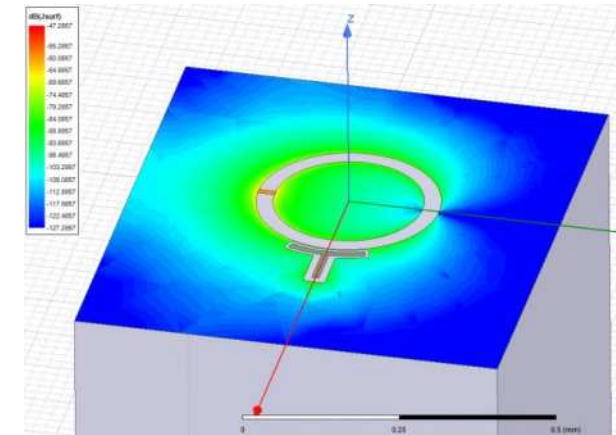
Chalmers 8Q



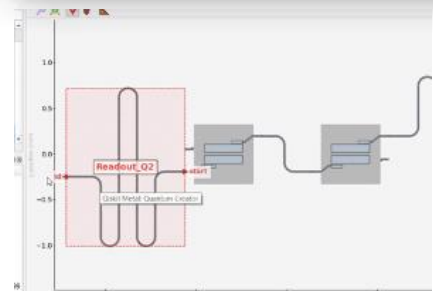
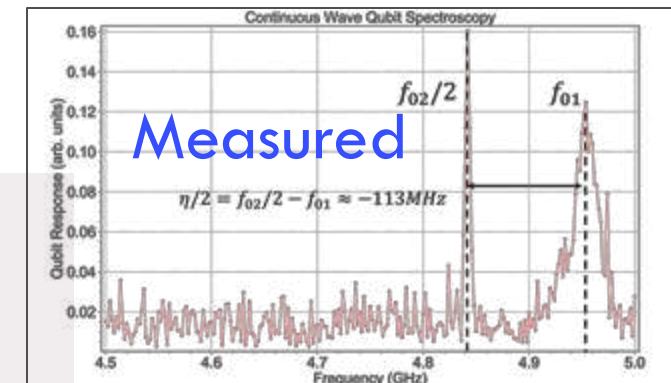
TU Delft Fluxonium
Fluxonium (teaser)



Simulated



Measured



APS physics **MARCH MEETING 2021**
MARCH 15-19 ONLINE

CEC/ICMC **21**
VIRTUAL CONFERENCE JULY 19-23
Cryogenics • Through • Collaboration

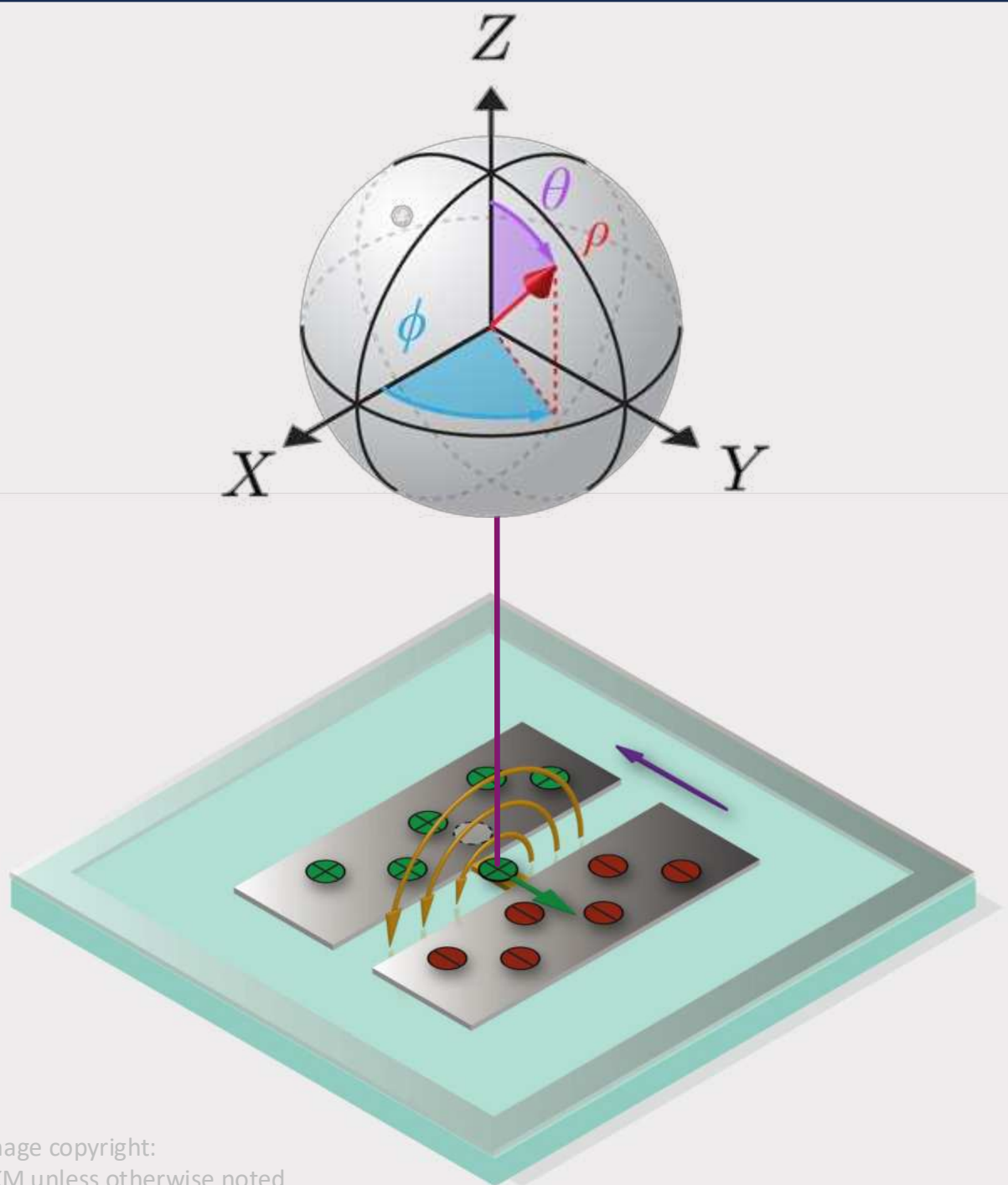


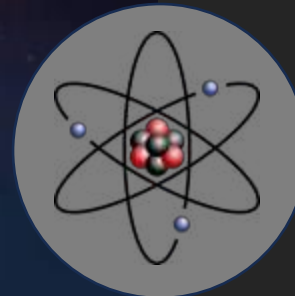
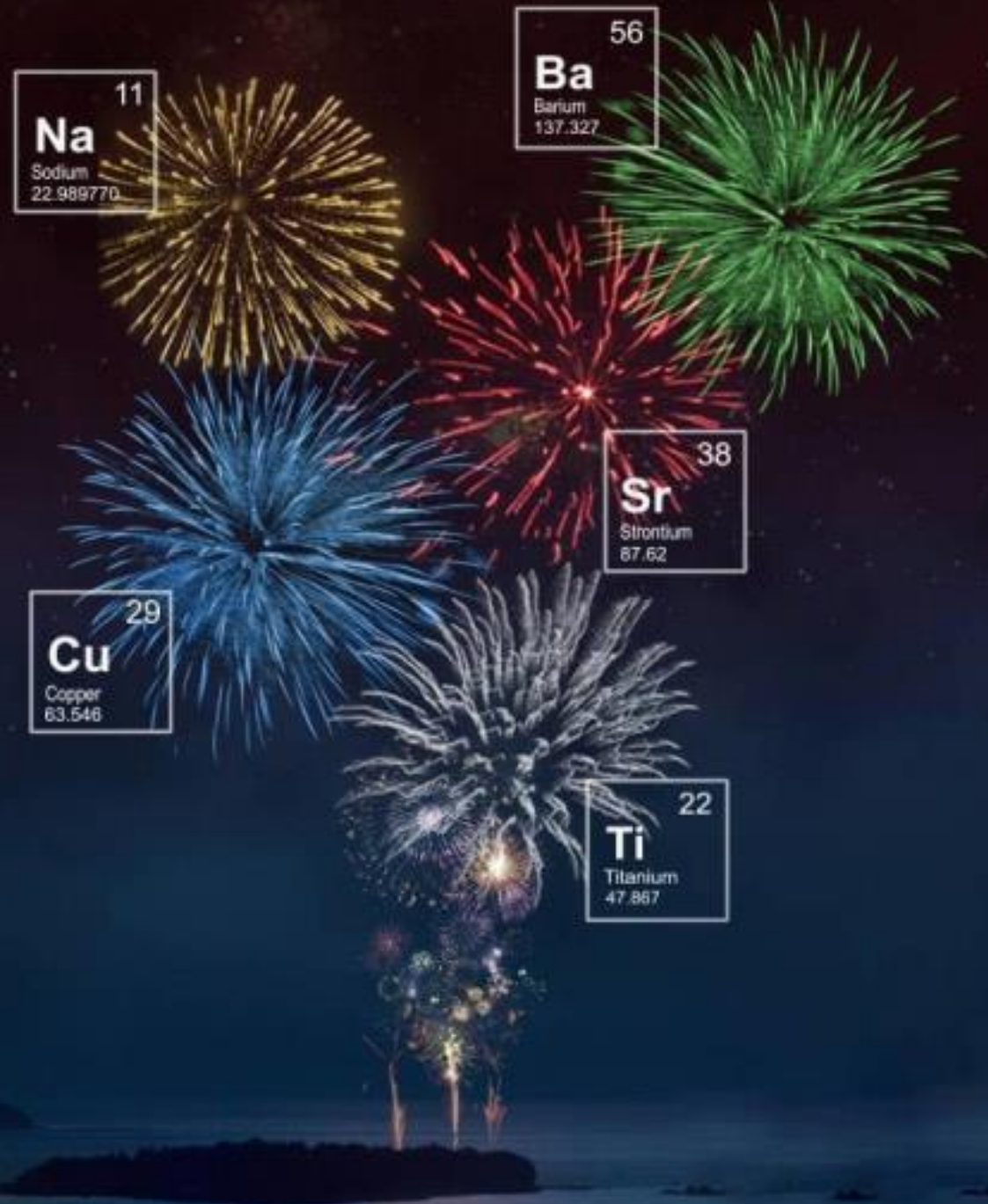
IEEE **QUANTUM WEEK** IEEE International Conference on Quantum Computing and Engineering — QCE21

Zlatko Mineev

Quantum bit

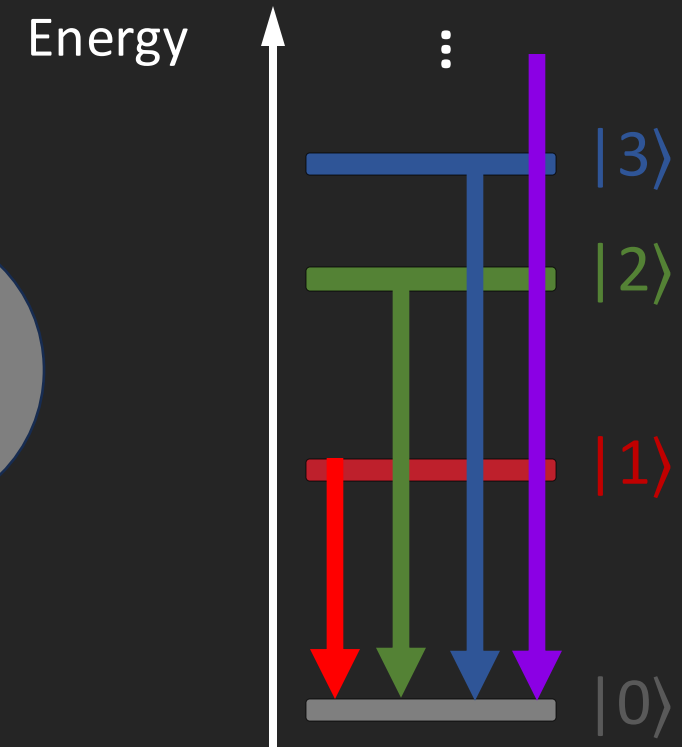
Reality to Abstraction





The Light of Atoms

Atoms are quantum:
discrete intrinsic energy levels*

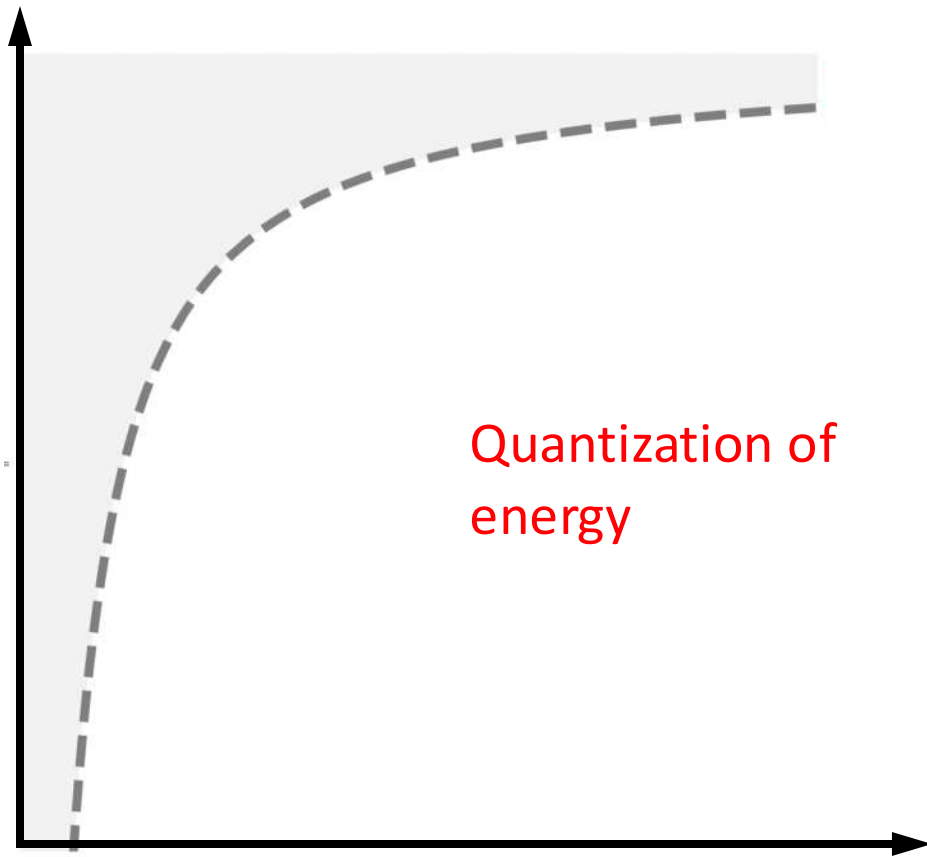


Quantization of energy

The notion of an energy level was proposed by Bohr in 1913.

Atomic energy levels and transitions

Electron potential-energy landscape



Radial distance from nucleus

Qubit from Atom

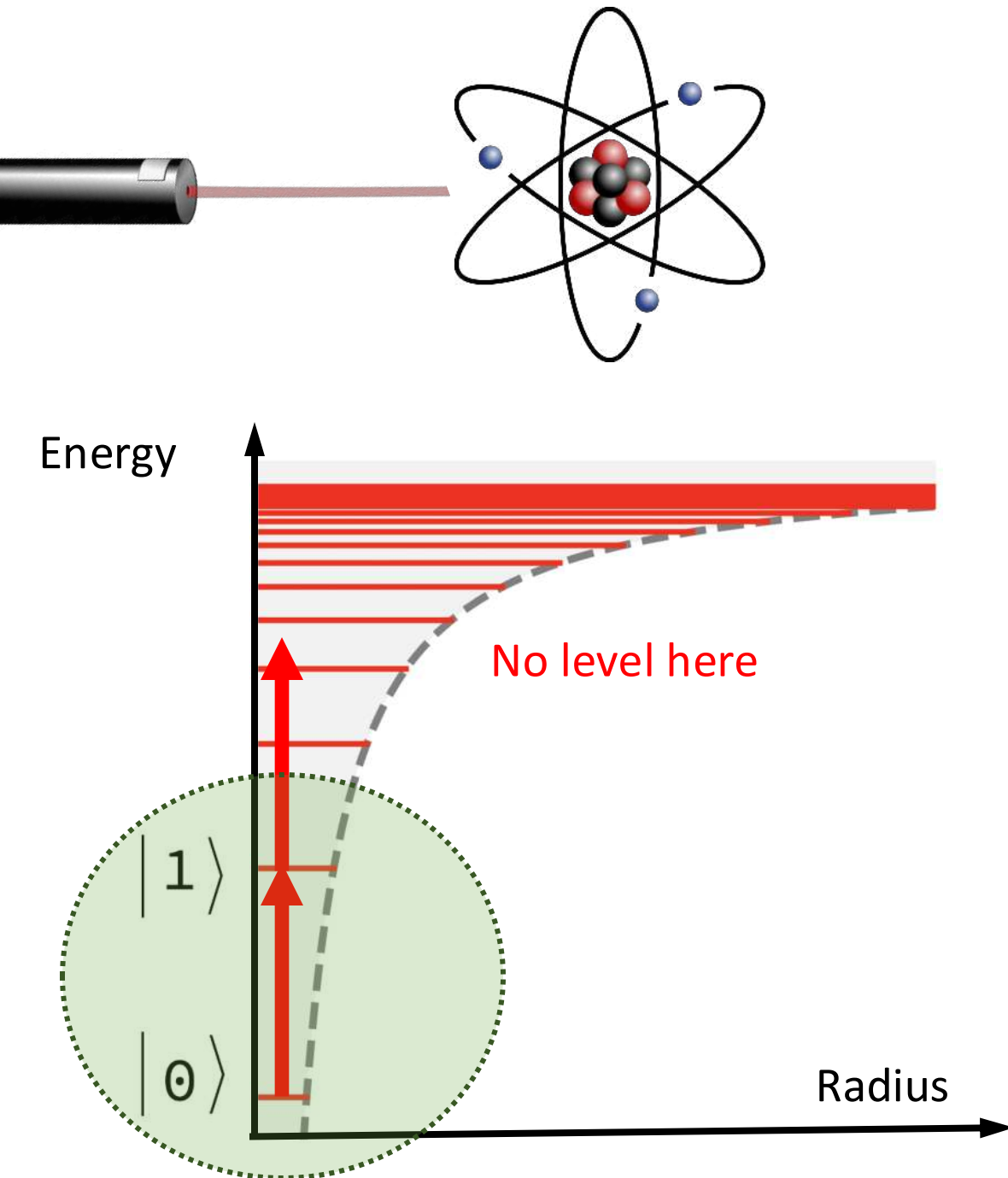
**Anharmonic degree of
freedom and spin**

**Isolated from environment and
thermal bath**

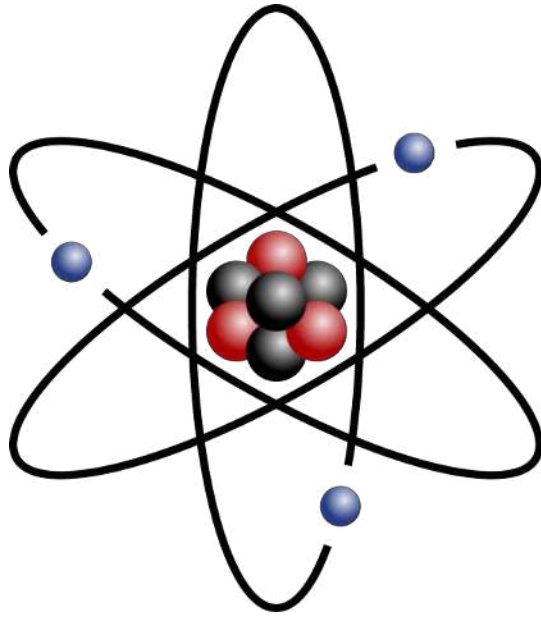
Low-loss

**Level diagram allows for qubit-
specific control and readout**

**There are always more than
two levels**

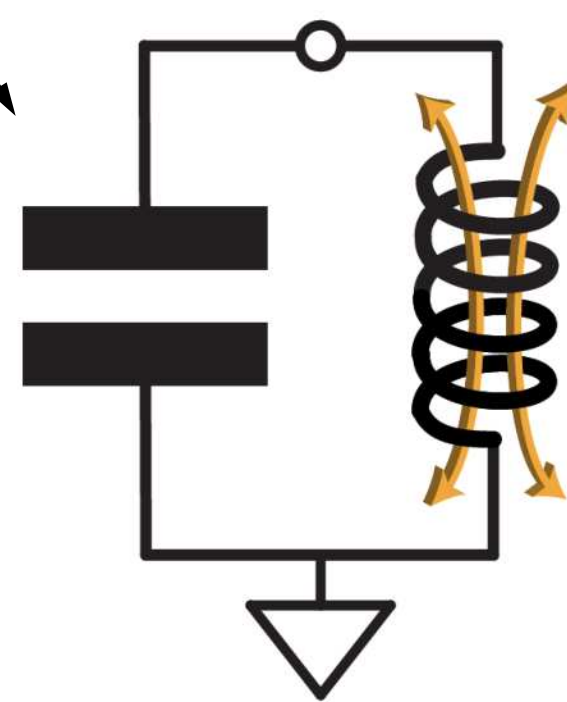


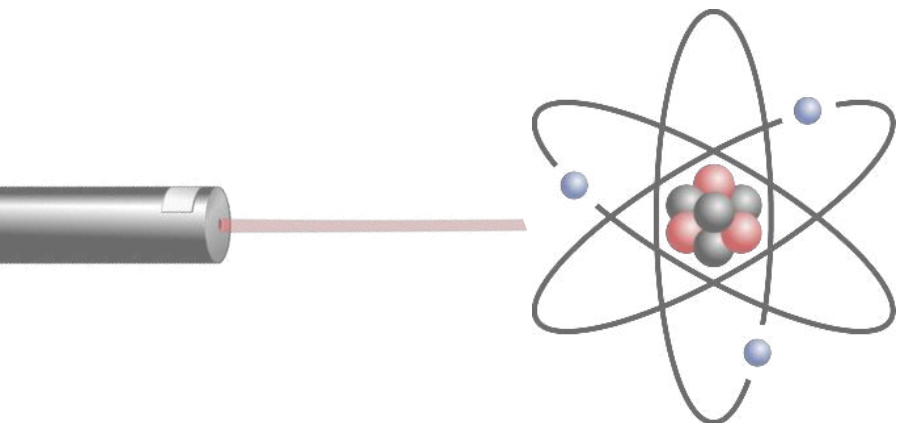
Artificial atoms



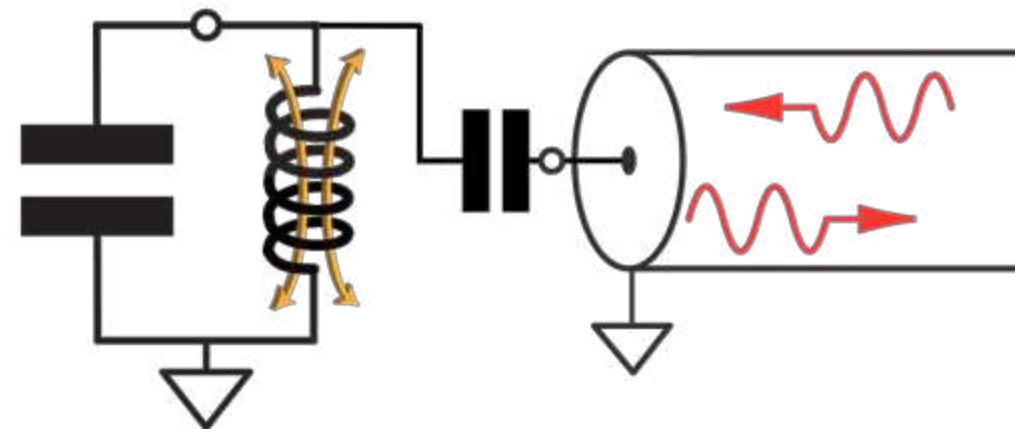
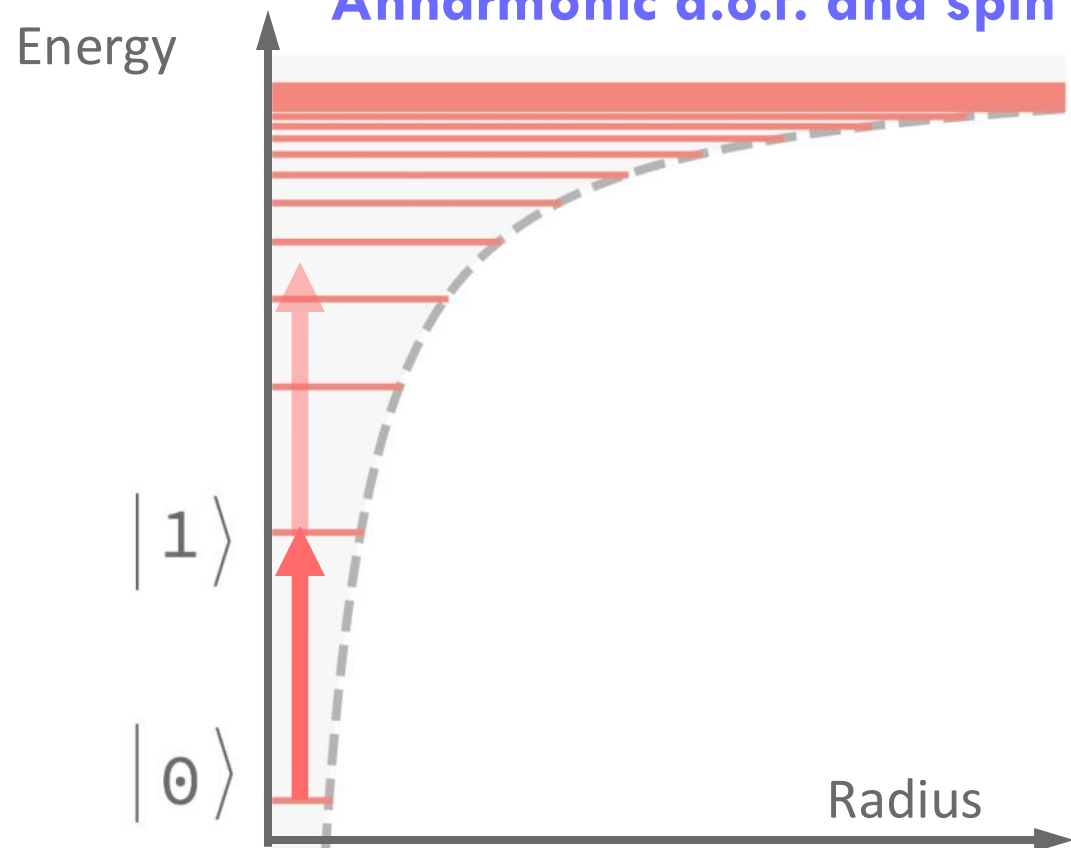
capacitor

inductor

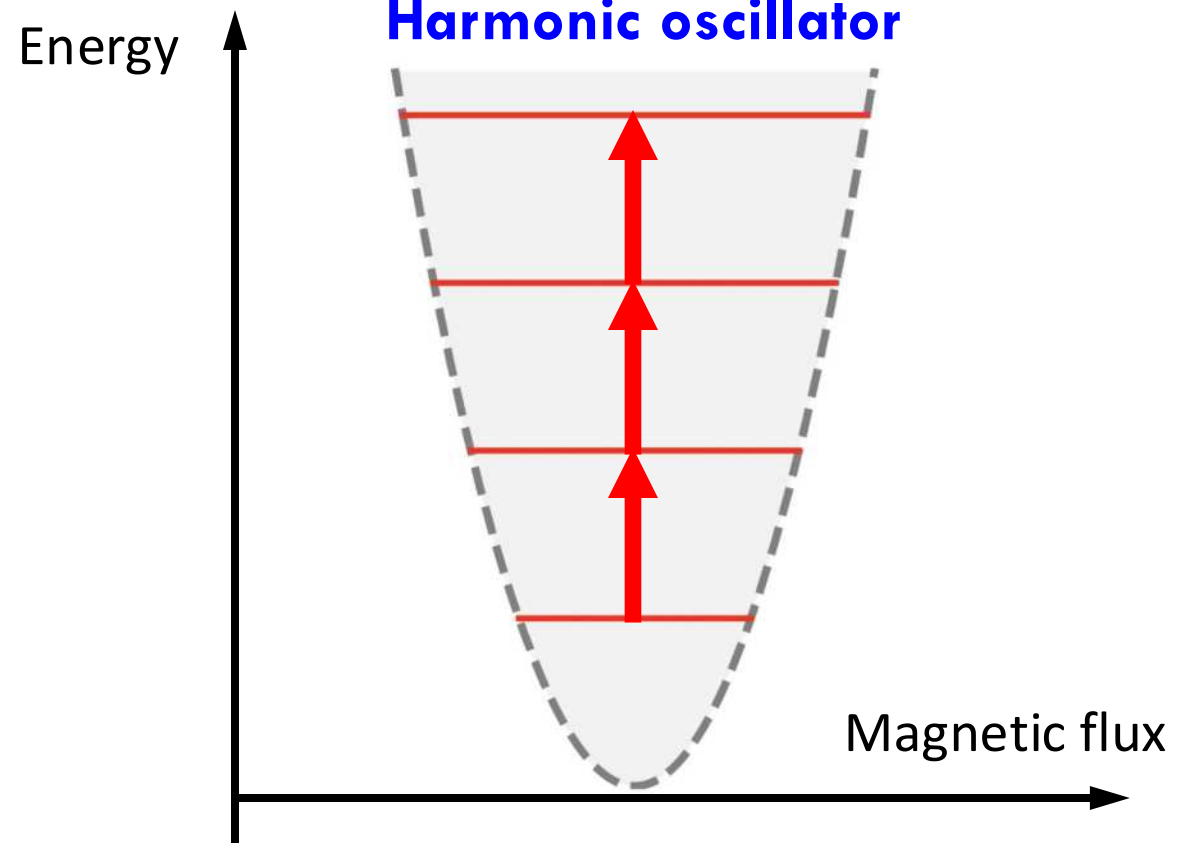


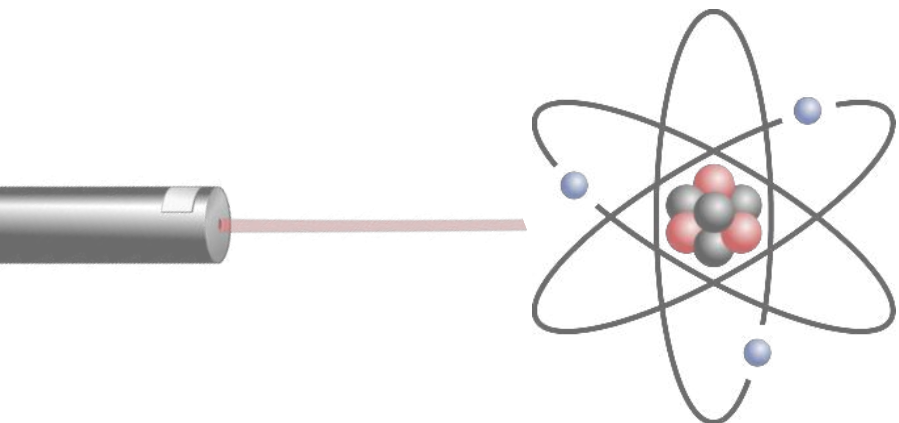


Anharmonic d.o.f. and spin

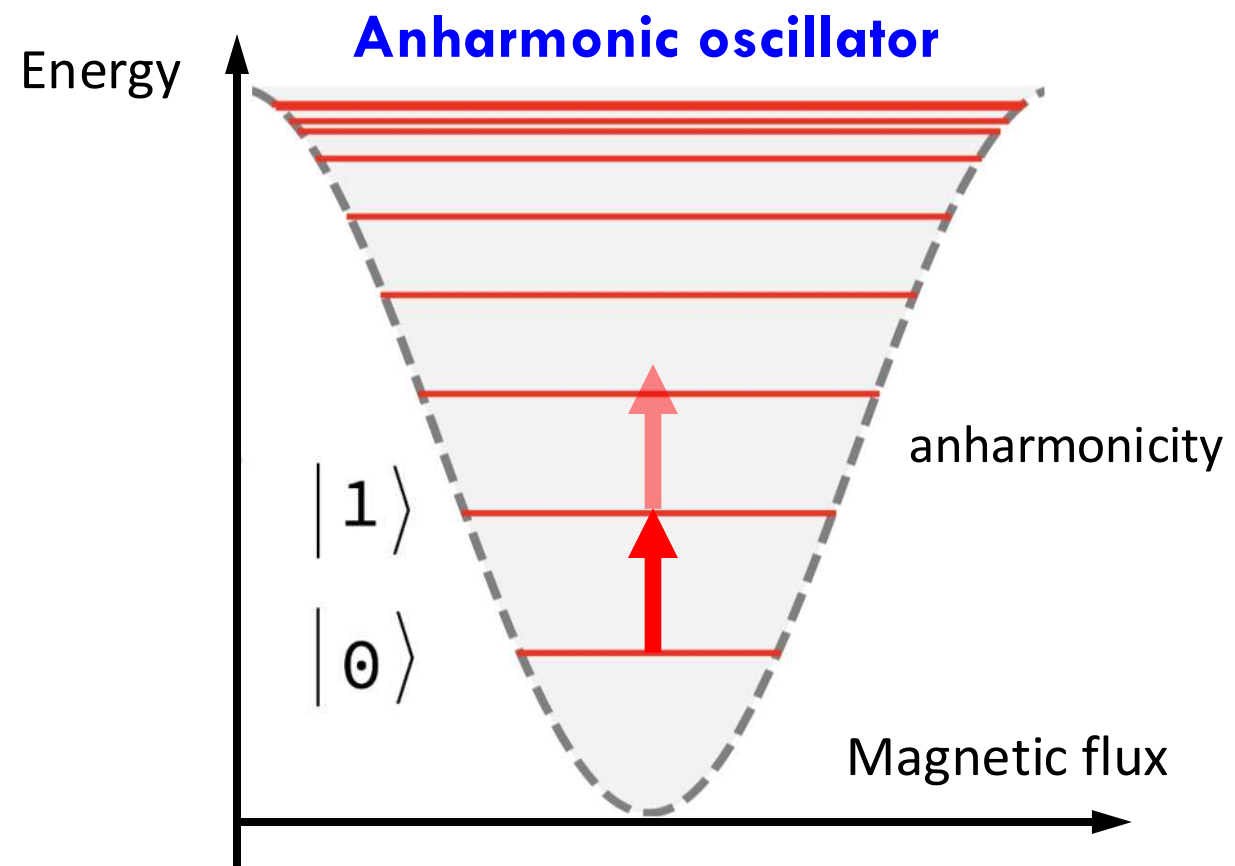
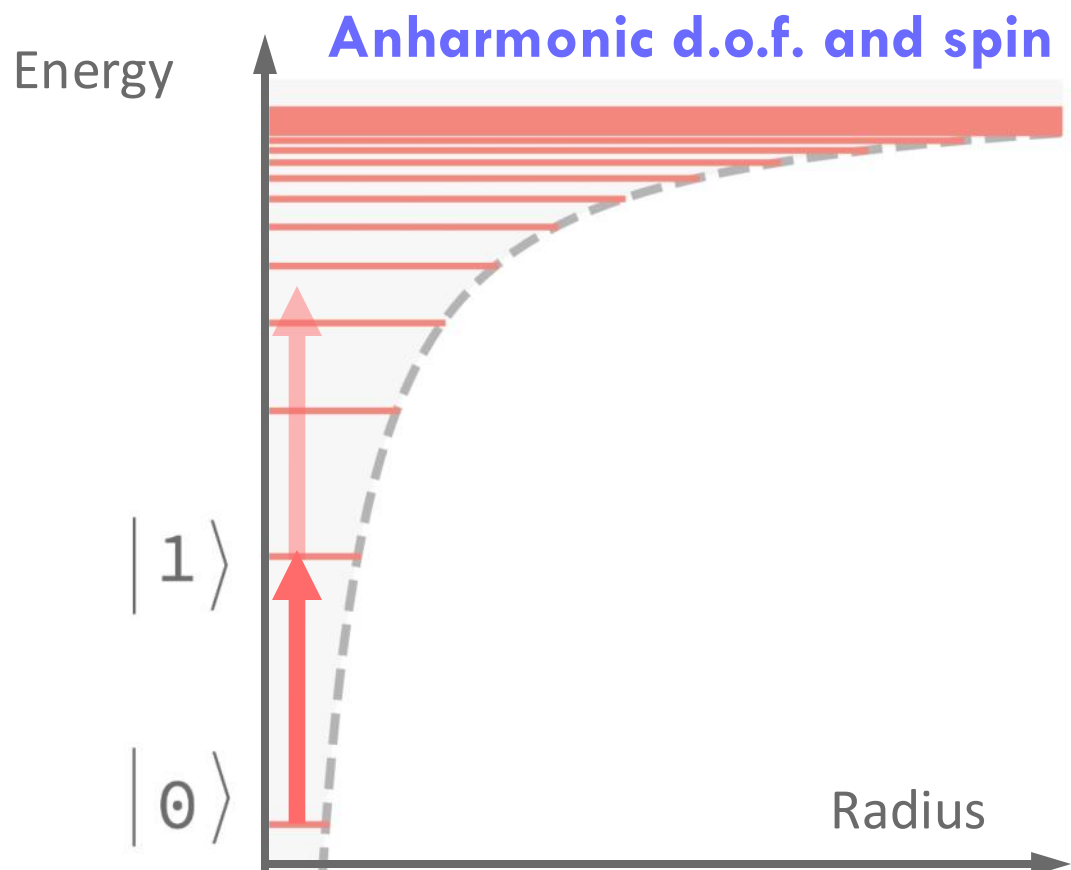
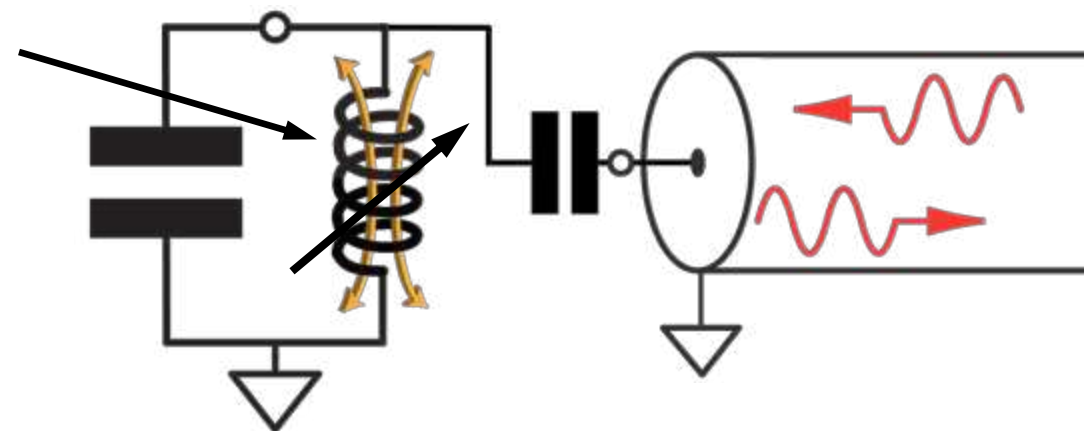


Harmonic oscillator



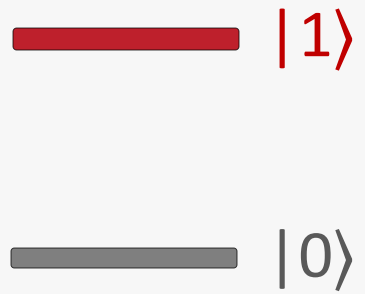


non-linear
inductor

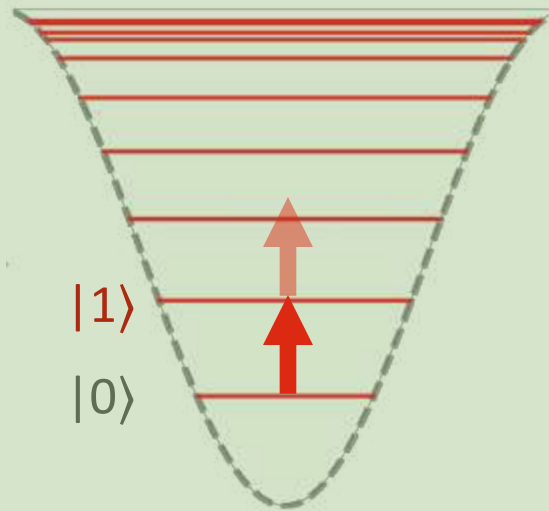


Big-picture connections

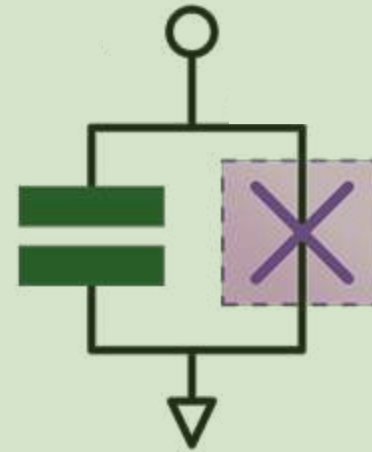
**Idealization of
qubit**



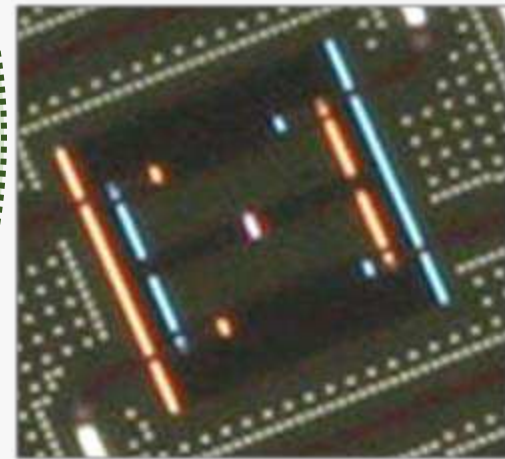
**Anharmonic
oscillator**



**Physical circuit
model**



Physical layout



Idealization



Physical reality

Zlatko Minev

Circuit Quantum Electrodynamics (cQED)

Macro
overview

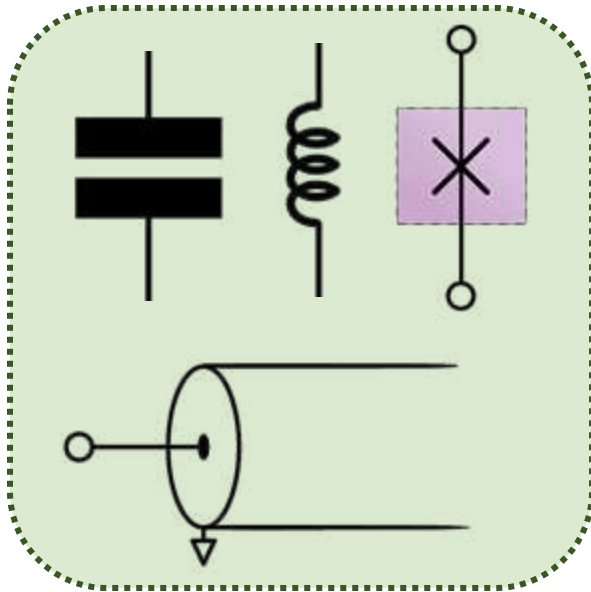
There are two kinds of physicists:

Those who believe all of physics is *spins*.

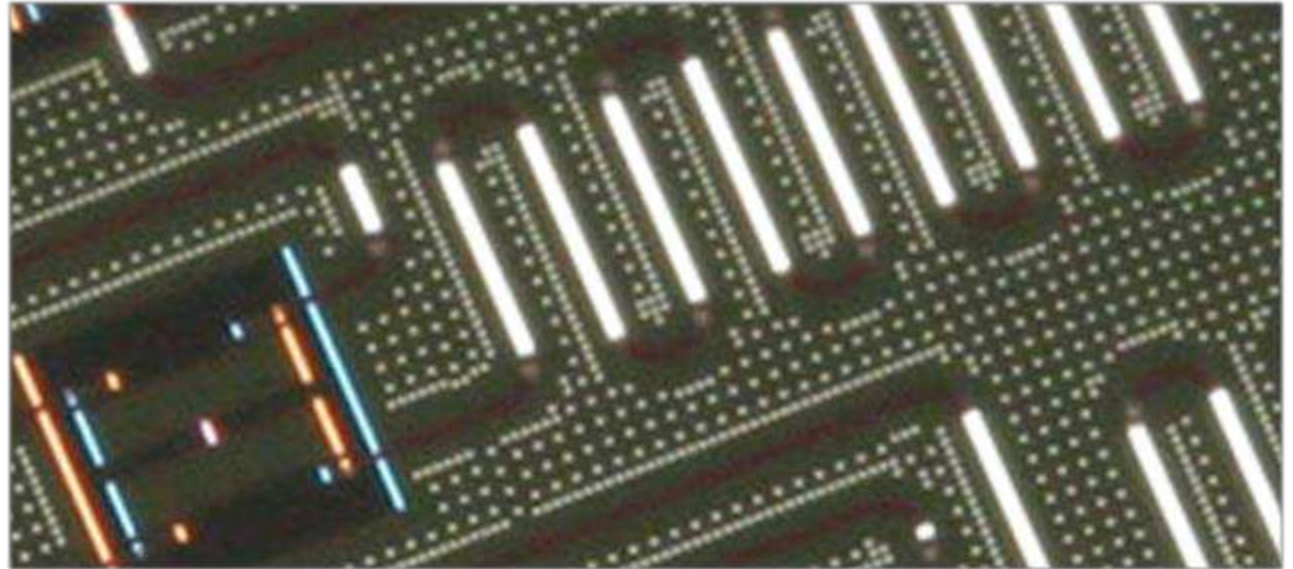
Those who believe all of physics is *oscillators*.

cQED Ingredients

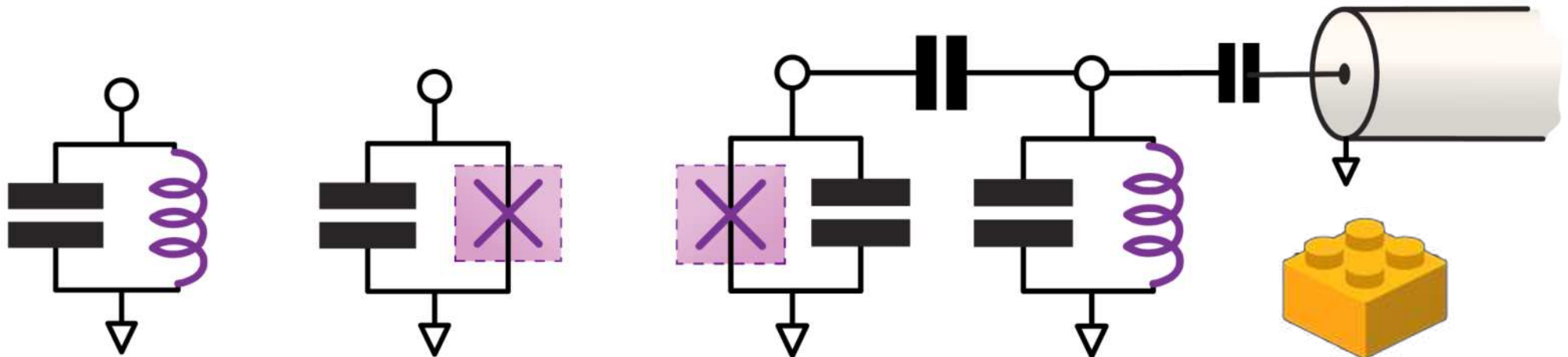
Circuit elements



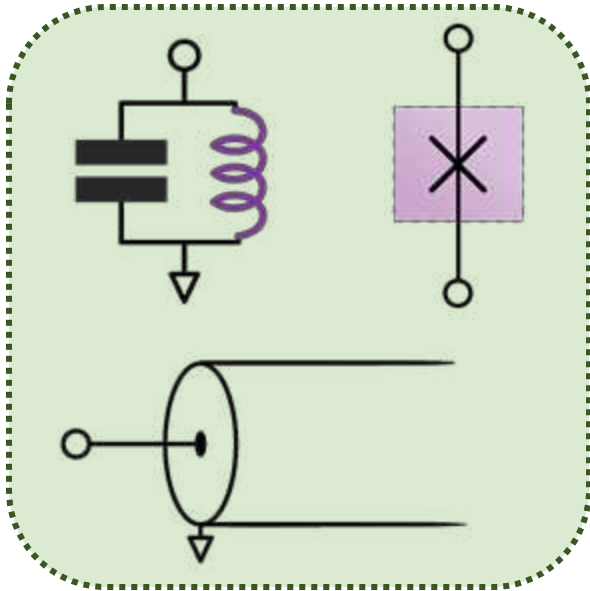
Microwave oscillators



Combine



cQED Ingredients

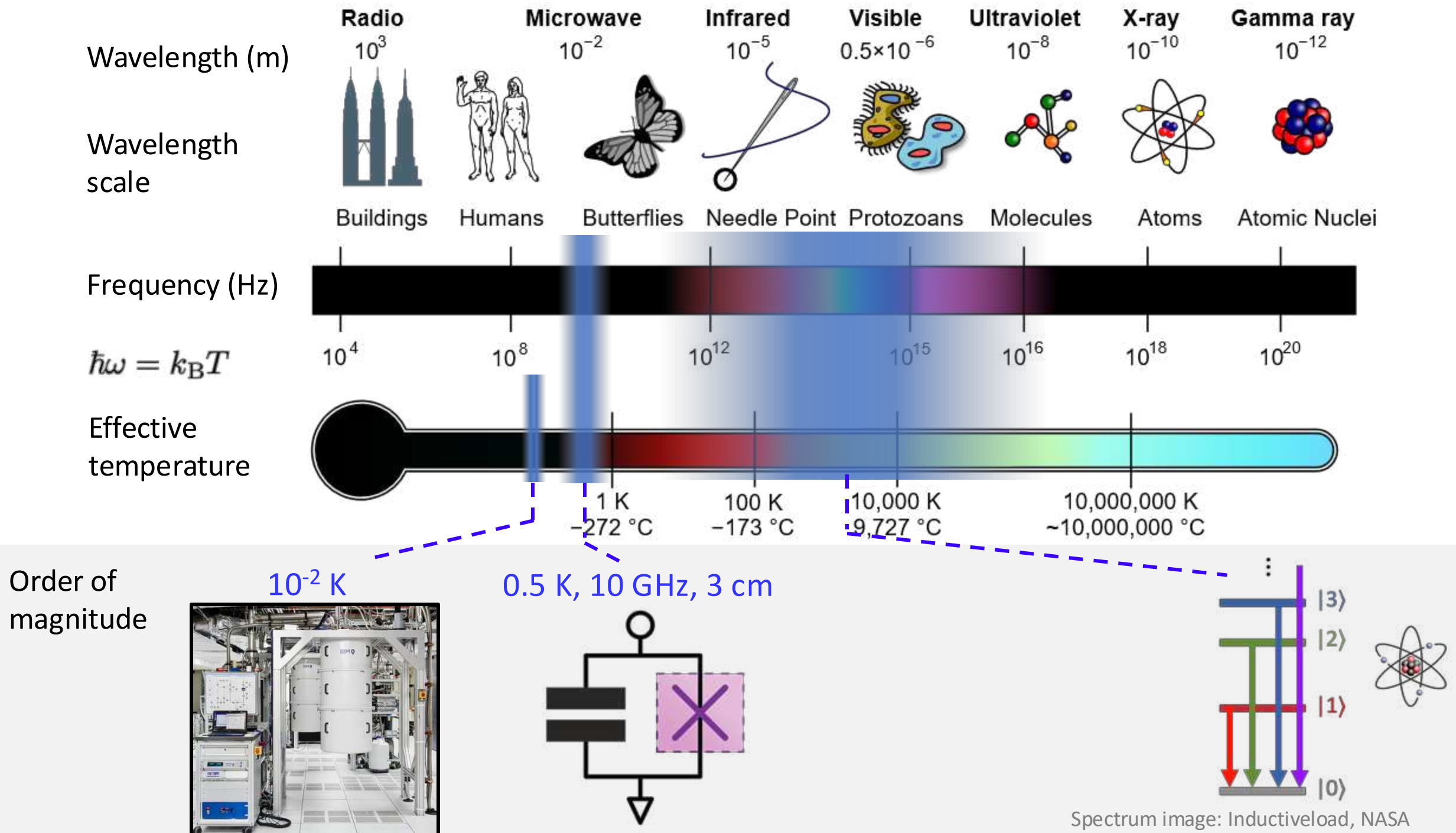


- Small dissipation
- Isolation from environment
- Low temperature
- Nonlinearity
- Large vacuum fluctuations

Superconductivity

- Nominally zero intrinsic dissipation and heat
- Nominal temperature far below energy level splitting
- Non-linear, robust Josephson tunnel junction effect





A few introductory reviews

And many more... check online or ask us for specific topic

Qiskit Textbook (2020; more chapters coming)

Blais, A., Grimsmo, A. L., Girvin, S. M., & Wallraff, A. (2020)
Circuit Quantum Electrodynamics (arXiv:2005.12667)

Kjaergaard, M., Schwartz, ... Oliver, W. D. (2020)
Superconducting Qubits: Current State of Play
Annual Reviews of Condensed Matter Physics 11, 369-395

Krantz, P., Kjaergaard, M., Yan, F., ... & Oliver, W. D. (2019)
A quantum engineer's guide to superconducting qubits
Applied Physics Reviews, 6(2), 021318

Corcoles, A. D., Kandala, A., ... Gambetta, J. M. (2019)
Challenges and Opportunities of Near-Term Quantum Computing Systems. *Proceedings of the IEEE*, 1–15.

Wendin, G. (2017)
Quantum information processing with superconducting circuits. *Reports on Progress in Physics*, 80(10), 106001

Gambetta, J. M., Chow, J. M., & Steffen, M. (2017)
Building logical qubits in a superconducting quantum computing system. *Npj Quantum Information*, 3(1), 2

Girvin, S. M. (2011) Circuit QED: superconducting qubits coupled to microwave photons. *Quantum machines: measurement and control of engineered quantum systems*, 113, 2.

Clerk, A. A., Girvin, S. M., Marquardt, F., & Schoelkopf, R. J. (2010)
Introduction to quantum noise, measurement, and amplification
Reviews of Modern Physics, 82(2), 1155–1208

Clarke, J., & Wilhelm, F. K. (2008)
Superconducting quantum bits. *Nature*, 453(7198), 1031–1042

Devoret, M. H. (1997)
Quantum Fluctuations in Electrical Circuits.
In *Fluctuations Quantiques/Quantum Fluctuations* (p. 351)

...

Where are all my slides?



Slides!

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- index.html

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zlatko-minev Initial Commit 7e

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Career Talks	Initial Commit
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Intro to Pauli Twirling (Tutorial)	Initial Commit
Intro to Quantum Error Mitigation ...	Initial Commit
Intro to Quantum Noise (Qiskit Gl...	Initial Commit
Intro to Quantum Noise (Qiskit Gl...	Initial Commit
Intro to Quantum Noise (Qiskit Gl...	Initial Commit
Superconducting Qubits 101 - Qu...	Initial Commit
A Prelude To Practical Quantum ...	Initial Commit

Zlatko Minev,
Ph.D.

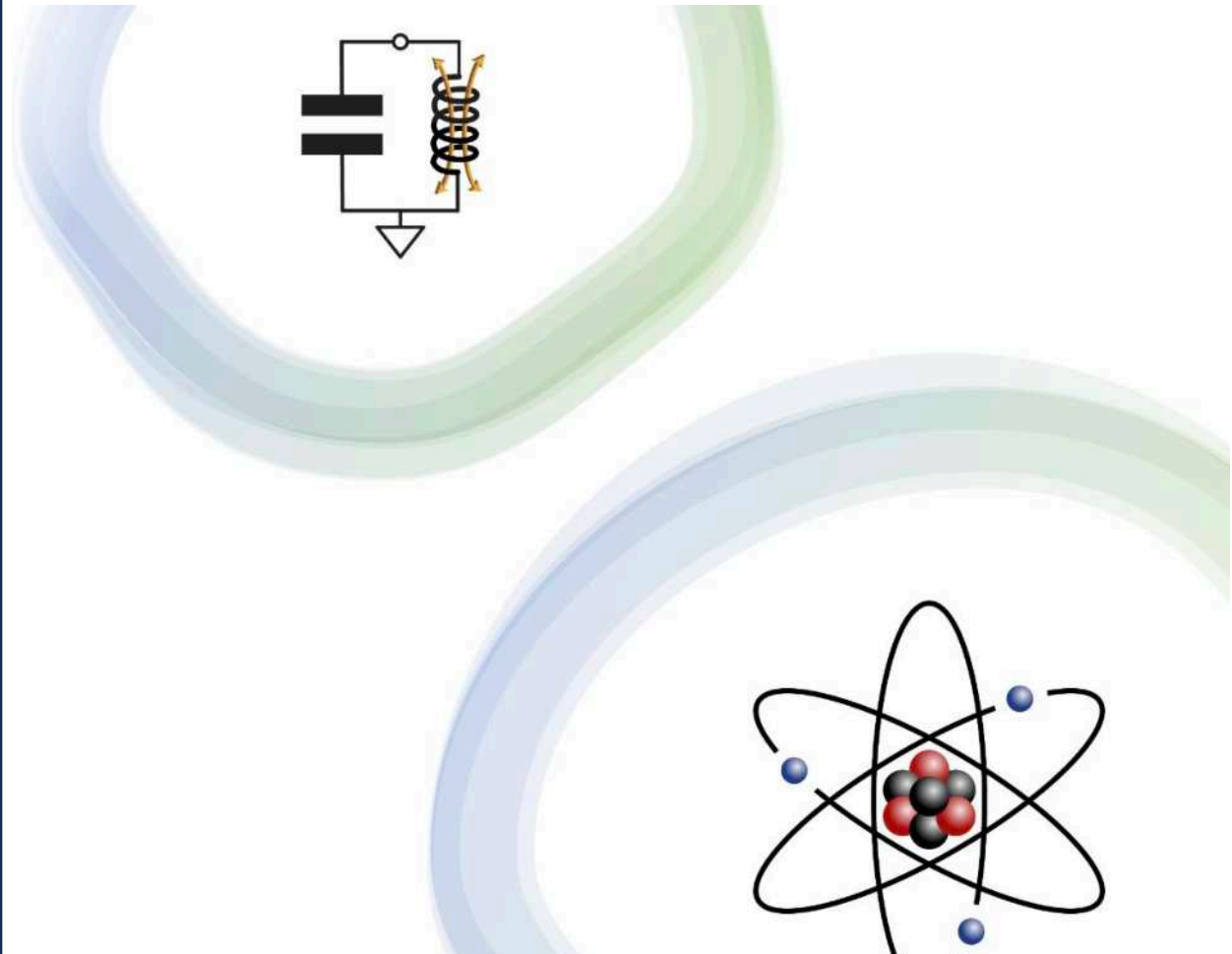
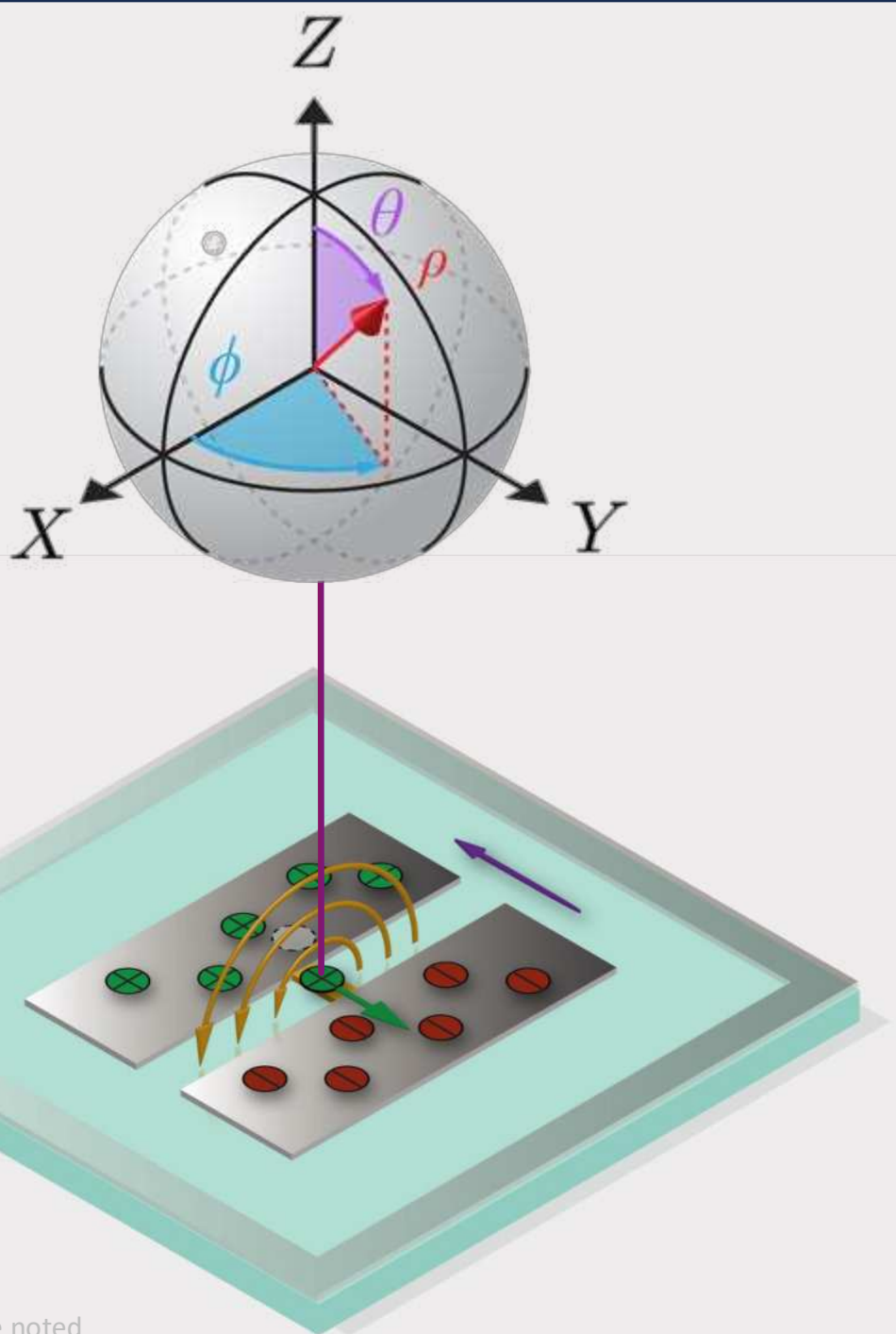
Zlatko's Blog

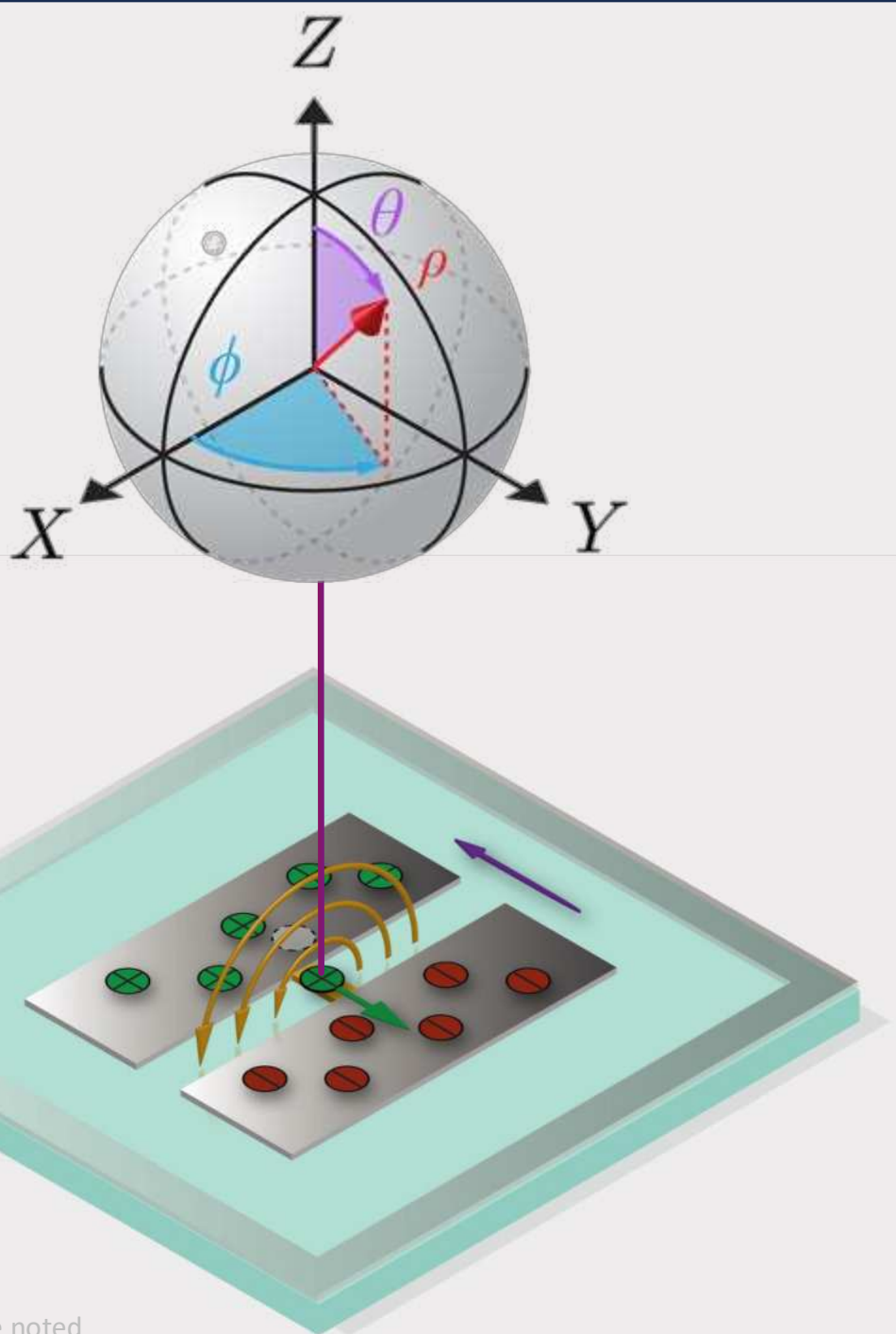
Casual notes, tutorials, and thoughts on quantum computing. Organized by categories:

- 📚 **Tutorial + Lecture** - Summer school style lectures / tutorials
- 🧐 **Tech Note** - Useful derivations and technical notes
- 🗣️ **Talk Slides** - Research paper talks, slides, videos
- 📝 **Non-Tech** - Non technical blog

zlatko-minev.com/blog

Circuit Quantum Electrodynamics



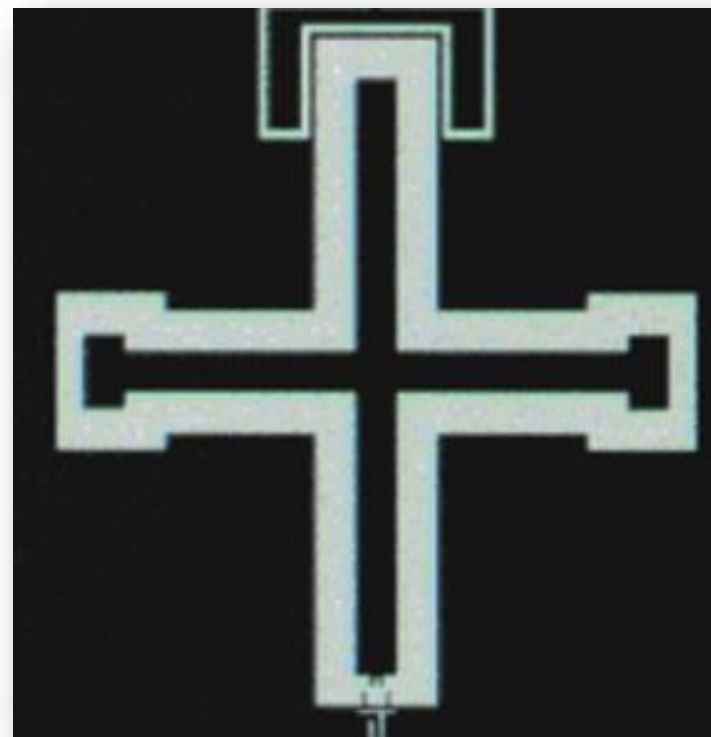
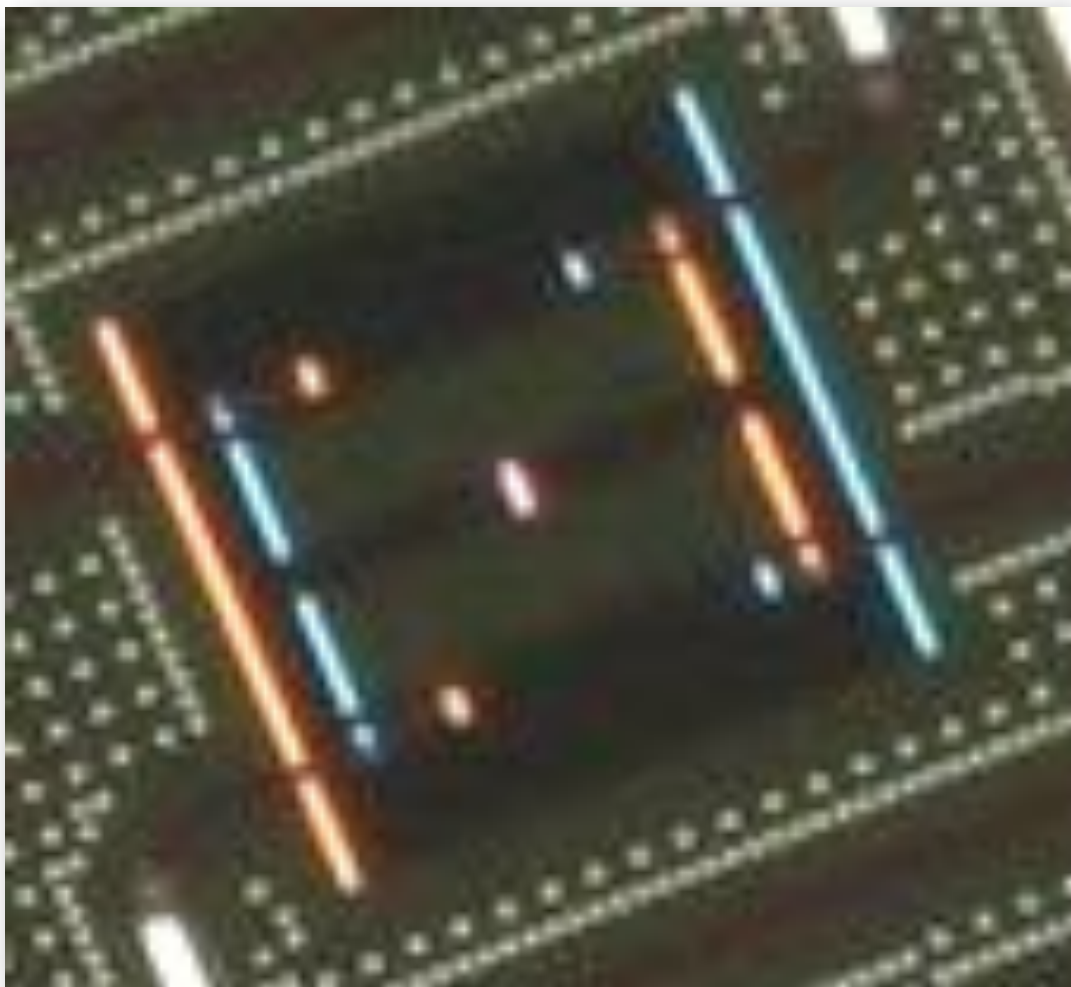


To quantum,

start classical

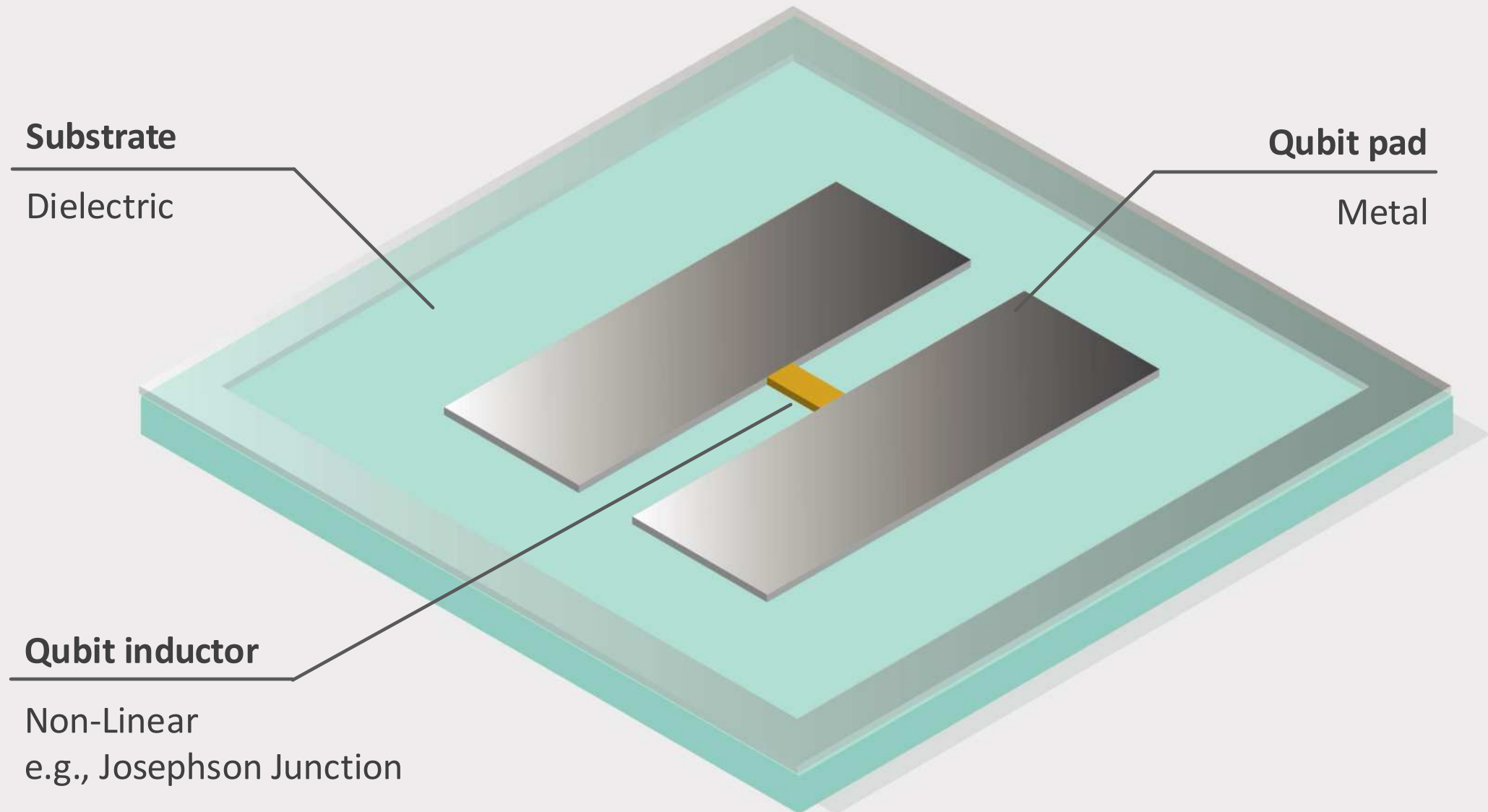
Classical oscillator

Transmon qubit

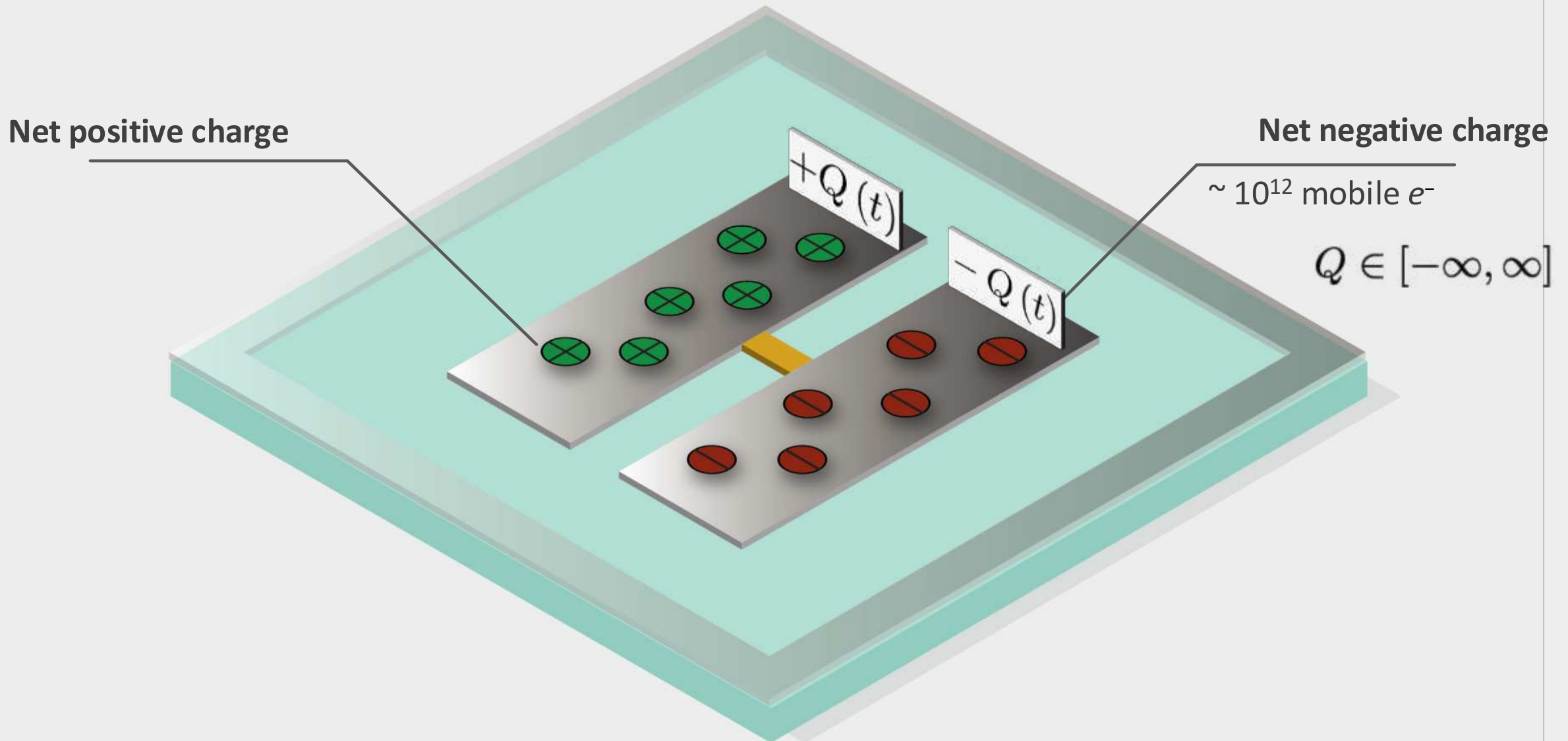


...

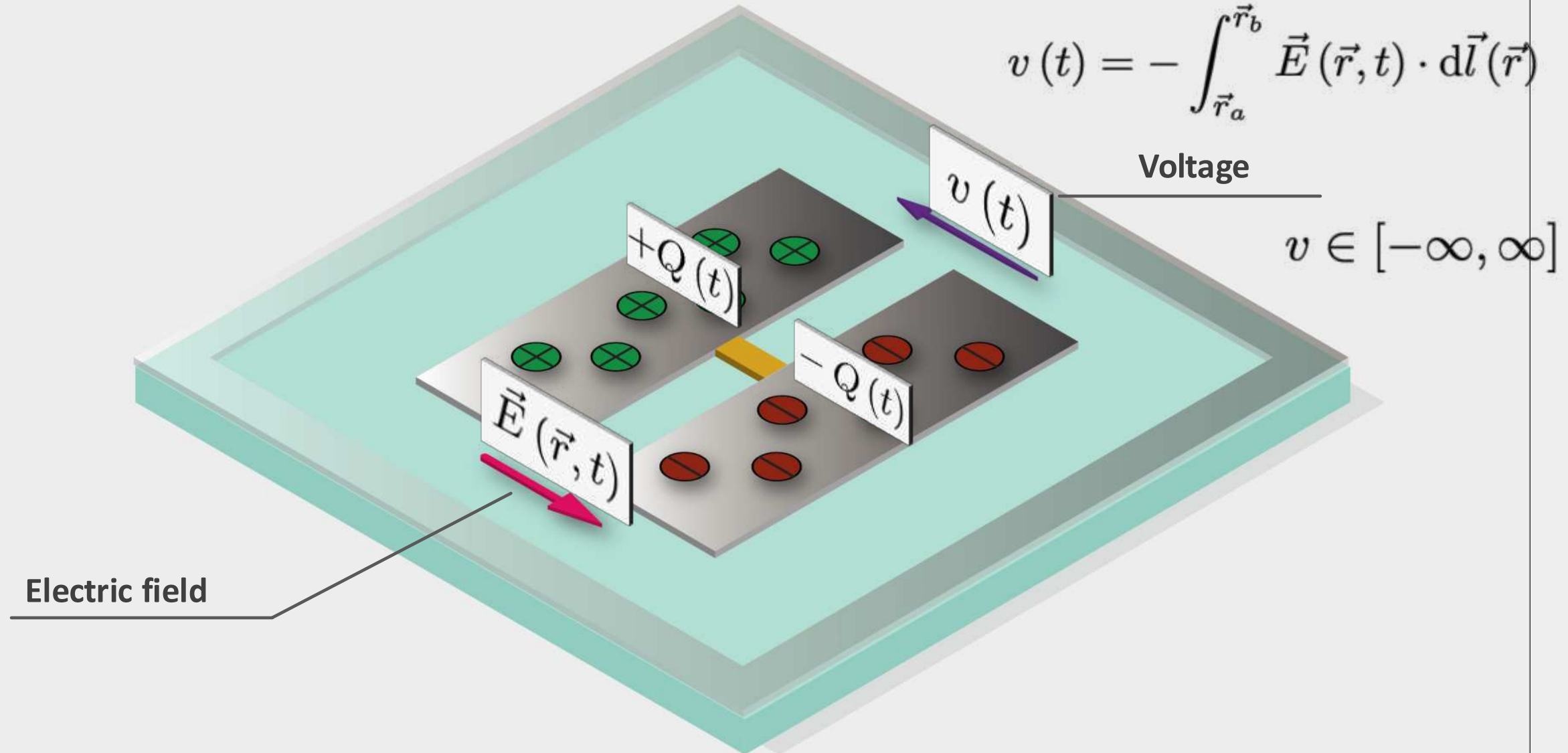
Transmon qubit



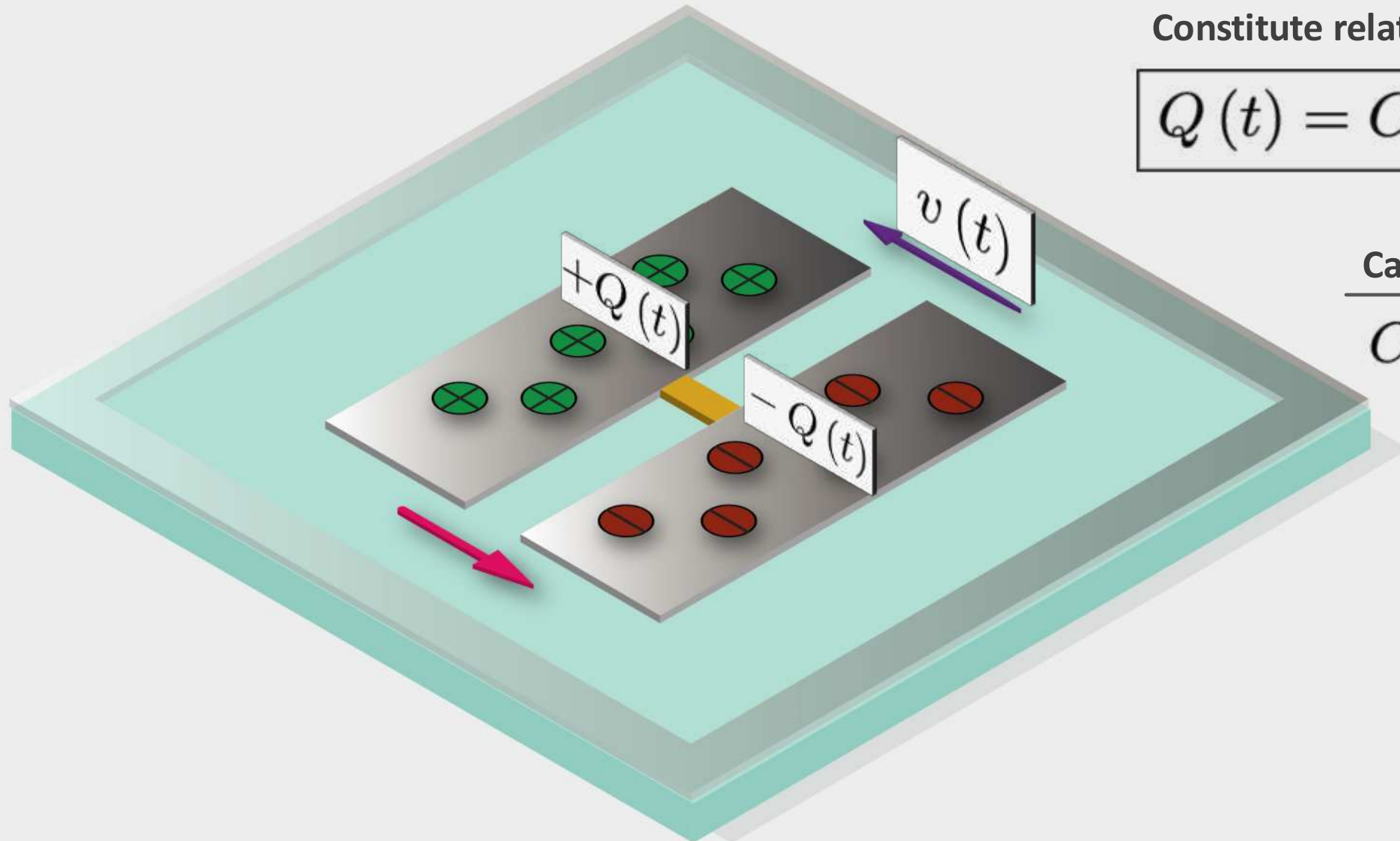
Transmon qubit: charge



Electric field and voltage



Charge and capacitance

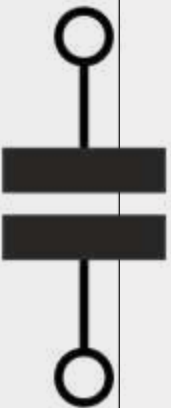


Constitute relationship

$$Q(t) = C v(t)$$

Capacitance

$$C \in (0, \infty)$$



For a good discussion, see "The Feynman Lectures on Physics Vol. II
Ch. 22: AC Circuits." Caltech.

Charge and current

Conservation of charge

Universal relationship

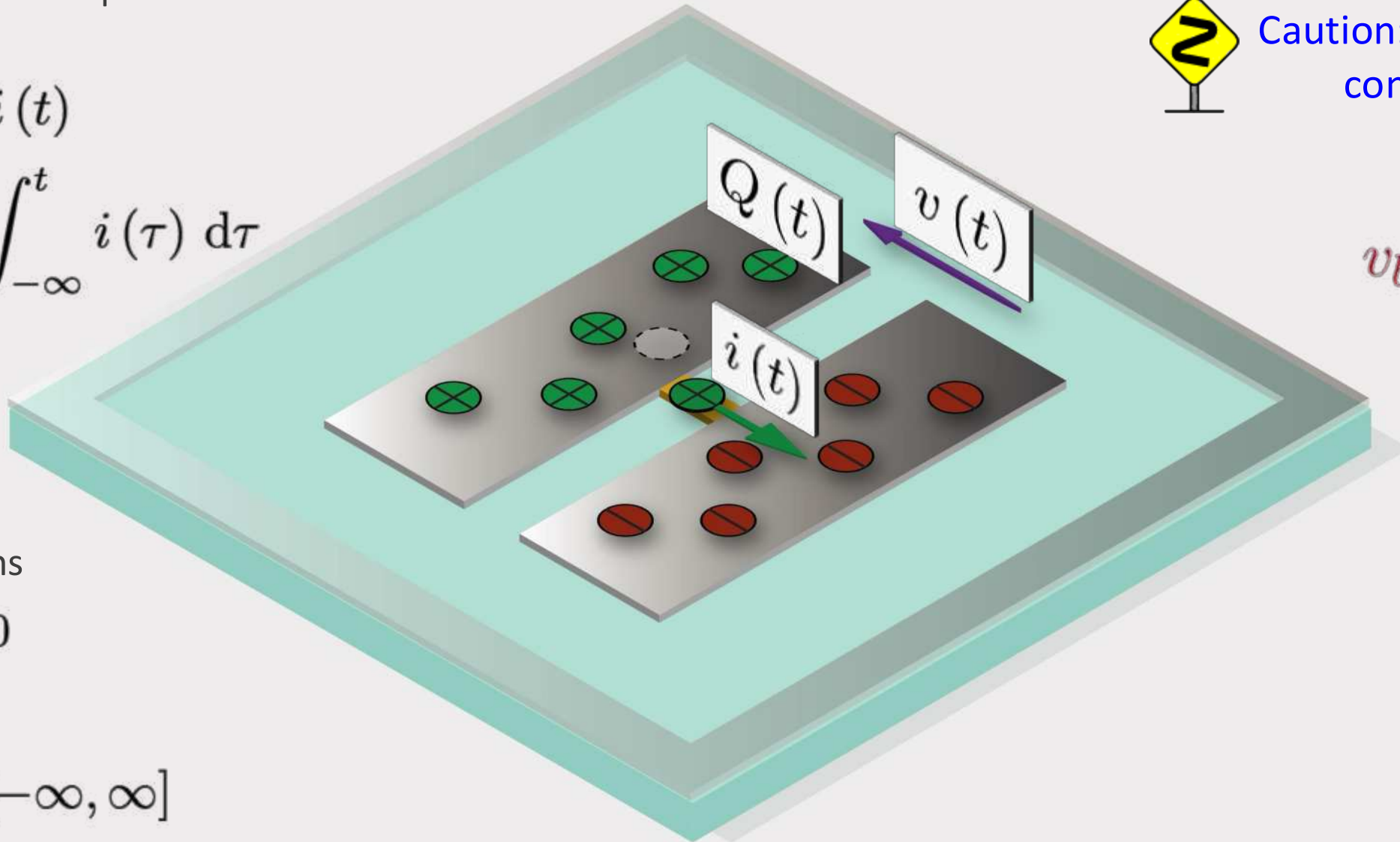
$$\frac{d}{dt}Q(t) = i(t)$$

$$Q(t) = \int_{-\infty}^t i(\tau) d\tau$$

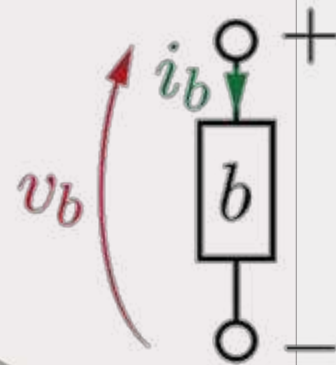
Initial conditions

$$Q(-\infty) = 0$$

$$i \in [-\infty, \infty]$$



Caution: Passive sign convention



Magnetic flux and inductance

Faraday's law of induction

Universal relationship

$$\Phi(t) = \int_{-\infty}^t v(\tau) d\tau$$

$$\frac{d}{dt}\Phi(t) = v(t)$$

Initial conditions

$$\Phi(-\infty) = 0$$

Magnetic flux

For kinetic inductors,
~98% of qubit inductive energy is not
in stored magnetic fields, but in
kinetic inductance

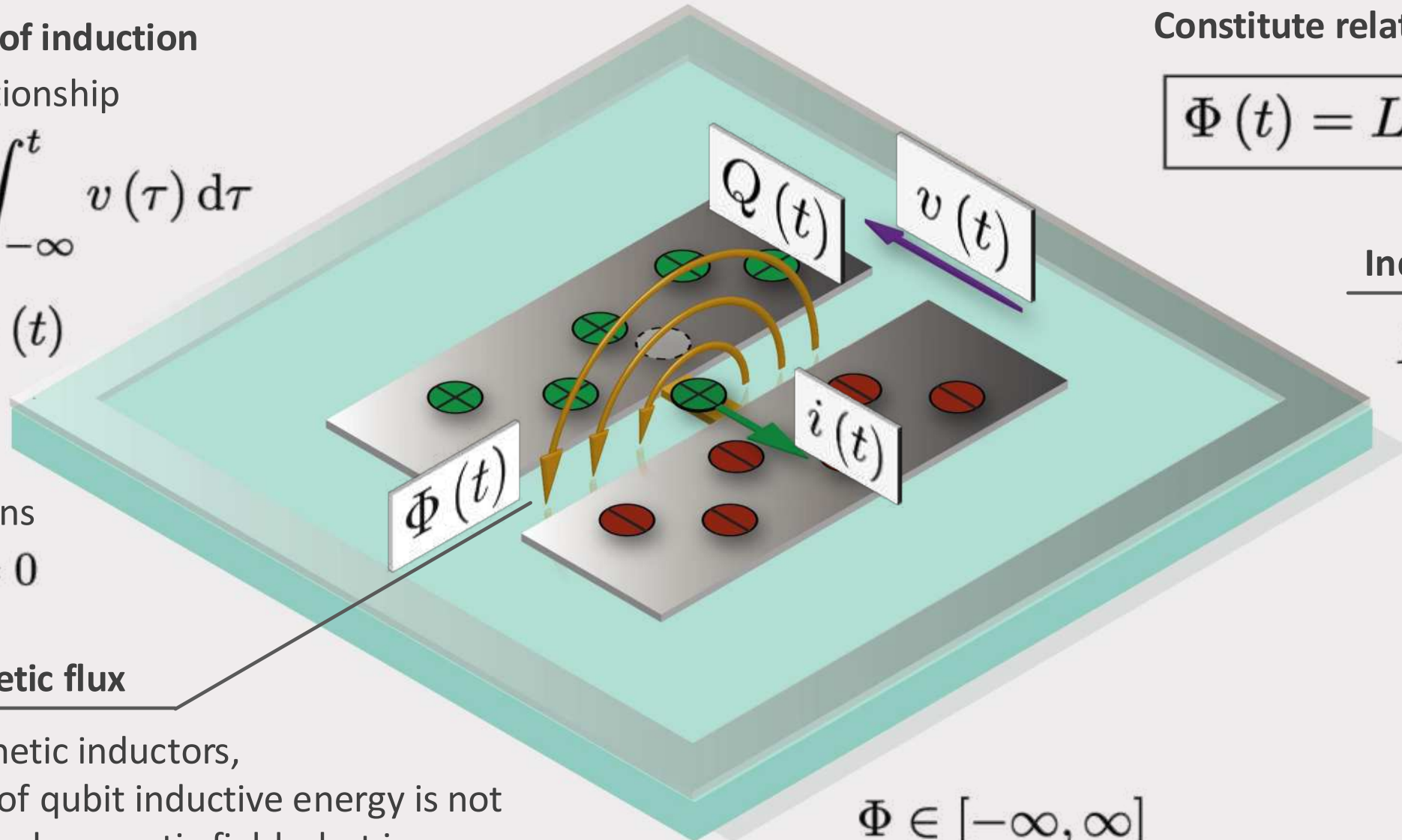
Constitute relationship

$$\Phi(t) = Li(t)$$

Inductance

$$L \in (0, \infty)$$

$$\Phi \in [-\infty, \infty]$$



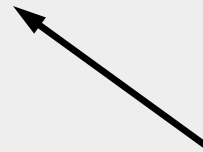
Power and energy

Universal

$$\frac{d}{dt}\mathcal{E}(t) = p(t) \equiv v(t)i(t)$$

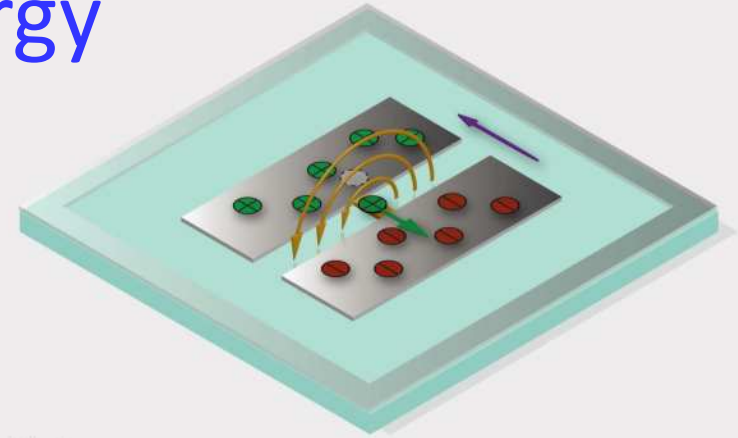


Energy stored *in*
(delivered *to*)
component



Instantaneous
power flowing *to*
component

$$\mathcal{E}(t) = \int_{-\infty}^t p(\tau) d\tau$$



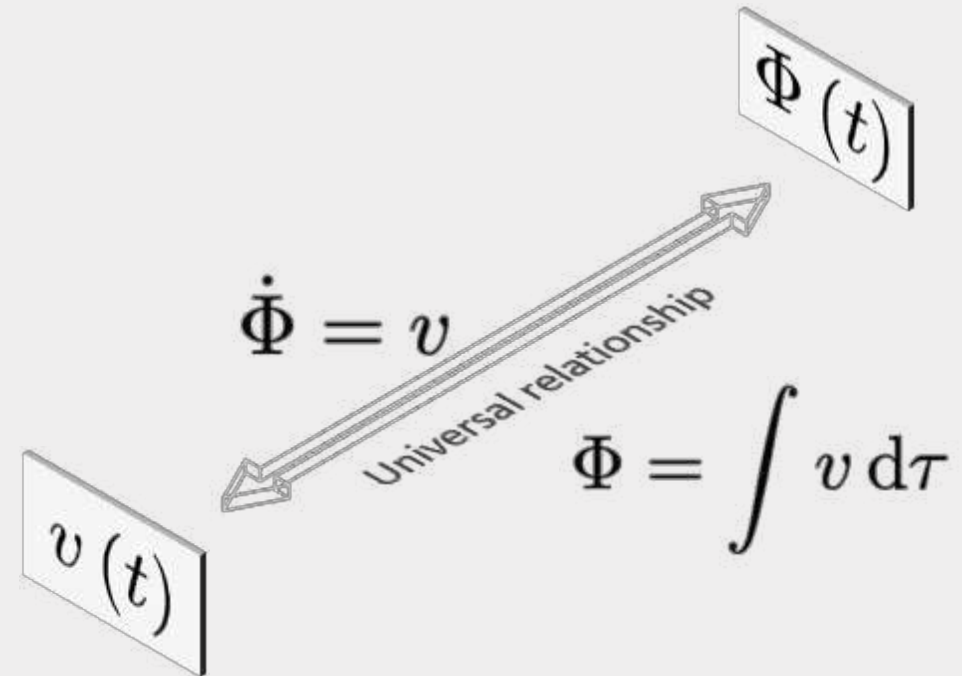
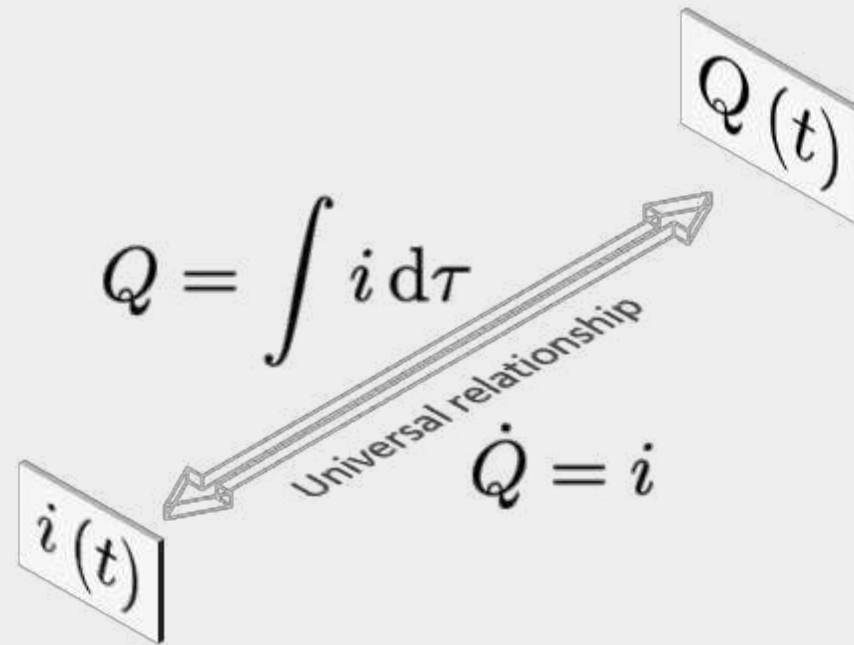
$$\mathcal{E}_{\text{cap}}(\dot{\Phi}) = \frac{1}{2}C\dot{\Phi}^2$$

$$\mathcal{E}_{\text{ind}}(\Phi) = \frac{\Phi^2}{2L}$$

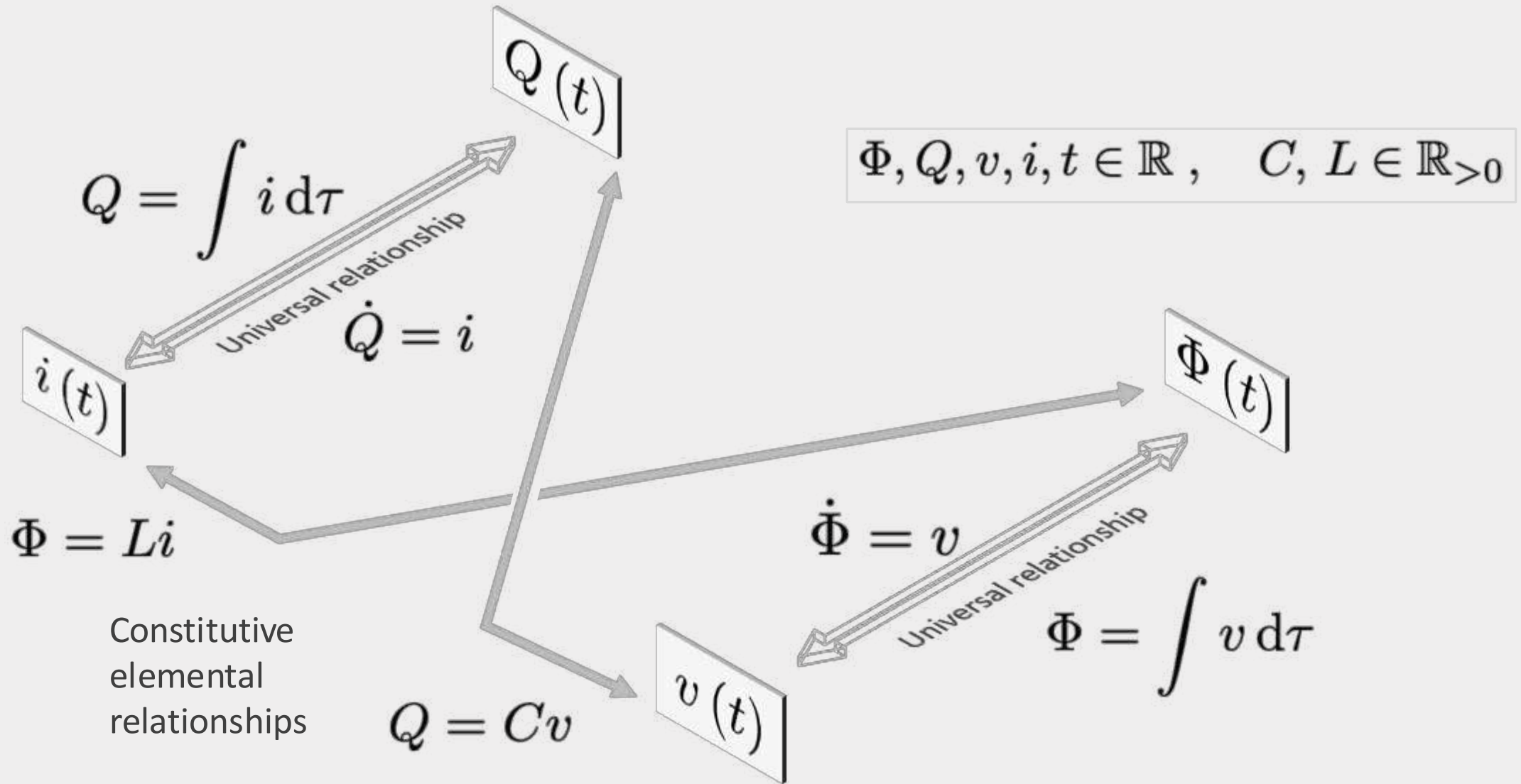
$$\mathcal{E}_{\text{cap}}(Q) = \frac{Q^2}{2C}$$

$$\mathcal{E}_{\text{ind}}(\dot{Q}) = \frac{1}{2}L\dot{Q}^2$$

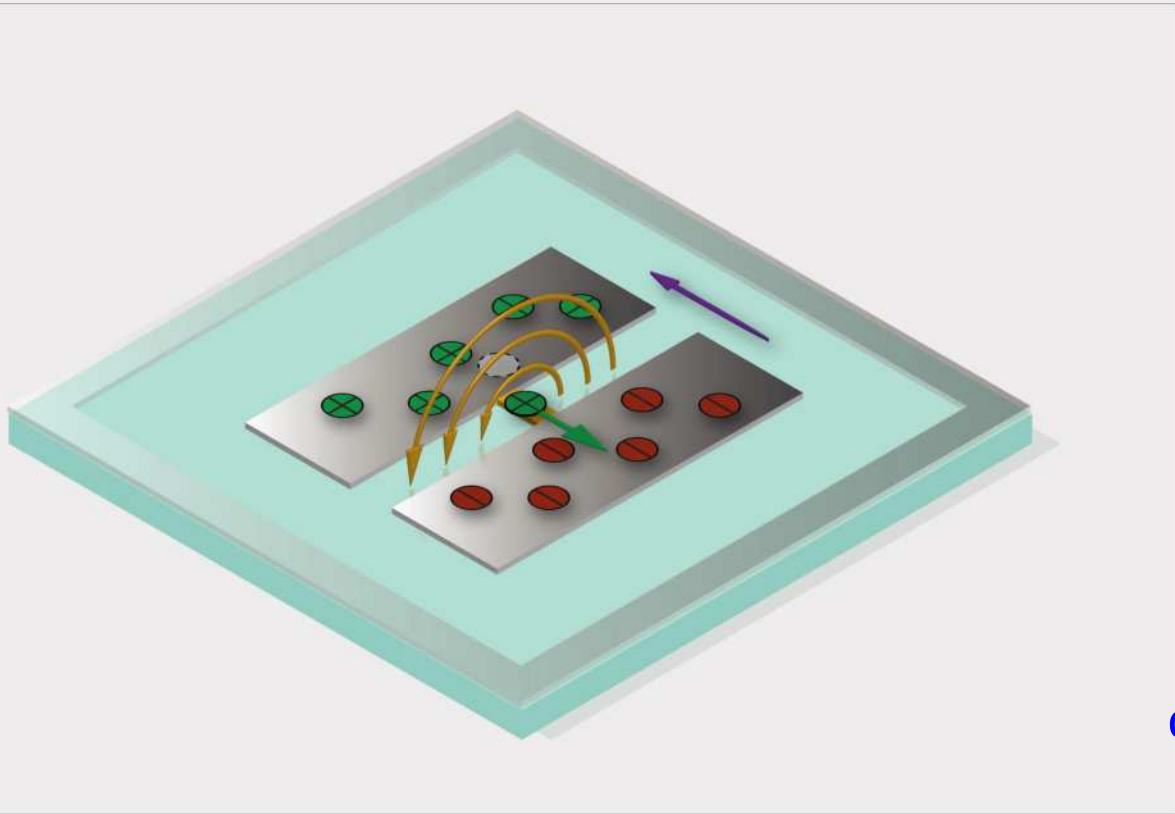
Four fundamental manifestations of electricity



Four fundamental manifestations of electricity



Electromagnetic oscillator



capacitance value

node of the circuit

C

L

inductance
value

inductor

capacitor

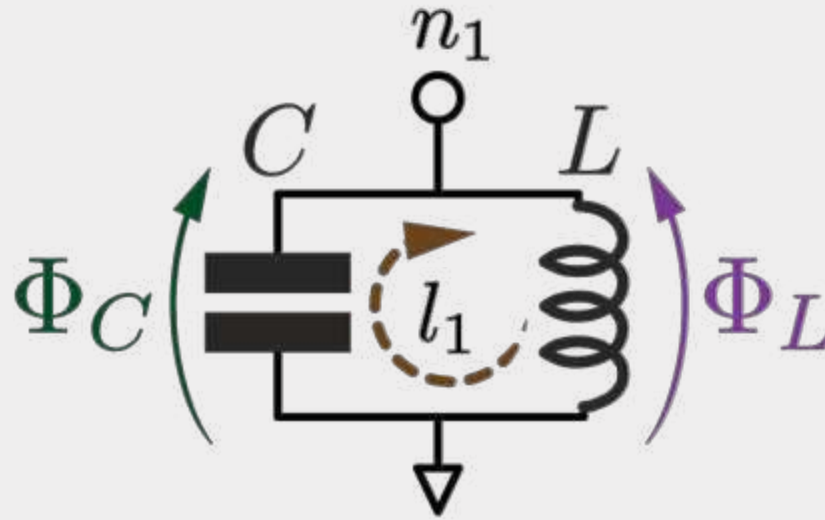
ground

Kirchhoff's network laws*

Conservation of charge

Kirchhoff's current law

$$\sum_{b \in \text{node}} \pm \dot{Q}_b(t) = 0$$



Faraday's law of induction

Kirchhoff's voltage law

$$\sum_{b \in \text{loop}} \pm \dot{\Phi}_b(t) = 0$$

$$n_1 : \quad \dot{Q}_C + \dot{Q}_L = 0$$

$$C\ddot{\Phi} + L^{-1}\Phi = 0$$

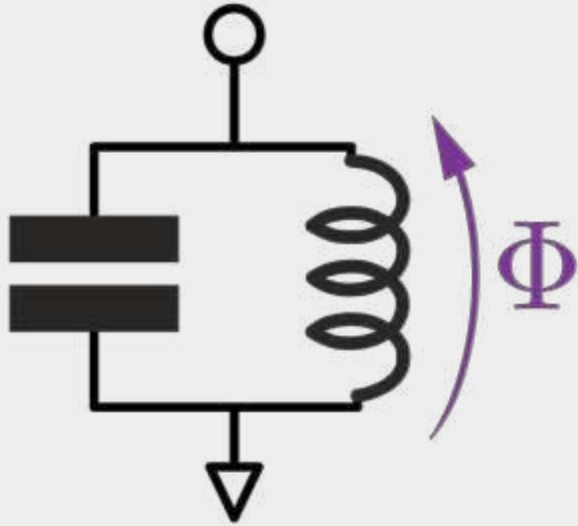
$$l_1 : \quad \dot{\Phi}_C - \dot{\Phi}_L = 0$$

$$\Phi_C = \Phi_L$$



As we will see later, for the Lagrangian description in flux basis, KVL acts as a set of holonomic constraints and KCL as the equations of motion

Oscillator analogy



$$C\ddot{\Phi} + L^{-1}\Phi = 0$$

$$\ddot{\Phi} = -\omega_0^2\Phi, \quad \omega_0 = \frac{1}{\sqrt{LC}}$$

Resonance
frequency



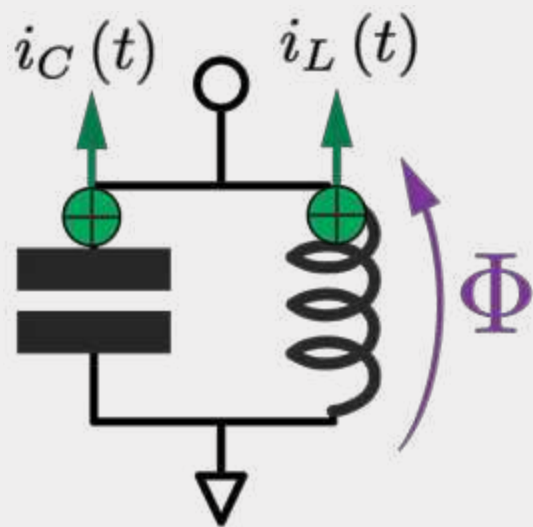
$\dot{\Phi}(t)$ Velocity

$\Phi(t)$ Deviation from equilibrium

Equilibrium position
 $x = 0, \quad \Phi = 0$



Lagrangian and Hamiltonian*



$$\underbrace{C\ddot{\Phi}}_{i_C(t)} + \underbrace{L^{-1}\Phi}_{i_L(t)} = 0$$

$$\mathcal{L}(\Phi, \dot{\Phi}) = \mathcal{E}_{\text{cap}}(\dot{\Phi}) - \mathcal{E}_{\text{ind}}(\Phi)$$

$$= \frac{1}{2}C\dot{\Phi}^2 - \frac{\Phi^2}{2L}$$

$$Q = \frac{\partial \mathcal{L}}{\partial \dot{\Phi}} = C\dot{\Phi}$$

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{\Phi}} = \frac{\partial \mathcal{L}}{\partial \Phi}$$

$$\mathcal{H}(\Phi, Q) = \frac{Q^2}{2C} + \frac{\Phi^2}{2L}$$

Kinetic energy – Potential energy

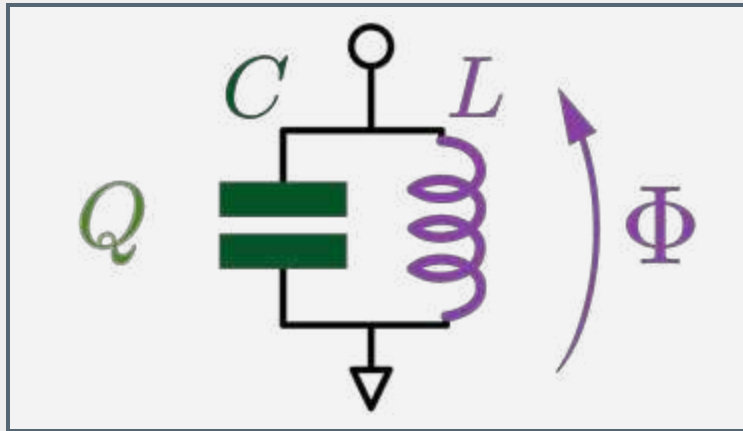
Canonically conjugate variable
(momentum; charge)

Euler-Lagrange equations, $F = ma$

Kinetic energy + Potential energy

* Temporarily going to assume some minimal knowledge of classical mechanics.
Since we eliminated the KVL constraints, this is a Lagrangian of the second kind.

The LC classical harmonic oscillator

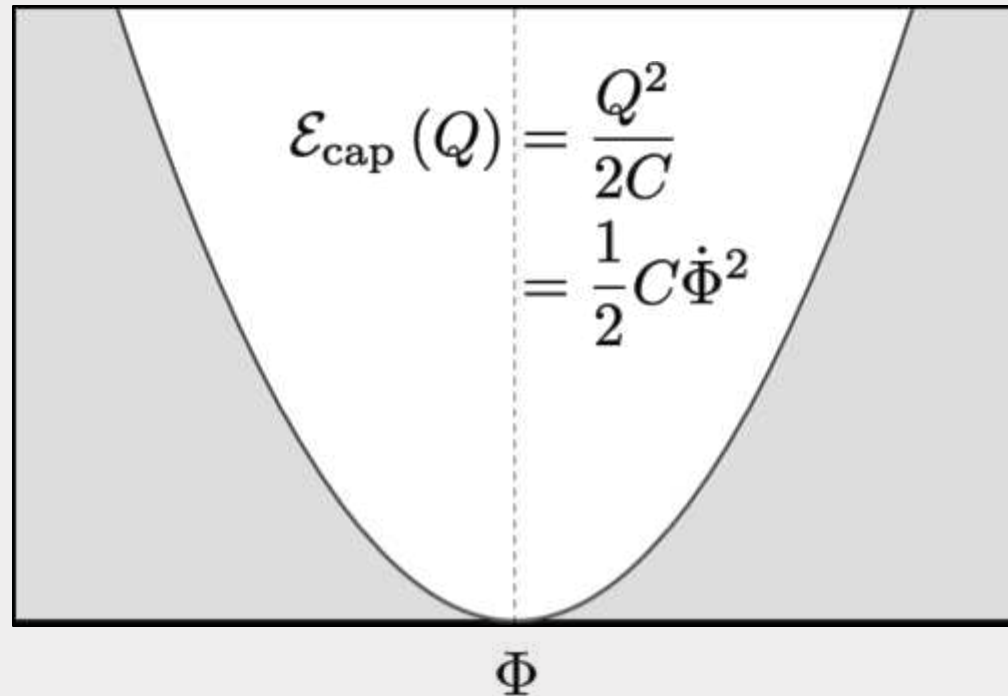


Position: $\Phi \mapsto x$
 Mass: $C \mapsto m$

Momentum: $Q \mapsto p$
 Spring constant: $L^{-1} \mapsto k$

$$\mathcal{E}_{\text{ind}}(\Phi) = \frac{\Phi^2}{2L}$$

Energy



$$\mathcal{H}(\Phi, Q) = \frac{Q^2}{2C} + \frac{\Phi^2}{2L}$$

$$\omega_0^2 = \frac{1}{LC} \quad Z_0 = \frac{L}{C}$$

“It is by *logic* that we prove,
but by *intuition* that we discover.”

Henri Poincaré

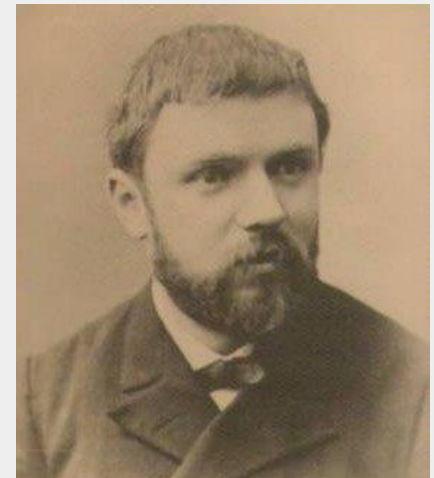


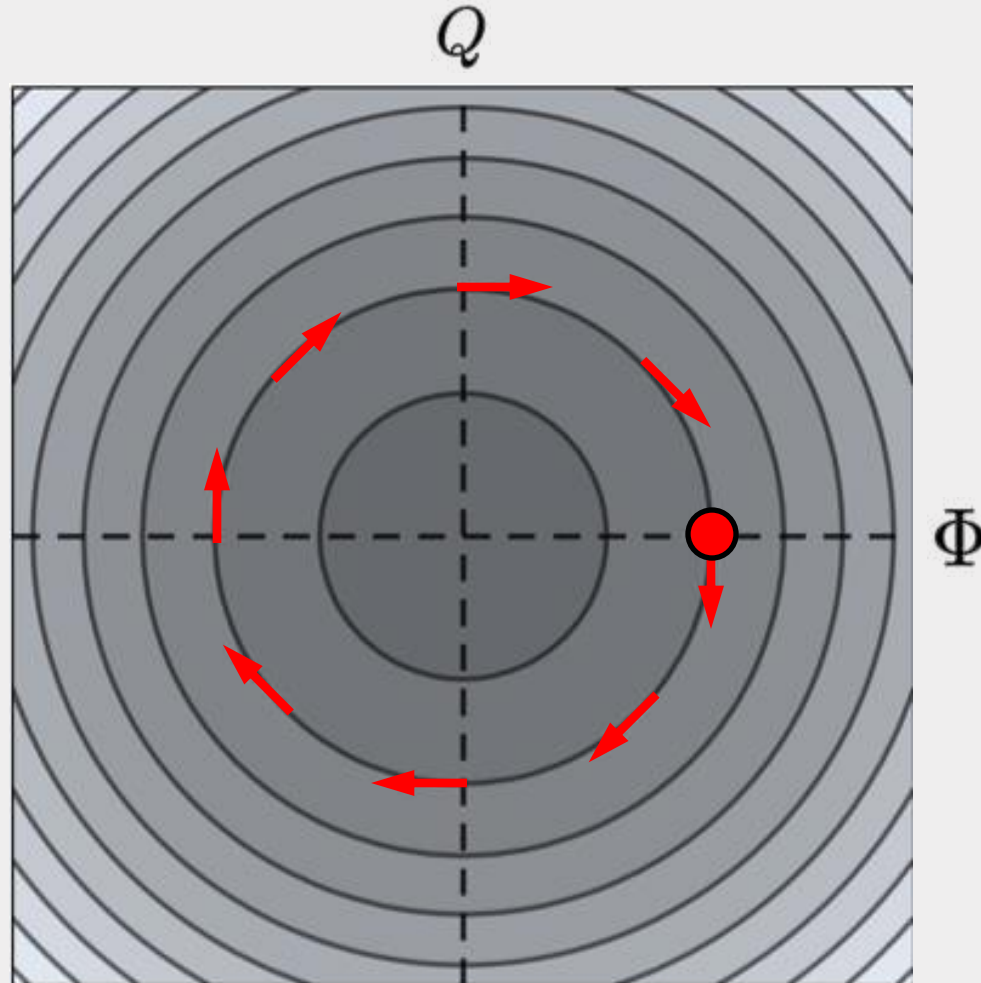
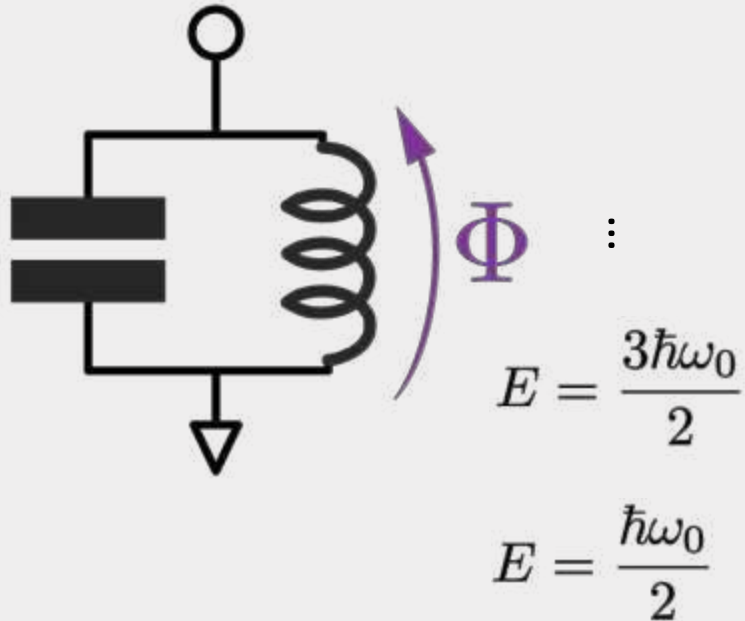
Photo by Eugène Pirou

Hamiltonian dynamics and phase space

$$\mathcal{H}(\Phi, Q) = \frac{Q^2}{2C} + \frac{\Phi^2}{2L}$$

with $L = C = 1$

$$E = \hbar\omega_0 \left(n + \frac{1}{2} \right) = \frac{1}{2} (Q^2 + \Phi^2)$$



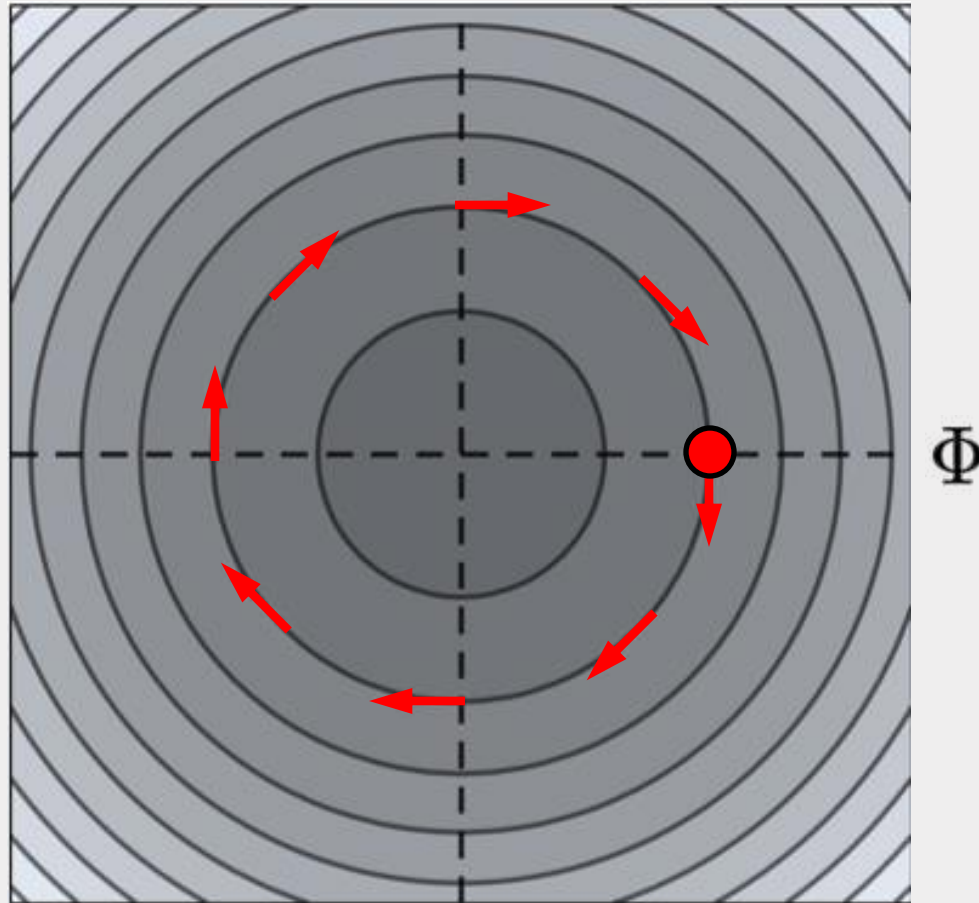
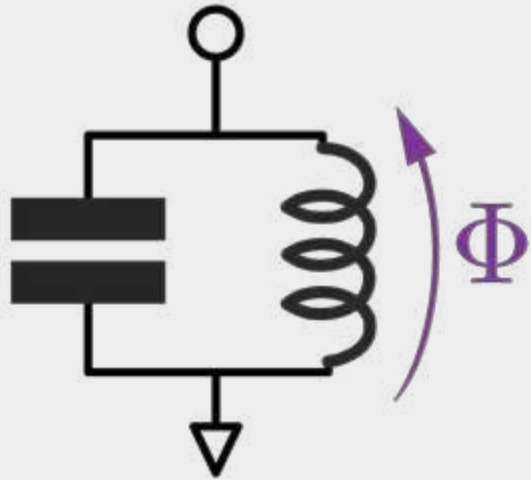
$$\dot{\Phi} = +\frac{\partial \mathcal{H}}{\partial Q} = +Q$$

$$\dot{Q} = -\frac{\partial \mathcal{H}}{\partial \Phi} = -\Phi$$

Complex action-angle variable

$$\mathcal{H}(\Phi, Q) = \frac{Q^2}{2C} + \frac{\Phi^2}{2L} \quad E = \hbar\omega_0 \left(n + \frac{1}{2} \right) \quad \alpha(t) = \sqrt{\frac{1}{2\hbar Z}} [\Phi(t) + iZQ(t)]$$

Q



Classical analog of the bosonic ladder operator

$$\alpha(t) = \alpha(0) e^{-i\omega_0 t}$$